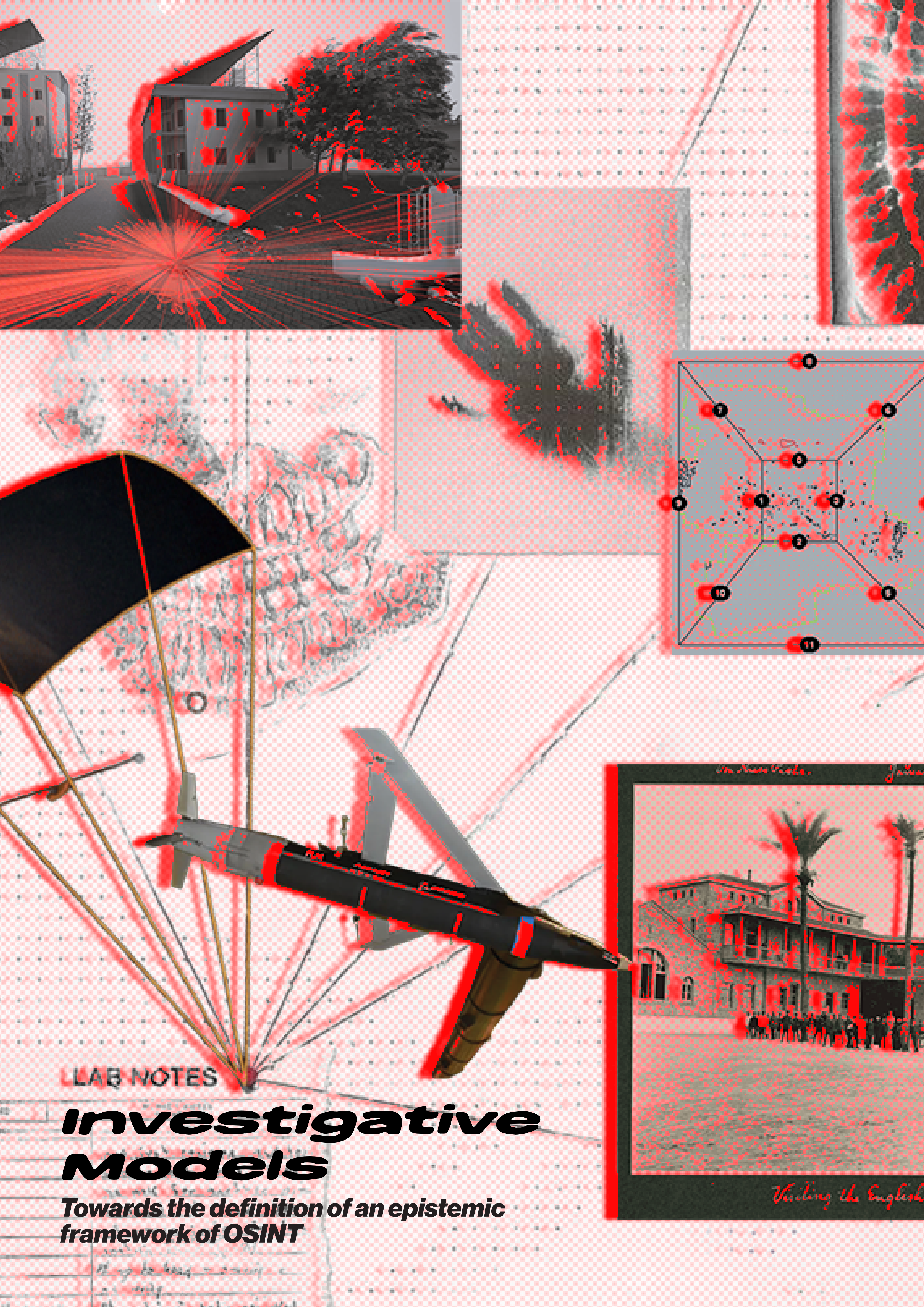

LAB NOTES

Investigative Models

Towards the definition of an epistemic framework of OSINT

Investigative Models

Towards the definition of an epistemic
framework of OSINT

Master Thesis
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Master Thesis by:
Engy El Shenawy - 4045625
Federico Zurani - 4044826

Supervised by:
Prof. Albert Lang
Prof. Dr. Marc Pfaff
Dr. Peter Polack

Declaration of Authorship

We hereby declare that the thesis submitted is our own, unaided work, completed without any external help. Only the sources and resources listed were used. All passages taken from the sources and aids used, either unchanged or paraphrased, have been marked as such. We further declare that we have not submitted the thesis in the same or similar form to any other examination authority.

Engy El Shenawy	Federico Zurani	Place, Date

Universität der Künste Berlin

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Glossary

This glossary defines key terms as they are used throughout the thesis. Definitions are intentionally concise, operational, and context-specific. The glossary functions as a working vocabulary for analysing evidential practices across historical, forensic, machinic, and open-source investigative regimes.

Term	Definition/Description
Evidential paradigm (Ginzburg)	A mode of knowledge production grounded in the interpretation of indicia – small, marginal signs that point beyond themselves. Knowledge emerges through conjectural, abductive reasoning rather than deduction or statistical generalisation. The paradigm assumes that traces pre-exist their interpretation and that epistemic work consists in reconstructing coherence across fragmentary signs.
Indicia (clues)	Indica often contain ambiguous details (footprints, stains, gestures, visual anomalies) that acquire meaning only through interpretation and cross-checking. In the evidential paradigm, indicia are not self-sufficient proofs but relational signs whose epistemic force depends on narrative synthesis and interpretive authority.
Trace (Ginzburg)	A material or perceptual remainder of an event that pre-exists its interpretation. The trace functions indexically, pointing retrospectively to a hidden cause. Its epistemic force derives from the investigator's capacity to read it conjecturally rather than from its intrinsic stability.
Trace (Rheinberger)	An inscription produced within an experimental system through the interaction of epistemic things and technical apparatuses. Traces do not precede inquiry but are generated by it. They initially appear as "asemic kernels" and gain epistemic force only once stabilised, stored, and transformed into data.
Trace (OSINT / digital investigations)	A hybrid epistemic entity that may originate externally (videos, sensor data, images) but becomes evidential only through system-level operations such as synchronisation, translation, and modelling. Unlike Ginzburgian traces, OSINT traces are often unstable, transient, and platform-dependent; unlike Rheinbergerian traces, they are rarely generated through controlled experimental interaction. Their epistemic force is therefore contingent on system design rather than discovery or experimentation alone.

Indexicality	The relation by which a trace bears a causal or material link to what produced it. In digital and OSINT contexts, indexicality is mediated, procedural, and often relational rather than direct, and it requires system-level reconstruction to be asserted.
Conjectural knowledge	A probabilistic mode of knowing that moves from fragments to hypotheses through abductive reasoning. Central to the evidential paradigm, it privileges interpretation over verification and remains structurally open to revision.
Abductive inference	Inference to the best explanation: reasoning from a puzzling detail toward a plausible causal account. Operative in both detective logic and OSINT trace interpretation, but insufficient on its own for system-based epistemic reliability.)
Detective figure	A paradigmatic interpreter of traces (e.g. Sherlock Holmes) whose authority depends on interpretive acuity and narrative performance. Serves as the historical model for conjectural epistemology.
Narrative performance	The rhetorical staging through which evidentiary coherence and certainty are produced. Narrative performance stabilises meaning but may foreclose epistemic openness and comparability.
Forensic gaze	A disciplined mode of seeing that seeks meaning in details, deviations, and marginal signs. Operates at the intersection of aesthetics and epistemology and precedes formal forensic science.
Era of the witness	A postwar evidentiary regime in which embodied human testimony becomes the primary bearer of truth. Knowledge is grounded in presence, voice, and lived experience rather than material traces alone.
Witness	A truth-bearing figure whose epistemic authority shifts historically – from embodied speaker to distributed, mediated, and machinic forms of observation.
Invisible witness	A condition in which witnessing is displaced into technical systems of vision (surveillance, sensors, algorithms) that record without memory, affect, or narrative agency.
Era of the thing	A forensic regime in which objects (bones, artifacts, digital files) function as "super-witnesses." Truth is produced through expert mediation, visualisation, and scientific rhetoric rather than testimony.
Forensic object	A material artefact rendered evidential through analysis, visualisation, and institutional framing. Its authority is constructed rather than intrinsic

Forensic aesthetics	The joint operation of material analysis and visual–rhetorical persuasion through which evidence is made to appear in public forums.
Forensic forum	A hybrid space (courtroom, laboratory, media platform) in which evidence is staged, contested, and validated across multiple epistemic communities.
Machinic vision	Automated seeing through sensors, satellites, drones, and algorithms that generate images beyond human perception and without phenomenological experience.
Operational images	Images are produced to act within systems (detect, target, predict) rather than to be seen as representations. Central to contemporary military and surveillance regimes
Threshold of detectability	The limit at which events become perceptible only through technical systems and expert interpretation rather than direct human observation.
Dark epistemology	A condition in which the infrastructures of seeing (algorithms, classified systems, proprietary platforms) are opaque, making visibility itself contestable.
OSINT (Open-Source Investigation)	Investigative practices using publicly available data to reconstruct events and build evidentiary claims through digital infrastructures.
Counter-forensics	Investigative practices that contest state or institutional narratives by reinterpreting operational images and data.
Investigative commons	A distributed infrastructure of tools, methods, and publics through which evidentiary practices are shared, iterated, and debated.
Forensic object	A material artefact rendered evidential through analysis, visualisation, and institutional framing. Its authority is constructed rather than intrinsic
Forensic aesthetics	The joint operation of material analysis and visual–rhetorical persuasion through which evidence is made to appear in public forums.
Forensic forum	A hybrid space (courtroom, laboratory, media platform) in which evidence is staged, contested, and validated across multiple epistemic communities.
Machinic vision	Automated seeing through sensors, satellites, drones, and algorithms that generate images beyond human perception and without phenomenological experience.

Aesthetic system	Definition given by Fuller and Weizman, where human and non-human actors participate in a complex network that is both “sensing”(registers information) and “sense-making” (reads and contextualises the information). From this network of actors (aesthetic system), new representations of facts emerge. These systems welcome new entities, communities, frameworks or elements, while also unsettling Western-centred epistemologies and reconfiguring the relationship between power and the construction of truth through narrative. Against an idea of objective and universal representation of facts, a new investigative system should help challenge these overarching mainstream narratives and give space to the marginalised ones. From a more technical perspective, these systems could be described as situated and perspectival.
Data translation	The transformation of heterogeneous traces into a shared representational environment (timelines, maps, models). Central to OSINT epistemic production and distinct from trace interpretation.
Model (Rheinberger)	A flexible scaffold in data space that allows variation, exposes dependencies, and supports comparison. Models are not representations of reality but epistemic instruments that may generate new questions and relations.
Model (OSINT / Forensic Architecture)	A constructed spatial, temporal, or relational environment (often 3D) in which heterogeneous data are aligned. Its epistemic value depends on openness, explicit assumptions, and comparability rather than visual realism.
Data space	A shared representational environment in which translated data can be manipulated, recombined, and compared.
Experimental system	A bounded techno-epistemic arrangement that generates knowledge through iterative manipulation rather than hypothesis verification. Epistemic reliability emerges through constrained openness and internal differentiation.
Epistemic thing	The open-ended object of inquiry that is not fully known in advance and becomes graspable only through system operations.
Technical object	A stabilised method, tool, or procedure that was once epistemically novel and now functions reliably within a system.
Internal objectivity	represents the processes or methods by means of which science investigates the world; so it might also appropriately be called “methodological objectivity”: the internal objectivity lies in the capability of a system to create a consistent and stable outcome.

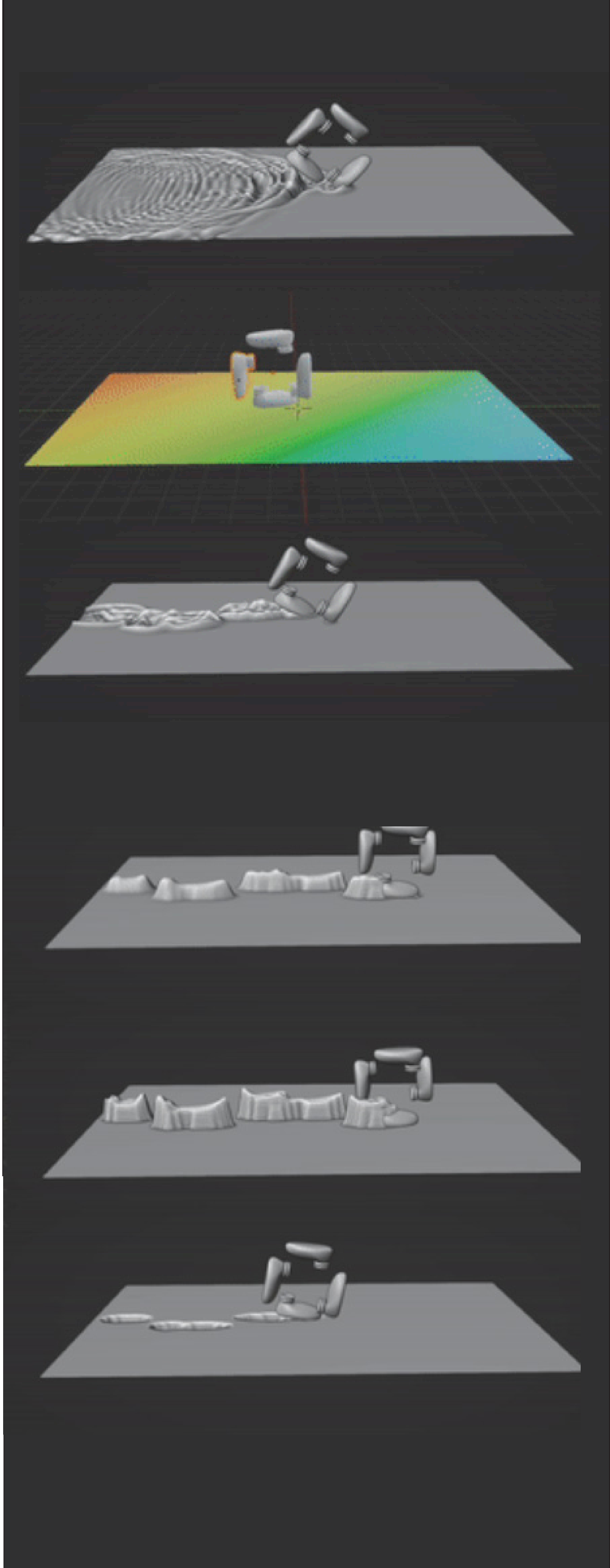
Joseph F. Hanna, “The Scope and Limits of Scientific Objectivity,” Philosophy of Science 71, no. 3 (2004): 340a approach,approach,.

External objectivity	represents “an entity, process, magnitude, construct which “existence and properties are independent of its representation” (the object is totally independent from subjective interferences)
Grafting	The hybridisation of methods or tools across domains to extend an investigative system’s epistemic capacity.
Investigative system/Hybrid system	Investigations that follow a model-led approach. The original traces (pieces of evidence) are of the type described by Guinzburg, while the system design coincides with scientific models in Rheinberger´s sense.
Reliability	We use this term to define how and when an investigation’s output can be considered valid, in relation to the epistemic paradigm in which it was generated. If we are dealing with a conjecture-led approach, reliability depends on the quality and quantity of the evidence, whereas if we are dealing with model-led approaches the reliability describes the limits and possibilities of the model.
Conjecture-led	Defines a type of investigation where conjectural reasoning and traces (indicial evidence) define its epistemic approach. Based on external objectivity.
Model-led	Type of investigation based on data translation and data mapping processes, where new data/findings emerge out of the process. Based on internal objectivity.
Micro epistemology	It is a “narrow-scope” or “localised” approach to the theory of knowledge that focuses on the specific, practical, and everyday processes by which knowledge is constructed, particularly in science and technology. Unlike traditional, top-down epistemology, which seeks universal, foundational laws of knowledge, microepistemology studies the detailed techniques, social interactions, and material tools (e.g., in a laboratory) used to create knowledge “from the ground up”.
STS science and technology studies	An interdisciplinary field that examines how social, cultural, and political contexts shape science and technology, and how these, in turn, shape society. It moves beyond viewing science as objective truth, instead analysing facts and technologies as socially constructed, co-produced, and inseparable from human values.

List of Figures

List of Abbreviations

Abbreviation	Full Term
AP	Associated Press
BBC	British Broadcasting Corporation
CGI	Computer-Generated Imagery
CNN	Cable News Network
CO19	Metropolitan Police Specialist Firearms Command
FA	Forensic Architecture
GBU-39	Guided Bomb Unit-39 (Small Diameter Bomb)
GBU-53	Guided Bomb Unit-53 (StormBreaker)
HRW	Human Rigwhts Watch
IDF	Israel Defence Forces
IPCC	Independent Police Complaints Commission
NGO	Non-Governmental Organisation
NYT	New York Times
OSINT/OSI	Open-Source Intelligence / Open-Source Investigation
PIJ	Palestinian Islamic Jihad
STS	Science and Technology Studies
UN	United Nations
VR	Virtual Reality
WaPo	Washington Post



Abstract

This thesis proposes a revised epistemological framework for *Open Source Intelligence*[1] (OSINT) by shifting the analytical focus from investigative outcomes to the conditions of knowledge production. It identifies two dominant epistemic paradigms within contemporary OSINT practice: *conjecture-led investigation*, which relies on abductive reasoning and the interpretation of discrete evidential traces; and *model-led investigation*, which constructs knowledge through data translation and data mapping, thereby generating a space in which new evidence can emerge. We argue that these paradigms correspond to two distinct understandings of objectivity: model-led approaches operate under what Joseph F. Hanna defines as *internal objectivity*[2], grounding reliability in the consistency and stability of the investigative system itself, while conjecture-led investigations rely on external objectivity, locating truth in the properties of the evidence independently of its representation. This difference has direct consequences for how reliability is asserted: in conjecture-led methods, reliability depends on the quality and quantity of evidence, whereas in model-led methods, it depends on the model's stability and explicit limits.

These claims are tested through two case studies. The Forensic Architecture case: *the killing of Mark Duggan*[3] investigation serves as a methodological proving ground, through which the applicability of Rheinberger's experimental systems framework to investigative practice is examined, mapping the points of convergence and divergence between scientific experimentation and model-led inquiry. The Al-Ahli Hospital explosion (Gaza, October 2023) then extends this framework into a comparative analysis of ten distinct investigations, testing its capacity to distinguish and evaluate competing epistemic paradigms operating on the same event.

The thesis concludes that model-led investigation represents a genuine epistemic advance but is fragile: without wider adoption across the field, its sophistication risks becoming an instrument of persuasion in favour of those adopting it, a superior narrative tool rather than a reliable approach to inquiry.

1 The term open-source intelligence (OSINT) is used throughout this thesis to refer to investigative practices that draw on publicly available data to reconstruct events and construct evidentiary claims. For the practitioner's account see Higgins, *Bellingcat: Truth in a Post-Truth World* (2021); for the theoretical framing operative in this thesis see Weizman, *Forensic Architecture* (2019), 64, and Fuller and Weizman, *Investigative Aesthetics* (2021).

2 This definition is taken from Joseph F. Hanna, "The Scope and Limits of Scientific Objectivity," Philosophy of Science 71, no. 3 (2004).
Hanna distinguishes between external and internal objectivity: the first representing "an entity, process, magnitude, construct which "existence and properties are independent of its representation" (the object is totally independent from subjective interferences), ; so it might also appropriately be called "methodological objectivity": he internal objectivity lies in the capability of a system to create a consistent and stable outcome.

3 Forensic Architecture. "The Killing of Mark Duggan: Report and Methodology." Forensic Architecture, Jan. 2020

Introduction

On 17 October 2023, an explosion at al-Ahli Hospital in Gaza-Palestine killed hundreds of people. In the days, weeks, months and years that followed, the incident became the subject of more than ten separate investigations by states, media organisations, human rights groups, and independent OSINT researchers. These inquiries drew on largely overlapping open-source material, yet arrived at very different and, in some cases, mutually incompatible conclusions about the cause of the blast and damage.

What makes this divergence remarkable is not the disagreement itself but the conditions under which it occurs. Many organisations involved in the investigation shared similar intentions, professional backgrounds, and normative commitments to human rights and evidentiary rigour. Yet despite this shared horizon, their analyses of the al-Ahli Hospital explosion produced not convergence but divergence: a proliferation of competing "truths," each internally coherent, evidence-based, and often methodologically sophisticated.

This thesis begins with the assumption that **contemporary visual investigations cannot be evaluated solely by classical notions of objectivity, neutrality, or factual correctness.** The problem is not simply whether an investigation is right or wrong, but whether it constructs the epistemic conditions under which truth claims can become reliable, comparable, and contestable.

The rapid expansion of OSINT practices—driven by digital media, satellite access, and networked platforms—has brought together a heterogeneous field of actors, including intelligence agencies, investigative journalists, human rights organisations, forensics collectives, and independent researchers. Despite this diversity, the epistemic differences between their methods remain largely *under-theorised*. Existing evaluative frameworks tend to assess investigations in terms of outcome—whether a claim is credible, plausible, or legally admissible, rather than examining the structural conditions under which different kinds of knowledge claims become possible.

This thesis intervenes here. It argues that the distinction between conjecture-led and model-led investigation is not methodological but fundamentally epistemological: it defines the conditions under which truth claims are produced, stabilised, and revised. Conjecture-led investigations operate through abductive inference and narrative coherence across discrete pieces of evidence. They remain vulnerable to collapse when individual elements are challenged, as they rely heavily on the narrative. Model-led investigations, by contrast, construct knowledge within structured, often computational environments that enable iterative testing, traceability, and continuous revision. In this sense, they produce what Hanna terms *internal objectivity*: a form of objectivity grounded in methodological accountability.

The thesis situates this distinction within two intersecting bodies of literature: The first concerns the epistemology of visual and forensic evidence, from Carlo Ginzburg's conjectural paradigm. The second draws on Hans-Jörg Rheinberger's concept of experimental systems. Read together, these traditions enable a shift in focus from what OSINT practitioners conclude to how their practices constitute knowledge in the first place and under what conditions that knowledge can be compared, trusted, or revised.

While OSINT scholarship often concentrates either on technical verification work-flows or on the ethical-political implications of citizen-led intelligence, it rarely offers a systematic epistemological comparison of investigative paradigms.[4] Studies of digital evidence standards in international law and adjacent fields address important questions of admissibility and verification, but stop short of analysing the underlying architectures of reasoning that structure different investigative approaches. Similarly, Forensic Architecture's extensive theoretical work provides a sophisticated account of counter-forensic practice, but the frame is largely based on socio-political institutional critique, directly inspired by science and technology studies (STS). This thesis builds on and extends these contributions by proposing an epistemological analytical approach.

Rather than treating OSINT as a unified practice, it conceptualises it as an epistemic field structured by distinct investigative paradigms that determine how an investigation is designed.

Methodologically, the thesis combines theoretical genealogy with comparative case analysis. It reconstructs a series of epistemic regimes—testimonial, forensic, machinic, and digital—not as linear progressions but as overlapping formations that continue to shape contemporary investigative practices.

The result of this approach is a revised epistemological framework for Open Source Intelligence (OSINT) investigations that shifts the focus from investigative outcomes to the conditions of knowledge production. The thesis identifies two dominant epistemic paradigms within contemporary OSINT practice: conjecture-led investigation, which relies on abductive reasoning and the interpretation of discrete evidential traces; and model-led

4 Berkeley protocol is an example of an attempt to normalize the OSINT with a set of technical-ethical rules. Office of the United Nations High Commissioner for Human Rights and Human Rights Center, University of California, Berkeley, *Berkeley Protocol on Digital Open Source Investigations: A Practical Guide on the Effective Use of Digital Open Source Information in Investigating Violations of International Criminal, Human Rights and Humanitarian Law* (Geneva: United Nations, 2022), https://www.ohchr.org/en/publications/policy-and-methodological-publications/berkeley-protocol-digital-open-source.

investigation, which constructs knowledge through essentially data-translation and data-mapping processes, creating a space where new evidence/data can emerge.

The empirical core of the thesis consists of two case studies. The first examines the Mark Duggan investigation through a comparative reading of forensic reports, inquest findings, and spatial reconstruction methods developed by forensic design practices. The second focuses on the al-Ahli Hospital explosion, comparing ten independent investigations conducted by actors including *Forensic Architecture, Human Rights Watch*, the *New York Times*, the *Washington Post*, *BBC Verify, the Israeli Defence Force, Bellingcat,* and independent researchers like *Michel Kobs* and *Maher Arar's blog*. Together, these cases demonstrate an essential difference, a rupture at the very heart of these approaches: the terms and conditions under which the results of an investigation are considered reliable are extremely different. For conjecture-led methods, reliability is defined by the quality and quantity of the evidence, whereas the reliability of model-led methods depends on the stability and limits of the model.

The thesis concludes that while model-led investigation represents a significant epistemic advance, its current scarcity risks producing epistemic centralisation: a condition in which a small number of technically sophisticated actors hold disproportionate authority over what counts as verified truth. The wider adoption of model-led OSINT is therefore necessary for the field to remain both reliable and democratically accountable.

The central research question of this thesis is therefore:

What are the epistemic foundations at the heart of Open Open Source Investigation?

What differentiates the newest approaches, such as Forensic Architecture's one, from the rest of the field?

To what extent can an epistemologically grounded approach generate substantive insights within this field?

An epistemological approach can help define future development in the field, strengthening the accountability of OSINT. Given that the discussion on the topic is broad and can be approached from many angles, we decided to limit our inquiry to the epistemic aspect only, neglecting the many others that are generally studied.

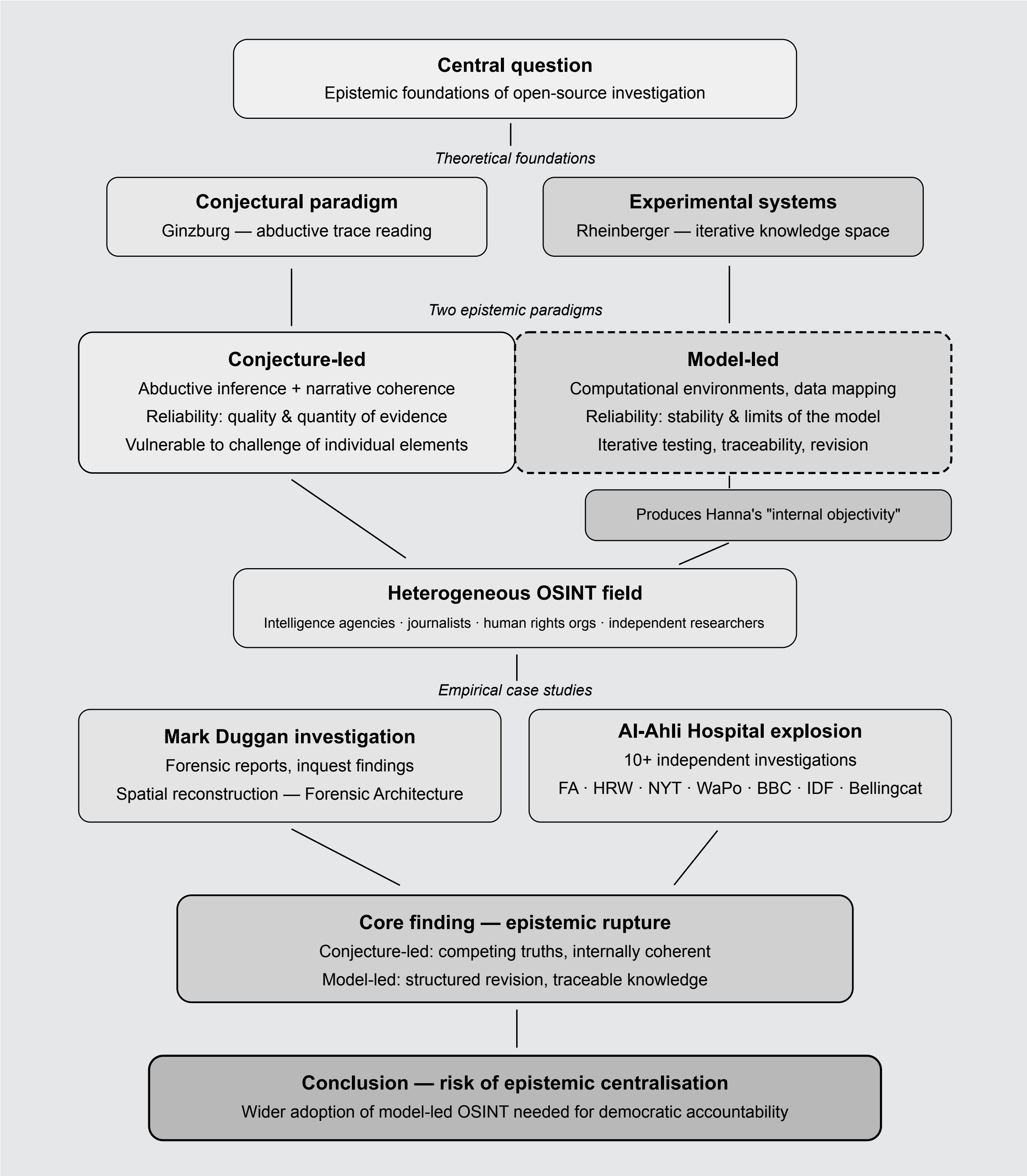


Figure 1 : Structure of Thesis

1
The Evidential Paradigm
A reflection on Joachim Harst’s Virtual Investigation

“To uncover the real, we must make the real.” - Eyal Weizman

I. The Evidential Paradigm: A reflection on Joachim Harst's Virtual Investigation

Joachim Harst's work on virtual investigations spans several key texts: the article "Virtuelle Investigationen: Transformationen des Indizienparadigmas zwischen Sherlock Holmes und Forensic Architecture" (2022), the programmatic essay "Virtual Investigations: Revising the Evidential Paradigm in Law, Literature and the Arts" (2024), and the edited volume Virtuelle Investigationen: Revisionen des Indizienparadigmas in Literatur und Kunst (2024), where he authored both the "Einleitung" and "Evidence Production: Spurenlesen in Open Source Investigations." This body of work originates in the Virtuelle Investigationen conference, which laid its conceptual foundation.

Today, visual investigation looks very different from what it did in the past. It used to rely on physical evidence such as footprints, fingerprints, and other direct traces of events. Nowadays, it is increasingly formed by virtual models, algorithms, and large-scale data systems. As a result of this shift, tools and methods such as simulation, interfaces, remote sensing, and networked databases have become more prominent subjects in investigations.[5]

This change marks a deeper epistemological reorientation in how knowledge is produced. It goes beyond adopting new technologies. The contemporary Western epistemology of evidence sprang from a semiotic logic that valued the interpretation of minute indications, traces, symptoms, or aberrations, as Carlo Ginzburg contended in his essay *Clues: Roots of an Evidential Paradigm* to a Virtual mode. Joachim Harst revisits this paradigm in his book "*Virtual Investigations: Transformations of the Evidential Paradigm Between Sherlock Holmes and Forensic Architecture*"(2023) to demonstrate how traces in the current digital environment correspond to data points whose meaning must be created through computation and visualisation rather than tangible remains. According to Harst, in the contemporary digital paradigm, the trace is digital and no longer refers to physicality but to relational patterns within systems of data and computation.

The evidential paradigm has thus shifted from the tangible and tactile to digital traces and systems. Investigators construct truth from assembled fragments scattered across digital environments rather than from physical remnants, necessitating computational work, team interpretation, and iterative visualisation. ***Eyal Weizman and Matthew Fuller*** **refer to this as the "production of evidence," in which producing and knowing coexist**. As a result, the process of investigation has become inextricably linked to the infrastructures that make evidence perceptible. Eyal Weizman's intervention is to insist that forensics is not only a technical procedure but also a public practice of truth. Re-activating the Latin root *forensis* ("of the forum"), Weizman reframes evidence as something that must be made apparent, argued, contested, and validated in front of audiences—not merely extracted from mute matter. In this sense, "forensic" refers to a political space in which objects, images, measurements, and testimonies acquire force only through their mediated presentation. The consequence is crucial for contemporary visual investigations: evidence is not simply *found* in the world but must be constructed for a forum, where credibility is negotiated across science, law, media, and civic publics.

5 Joachim Harst, "Virtual Investigations: Transformations of the Evidential Paradigm Between Sherlock Holmes and Forensic Architecture," Medienkomparatistik (2022): 23–25

Aesthetic System: Definition given by Fuller and Weizman, where human and non-human actors participate in a complex network that is both "sensing"(registers information) and "sense-making" (reads and contextualises the information). From this network of actors (aesthetic system), new representations of facts emerge. These systems welcome new entities, communities, frameworks or elements, while also unsettling Western-centred epistemologies and reconfiguring the relationship between power and the construction of truth through narrative. Against an idea of objective and universal representation of facts, a new investigative

To understand the need to revisit this paradigm, we will examine the role of the witness and how it has been reconfigured. Mid-twentieth-century paradigms of truth-telling, particularly in the aftermath of genocide and mass political violence, privileged *embodied human testimony* as the primary bearer of historical and juridical truth. As Thomas Keenan and Eyal Weizman recount in *Mengele's Skull (2012)*, the *Eichmann* trial inaugurated the "era of the witness" in which subjective testimony, despite its fragility, became the central evidentiary form[6]. Today, however, modes of witnessing increasingly involve disembodied sensors, surveillance systems, and algorithmic observers that generate their own visual records. Machines "see" in ways that exceed human perception, yet they do so without memory, affect, or narrative; a condition Jane Blocker describes as the emergence of the "invisible witness" in contemporary politics of vision.[7]

These machinic forms of seeing raise fundamental questions: What does it mean to witness in an age when machines can see more than humans can? How do we ascribe credibility to images without trusted, identifiable authors? What subject or object constitutes a witness now?

Witnessing today is fragmented and aestheticised. It is shaped not only by who or what observes, but by how observation is framed, mediated, and circulated. It is now hard to trust what we see every day; witnessing becomes a function of design and emotion, which refers to what is observed, who sees it, and how it is made readable.[8] As visual investigations evolve from analogue to digital, and from linear narratives to networked assemblages, they reveal new tensions between visibility and obfuscation, objectivity and narration, immediacy and construction.

In what follows, we trace these transformations through a comparative, historical, and epistemological lens. Drawing on Ginzburg's notion of the evidential paradigm, the testimonial culture emerging from the *Eichmann* trial, the forensic regime inaugurated by the *Mengele* investigation, and contemporary practices such as Harun Farocki's machinic images and Forensic Architecture's digital counter-forensics, and lastly Rheinberger's Experimental Systems, we argue that the history of visual investigation is not a linear progression toward ever-greater certainty. Instead, it is a history of ruptures, displacements, and reconfigurations, each shaped by new ways of seeing, knowing, and claiming truth and by shifting figures of the witness.

Fig.2: Arthur Conan Doyle Encyclopedia, "New Adventures of Sherlock Holmes (comics Dell 1961).

6 Thomas Keenan and Eyal Weizman, *Mengele'estheticisedanalogues Skull: The Advent of a Forensic Aesthetics* (Frankfurt: Sternberg Press, 2012), 11–13

7 Jane Blocker, *Seeing Witness: Visuality and the Ethics of Testimony* (Minneapolis: University of Minnesota Press, 2009), xiii–xvi.

8 Harst, "Virtual Investigations," 40–41.


SHERLOCK HOLMES
DETECTION METHODS
IN SHERLOCK HOLMES' DAY THERE WERE NO MODERN DETECTION TOOLS SUCH AS FINGER-PRINTING TO SOLVE CRIMES. HOLMES USED HIS DEDUCTIVE POWERS TO CATCH CRIMINALS.
BY MEASURING SHOE TRACKS OR FOOTPRINTS AND NOTING THE LENGTH OF STRIDE BETWEEN PRINTS, HOLMES COULD DETERMINE HOW TALL A PERSON WAS.
IF THE IMPRESSION OF ONE FOOT WAS HEAVIER THAN THE OTHER, IT TOLD HOLMES THAT THE PERSON HE SOUGHT LIMPED OR WAS WOUNDED IN ONE LEG.
SPOTTING A DISCARDED CIGAR, HOLMES LOOKED TO SEE IF THE END WAS CHEWED, OR IF THE SMOKER WAS A PERSON WHO USED A CIGAR HOLDER.
HE CLAIMED NO TWO TYPEWRITERS WERE EXACTLY ALIKE AND EASILY MATCHED UP EVI-DENCE WRITTEN BY THE SAME TYPEWRITER BY COMPARING WORN LETTERS AND SPACING OF WORDS.
HOLMES COULD TELL A MAN'S JOB BY A GLANCE AT HIS HANDS. SAILORS', PRINTERS', WEAVERS' OR DIAMOND CUTTERS' HANDS ARE SHAPED OR STAINED BY THEIR TRADES.

II. The Classical Evidential Paradigm By Ginzburg: Traces, Certainty, and the Detective

"Evidence derives from the Latin indicare: to indicate."[9]

***"Often associated with this basic meaning is the notion that evidence does not prove anything, but only points to something. Associated with evidence and speculative, almost dubious ambiguity. This widespread understanding of evidence, which is still prevalent today, stems from a criminal law debate that, in the 18th century, discussed the status of evidence in a contradictory manner, but redefined evidence as a privileged legal instrument by the mid-19th century."*[10].**

The 19th-century detective, Sherlock Holmes, crystallises what Carlo Ginzburg famously called the *"evidential paradigm"*: a mode of knowledge grounded in the interpretation of traces, small signs, and marginal details that reveal a hidden reality otherwise unavailable to direct observation[11]. Rooted in semiotics, psychoanalysis, and the emerging human sciences, this paradigm marked a decisive shift away from general laws toward *conjectural reasoning*, a form of knowledge that privileges the exceptional and the fragmentary over the systematic and the universal. In other words, it's a way of building knowledge that is probable, interpretive, and always open to revision.

In his essay *Clues: Roots of an Evidential Paradigm*, Ginzburg identifies a constellation of figures: the detective, the doctor, and the artist, united by their capacity to infer causes from traces. The detective follows footprints or cigar ashes; the doctor reads the body's symptoms; the art historian attributes authorship through brushstrokes or pigment use. He maps his Conjectural Paradigm by citing three chief representatives to explain its emergence and subsequent expression within society. These three representatives are Giovanni Morelli, an Italian art critic; Sigmund Freud, an Austrian neurologist; and Conan Doyle's fictional character, Sherlock Holmes.

As illustrated in Figure 3, the evidential paradigm operates by collecting fragmentary, marginal signs and synthesising them through a conjectural leap to reconstruct a coherent yet structurally open narrative. Each of these figures practices what Ginzburg calls *conjectural knowledge*, which is a form of reasoning based on *abductive inference*, the capacity to move from the smallest, most insignificant detail to a larger truth[12]. Knowledge here is never absolute but interpretive: an act of reading traces, translating the imperceptible into the visible.

9 Carlo Ginzburg, *Clues, Myths, and the Historical Method*, trans. John and Anne C. Tedeschi (Baltimore: Johns Hopkins University Press, 1989).
10 Harst,"Virtual Investigations," pp.20–21
11 Ginzburg, *Clues, Myths, and the Historical Method* (1979), 101–103
12 Carlo Ginzburg, Clues, Myths, and the Historical Method, trans. John Tedeschi and Anne C. Tedeschi (Baltimore: Johns Hopkins University Press, 1989), 103–105.

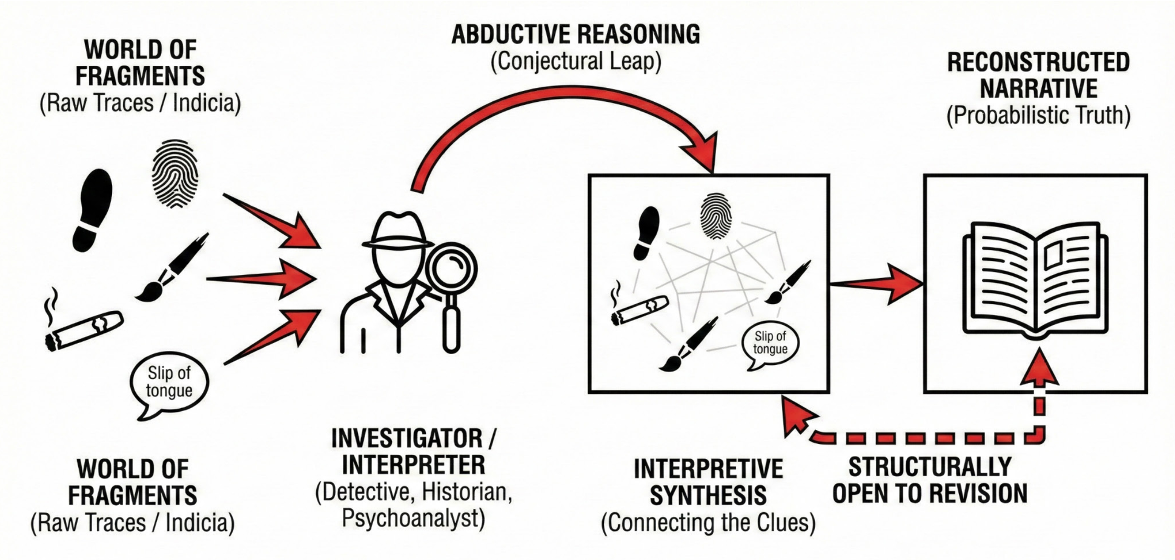


Fig. 3 Conjectural Reasoning based on Ginburg

Sherlock Holmes stands as the paradigmatic embodiment of this evidential logic. His methods, grounded in observation and reconstruction, translate sensory details into narrative coherence. The trace, in Holmes's epistemology, guarantees the past: each footprint, fingerprint, or strand of ash carries within it an *indexical link* to a prior event. As Joachim Harst notes, Holmesian forensics embodies the 19th-century belief in the *"mathematical certainty of deduction"*— the idea that truth can be stabilised through meticulous attention to traces that "speak" for themselves.[13]

However, Ginzburg's recovery of this paradigm is fundamentally anti-positivist. Ginzburg himself cautions that the evidential paradigm, while epistemologically powerful, is also qualitative. It depends on narrative framing and interpretation rather than universal laws.[14] Harst elaborates that even in Holmes's seemingly airtight deductions, interpretive gaps are concealed through narrative devices such as Dr Watson's astonishment, which naturalises Holmes's inferential leaps.[15] The detective's "objectivity" thus depends not only on material evidence but also on the rhetorical performance of certainty.

This intertwining of observation and narration reveals the aesthetic dimension of evidence. The trace functions both as an epistemic object and as a narrative device, enabling stories of reconstruction, revelation, and closure. The detective story, as Eyal Weizman later observes, establishes a model of knowledge production that continues through the history of contemporary forensic practices: an aspiration to recover the real through the patient reading of marks and residues.[16]

13 Harst, "Virtual Investigations," 27.
14 Carlo Ginzburg, *Clues, Myths, and the Historical Method*, trans. John Tedeschi and Anne C. Tedeschi (Baltimore: Johns Hopkins University Press, 1989), 107
15 Harst, 2023, pp. 28–29
16 Eyal Weizman, *Forensic Architecture: Violence at the Threshold of Detectability* (New York: Zone Books, 2019), 60–62.

In its classical form, the evidential paradigm may be described through three interrelated dimensions:

- Ontological: Traces function as material remnants of a past event, preserving an indexical relation to what caused them.
- Epistemological: Knowledge is produced through abductive reasoning, that is, through the interpretation of minor details to reconstruct hidden realities.
- Aesthetic: Evidence is inseparable from narrative performance, since it must be staged, interpreted, and rendered persuasive before an audience.

As Harst argues, the detective's world embodies a moment of *"epistemological optimism,"* in which reality was thought to be fully recoverable through material traces. It is the belief that knowledge is attainable and progress is driven by learning, and that reality is ultimately comprehensible. Yet this optimism will be repeatedly challenged in the 20th century: first by the testimonial turn, which centres the embodied witness as the locus of truth, and later by the forensic and machinic regimes, in which traces are mediated by technologies, probabilities, and aesthetic persuasion. The detective's certainty hus foreshadows its own undoing and the eventual recognition that every trace is already an interpretation.

III . Between the Eye and the Trace: The Aesthetic Origins of Forensic Vision

The emergence of forensic thinking in modernity was not confined to the courtroom or the laboratory. As Antonia Eder argues in *Virtuelle Investigationen,* [17]The roots of investigative epistemology lie equally in the visual arts and the interpretive practices that shaped modern perception. When detached from its juridical frame, the forensic method reveals itself as an aesthetic of attention and a disciplined form of looking that seeks meaning in traces, details, and deviations.

At the end of the nineteenth century, the act of seeing became infused with suspicion and analytical precision. The world appeared as a surface of indicators (Indizien) awaiting interpretation. Artists, scientists, and detectives shared this sensibility: all treated the visible as a field of signs through which the invisible might be inferred. The investigator was not merely a judge of evidence but a *reader of appearances*. Eder thus recasts the forensic gaze as a semiotic practice, situated between epistemology and aesthetics.

17 Harst, Joachim, und Nursan Celik. 2024. „Virtual Investigations: Revising the Evidential Paradigm in Law, Literature and the Arts". Wiener Digitale Revue, Nr. 5 (September). https://doi.org/10.25365/wdr-05-04-05.

Like Carlo Ginzburg's "evidential paradigm," this visual logic depends on minute details that stand in for absent causes, such as a stain, a shadow, a gesture, or a misalignment. But Eder pushes beyond Ginzburg's archival model to suggest that these traces are not inert residues of the past; they are relational phenomena that take meaning only through acts of perception and interpretation. The trace is less of a "thing left behind" than a threshold of knowledge, a point where visibility and invisibility intersect.

This shift relocates the site of truth from the courtroom to the surface of the image itself. The image becomes both evidence and method and a medium in which seeing and knowing converge. Photography embodies this dual function. Every photograph is an experiment of evidence: it records light, time, and gesture, yet demands interpretation to yield meaning. Thus, the aesthetic act of framing or focusing parallels the forensic act of isolating a clue.

Eder's argument illuminates a crucial transition: from a positivist belief in objective traces to a reflexive awareness that every act of observation is also an act of construction. The forensic gaze no longer promises certainty; it cultivates *sensitivity* and readiness to notice **what escapes dominant narratives or institutional attention**. In this sense, the investigator and the artist share an ethical vocation: to attend to the marginal, to give form to the barely perceptible.

Placed within the broader genealogy of visual investigation, Eder's analysis reveals the continuity between Ginzburg's detective, the modern artist, and today's digital investigator. Each operates within a field where truth is not simply found but shaped through mediation. What begins as a way of reading traces becomes, in time, a way of constructing visibility itself, a process that will underpin both the testimonial culture of the mid-twentieth century and the forensic aesthetics of the digital age.

The aestheticisation of the forensic gaze that emerged in the early twentieth century, what Eder describes as a mode of *seeing that reads the visible for what escapes it*, laid the groundwork for a new politics of truth. Yet this epistemology of the eye undergoes a profound transformation. In the aftermath of the Holocaust, the relation between vision and knowledge was not abandoned but reconfigured through embodiment and speech. The witness did not replace the image but rather made *seeing itself an ethical act*: to bear witness was to be seen and heard simultaneously. Testimony thus folded the visual into the performative, transforming observation into a shared scene of memory, vulnerability, and moral address. The act of interpretation, once belonging to the detective or the artist, now belonged to the witness, whose voice and presence extended the forensic logic of the trace into the domain of lived experience.

IV. The Postwar Testimonial Turn: The “Era of the Witness”

The Eichmann trial in Jerusalem (1961) marks a decisive break in modern politics of truth and evidence. At the same time, the Nuremberg Trials (1945–46) had largely privileged *documentary and archival evidence*, bureaucratic records, official orders, and captured documents. The Eichmann trial foregrounded survivor testimony as its central evidentiary form. This transformation, as Shoshana Felman and Annette Wieviorka argue, inaugurated what would later be called *“the era of the witness”*.[18]

Thomas Keenan and Eyal Weizman read this event as an epistemological turning point. In *Mengele’s Skull*, they note that the Eichmann trial “opened up a new forum” in which the human witness became both the bearer and producer of truth.[19] Unlike the positivist evidential logic of 19th-century forensics, this juridical space embraced subjectivity and emotion as constitutive of historical knowledge. The witness’s *voices* and *presence* displaced the earlier authority of the object or the document.

The chief prosecutor, Gideon Hausner, structured the trial as a form of historical pedagogy rather than a narrowly legal proceeding. His goal was not only to convict Eichmann but to create what he called a “living record” of catastrophe, transforming the courtroom into a stage for collective memory[20]. As Jane Blocker observes, this moment “transformed witnessing into both a juridical and aesthetic performance,” where seeing and speaking became inseparable acts of public pedagogy.[21]

Felman, in The Testimony of Crises, insists that this transformation did not occur despite the fragility of testimony but through it. Survivor narratives were often marked by silence, contradiction, confusion, or error and precisely the signs of trauma’s disruptive force. Rather than disqualifying the witness, these ruptures became central to the trial’s moral and epistemic authority. The courtroom, Felman argues, “embraces the fragility of the human witness” and turns it into a space where failure, breakdown, and excess can themselves signify truth[22].

This new evidentiary paradigm of embodied witnessing carries its own contradictions. The most haunting example is that of *K-Zetnik 135633*, a witness who refused to speak his given name and identified himself only as a “concentration camper.” His testimony collapsed under the weight of representation; fainting mid-sentence, he became an emblem of what

18 Shoshana Felman and Dori Laub, Testimony: Crises of Witnessing in Literature, Psychoanalysis, and History (New York: Routledge, 1992); Annette Wieviorka, The Era of the Witness, trans. Jared Stark (Ithaca: Cornell University Press, 2006).

19 Thomas Keenan and Eyal Weizman, Mengele's Skull: The Advent of a Forensic Aesthetics (Berlin: Sternberg Press, 2012), 11–13.

20 Ibid., 13.

21 Jane Blocker, Seeing Witness: Visuality and the Ethics of Testimony (Minneapolis: University of Minnesota Press, 2009), xiii–xv.

22 Shoshana Felman and Dori Laub, Testimony: Crises of Witnessing in Literature, Psychoanalysis, and History (New York: Routledge, 1992), 57legitimised.

Felman calls *"frontier evidence"* where the testimony that occurs at the limit between speech and silence, life and death[23]. Keenan and Weizman echo this insight, noting that "the witness's collapse was itself a form of evidence," a performance of trauma that revealed the inadequacy of legal discourse to contain historical pain[24].

The ethics of witnessing that emerged from Eichmann's trial profoundly reshaped the cultural landscape. As Wieviorka writes, "the witness became a key actor in public life, embodying a new moral authority that extended beyond the courtroom"[25]. Institutions such as Holocaust memorials, oral history archives, and museums institutionalised this testimonial paradigm, transforming the singular act of testimony into a global *infrastructure of memory*. The witness figure, once a legal participant, now stood as a moral compass for historical consciousness.

Yet, as Joachim Harst notes, this *testimonial regime* also introduced new hierarchies and burdens[26]. Certain voices that were able to communicate trauma within logical, emotionally resonant tales were privileged, while others, defined by silence, incoherence, or refusal, were marginalised. This dynamic created what Geoffrey Hartman calls a *"testimonial economy,"* where suffering could be legitimised only through its communicability.[27]

From an epistemological perspective, the testimonial turn altered the link between subjectivity and truth. The classical evidential paradigm had grounded knowledge in the objectivity of the trace; now, truth resided in the body, in lived experience, and in the *act of speech itself*. The witness became both *evidence and event*, embodying the tension between memory and mediation. Keenan and Weizman describe this as a transformation in the very ontology of evidence: "the witness stands not for what can be verified, but for what cannot be fully known—what must still be spoken despite its impossibility"[28].

However, this very centrality of the human witness would be challenged by a new evidentiary regime in which the forensic turn of the late 20th century saw objects, bones, and technical images begin to "speak" in the courtroom. The "era of the witness" thus gave way to the "era of the thing," as the domain of truth shifted once again from embodied memory to material and technological evidence.

23 Ibid., 59–60.
24 Thomas Keenan and Eyal Weizman, Mengele's Skull: The Advent of a Forensic Aesthetics (Berlin: Sternberg Press, 2012), 14.
25 Annette Wieviorka, The Era of the Witness, trans. Jared Stark (Ithaca: Cornell University Press, 2006), 85.
26 Joachim Harst, "Virtual Investigations: Revising the Evidential Paradigm in Law, Literature and the Arts," *Wiener Digitale Revue* 5 (2024): 33–34.
27 Geoffrey H. Hartman, *The Longest Shadow: In the Aftermath of the Holocaust* (Bloomington: Indiana University Press, 1996), 153–156.
28 Thomas Keenan and Eyal Weizman, Mengele's Skull: The Advent of a Forensic Aesthetics (Berlin: Sternberg Press, 2012), 15.

V. Forensic Aesthetics and the "Era of the Thing"

The discovery and identification of Josef Mengele's remains in Brazil in 1985 marked another major shift in the epistemology of war crimes investigations. As Thomas Keenan and Eyal Weizman argue in *Mengele's Skull: The Advent of a Forensic Aesthetics*, this event signalled the emergence of what they call *"the era of the thing"*, which is a regime in which material objects, rather than human witnesses, become central agents in the production of truth[29].

In this new evidentiary order, the forensic object, such as the bone, the photograph, and the digital file, functions as both subject and witness. The interdisciplinary inquiry that confirmed Mengele's identity relied on the convergence of osteobiography, dental records, and innovative visual technologies such as face–skull superimposition. These techniques transformed bones into *legible archives of a life*, capable of being "read" by forensic experts. As anthropologist Clyde Snow[30] famously declared, "bones make good witnesses… although they speak softly, they never lie, and they never forget"[31].

This rhetorical personification of objects encapsulates the paradox of forensic modernity: matter itself appears to testify. The object becomes a "super-witness", a mute yet infallible guarantor of truth, seemingly free from the biases and traumas that affect human memory. Yet, as Keenan and Weizman demonstrate, the forensic regime does not eliminate interpretation; it merely relocates it. "Every object," they write, "is always already entangled in aesthetic and rhetorical work, requiring an act of mediation to be legible as evidence"[32].

The Mengele investigation exemplified this entanglement between science and aesthetics. The process of face–skull superimposition, developed by forensic scientist Richard Helmer, employed two high-resolution cameras to align images of the skull and photographs of Mengele's face. The resulting composite—half bone, half face—produced what Keenan and Weizman describe as "a spectral presence, present and represented at one and the same time"[33]. This hybrid image was not a simple scientific proof; it was a persuasive visualisation designed to convince both scientific peers and the broader public. In this sense, forensic evidence was not discovered but performed, enacted through the aesthetics of visibility.

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29 Thomas Keenan and Eyal Weizman, *Mengele's Skull: The Advent of a Forensic Aesthetics* (Berlin: Sternberg Press, 2012), 13verification process that blurred the boundaries among.
30 "Clyde Snow," *Wikipedia*, last modified September 5, 2025, https://en.wikipedia.org/wiki/Clyde_Snow.
31 Thomas Keenan and Eyal Weizman, *Mengele's Skull: The Advent of a Forensic Aesthetics* (Berlin: Sternberg Press, 2012), 20.
32 Ibid.32.
33 Ibid.23–24

The forensic forum that emerged in São Paulo was itself a hybrid space, part laboratory and part tribunal. Journalists, scientists, and officials gathered to witness the bones' identification, a process of verification that blurred the boundaries between legal, scientific, and media domains. Keenan and Weizman describe this as a prototype of the *forensic forum*, defined by the dynamic interplay of object, mediator, and audience[34]. In such forums, truth is not inherent in the object but constructed through collective negotiation, expert mediation, and public persuasion.

Joachim Harst connects this moment to what he calls "the aesthet-icisation of evidence." In the Mengele case, he notes that scientific visualisation and narrative performance merged into a single apparatus of conviction. The evidentiary process was less about *discovering* facts than about producing an image of truth that could circulate and convince across different communities. Harst situates this shift as a precursor to contemporary digital investigations, in which aesthetic form and forensic verification are inseparable[35].

From a broader perspective, Weizman interprets this transformation as a redefinition of what counts as truth in political and juridical contexts. In *Forensic Architecture: Violence at the Threshold of Detectability*, he observes that "forensics came to mean not only the application of science to law but the staging of truth in public"[36]. Here the forensic turn thus reconfigures the relationship between objectivity and performance, establishing a new politics of evidence that operates through the visual and rhetorical construction of fact.

Within this "era of the thing," several epistemological and aesthetic shifts are especially significant:

- From testimony to objecthood: Truth is increasingly located in things rather than persons, as the human witness gives way to the forensic artefact.
- From narrative to visualisation: The persuasive force of evidence lies less in spoken narration than in visual form and its capacity to compel belief.
- From discovery to production: Evidence is no longer treated as something merely found, but as something assembled, mediated, and staged through technical processes.
- From fixed to distributed forums: The juridical forum expands beyond the courtroom into laboratories, media platforms, and transnational circuits of verification.

34 Thomas Keenan and Eyal Weizman, *Mengele's Skull: The Advent of a Forensic Aesthetics* (Berlin: Sternberg Press, 2012), 29–30.
35 Ibid.,33–35.
36 Eyal Weizman, *Forensic Architecture: Violence at the Threshold of Detectability* (New York: Zone Books, 2019), 65.

Mengele's Skull

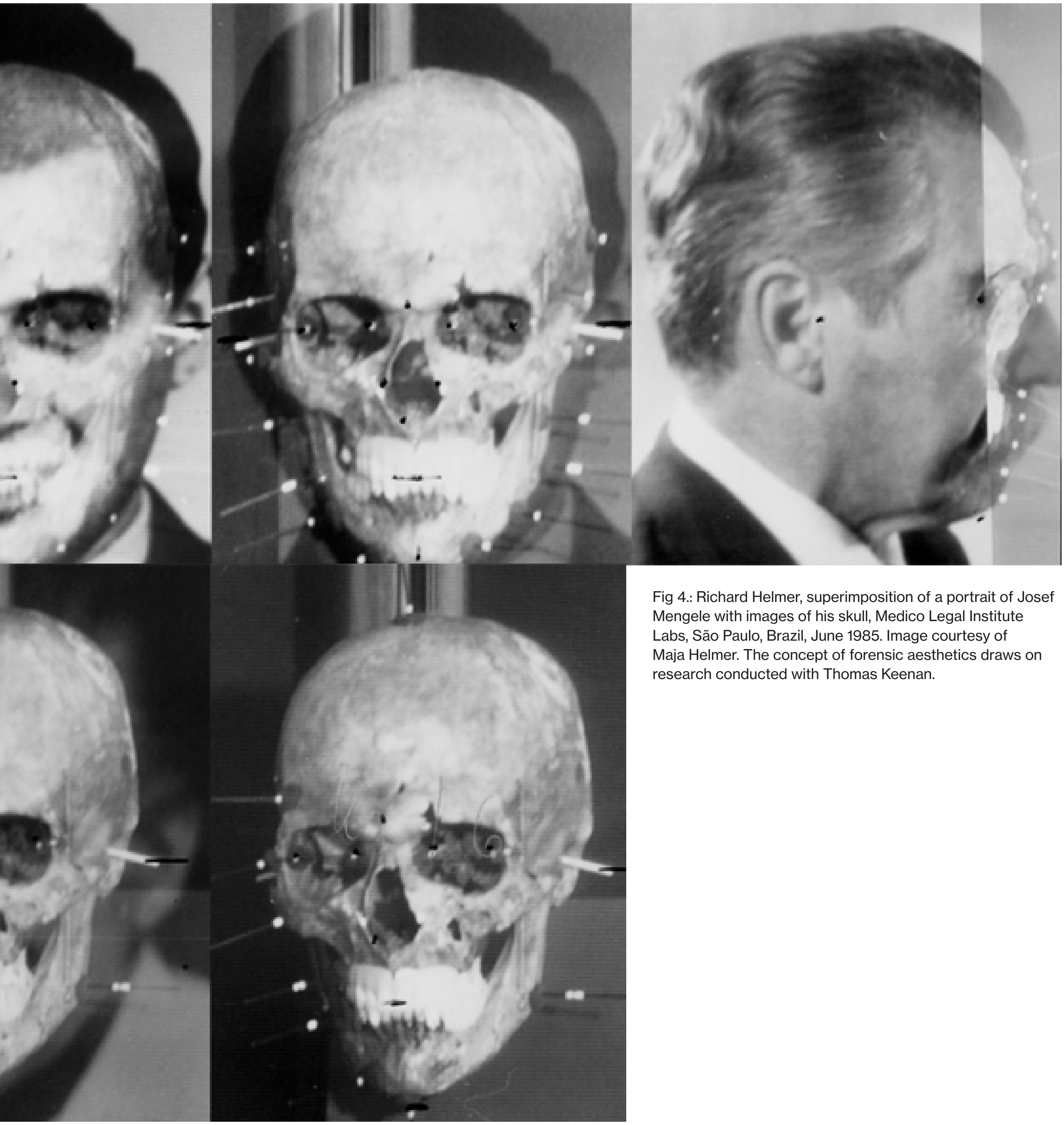

Fig 4.: Richard Helmer, superimposition of a portrait of Josef Mengele with images of his skull, Medico Legal Institute Labs, São Paulo, Brazil, June 1985. Image courtesy of Maja Helmer. The concept of forensic aesthetics draws on research conducted with Thomas Keenan.

Keenan and Weizman describe this new configuration as a forensic aesthetic—a mode of truth production that combines material analysis, visual persuasion, and public performance (Keenan & Weizman, 2012, p. 62). The forensic aesthetic does not oppose narrative or affect; rather, it depends on them to transform inert matter into meaningful evidence.

The "era of the thing" therefore complements, rather than displaces, the "era of the witness." It retains the juridical aspiration to reconstruct events, but it replaces the embodied voice with material remains and visual models. Both regimes depend on mediation: the witness through speech, the thing through visualisation. And both reveal the same paradox—that truth, to be believed, must be performed.

VI. Machinic Vision and the Crisis of Human Testimony

During this chapter, we're going to look at another epistemic shift that occurs as vision itself becomes automated. In the late twentieth and early twenty-first centuries, the proliferation of satellite imaging, drones, CCTV networks, and algorithmic recognition systems has transformed how events are observed, recorded, and verified. These technologies generate images that often bypass human perception altogether - what Harun Farocki famously termed *"operational images"*: images produced "not to be seen, but to act"[37].

Operational images are not primarily used to document or bear testimony, unlike the testimonial photos associated with atrocities or the evidentiary photos of classical forensics. They are designed to operate within technical systems of control, targeting, and computation. They identify, categorise, and forecast rather than observe events. Such images operate at the "threshold of detectability," or at the boundaries of what human perception can fully register or interpret, according to Eyal Weizman. Under this system, machine vision not only records violence but also increasingly mediates—and occasionally enacts—it, producing evidence that no longer relies on human witnesses.

Here, the very definition of the witness begins to unravel. As Joachim Harst notes, "the locus of observation has migrated from the human eye to computational infrastructures," displacing the ethical and sensory dimensions of witnessing with machinic perception.[38] These systems do not remember or empathise; they process. Their authority lies in their apparent neutrality—their capacity to record continuously, to see from above, and to generate visual data at scales and speeds inaccessible to human cognition.

37 Harun Farocki, *Eye/Machine I* (Germany, 2001), video.
38 Joachim Harst, "Virtuelle Investigationen: Transformationen des Indizienparadigmas zwischen Sherlock Holmes und Forensic Architecture," in *Medienkomparatistik*, vol. 4 (Bielefeld: Aisthesis, 2023), 38.

Algorithmic vision is not a passive record of reality but a performative act shaped by institutional agendas, technological parameters, and biases.[39]

The political and epistemic implications of this shift became painfully visible in moments such as Colin Powell's presentation to the UN Security Council on 5 February 2003, where satellite images were displayed as incontrovertible proof of Iraq's alleged weapons facilities. Powell's speech emphasised the expertise required to interpret these images that were the "painstaking work" of analysts deciphering visual "signatures" invisible to the untrained eye. The image was thus framed as omniscient yet inaccessible: a divine, authorless witness. Yet when the intelligence was later revealed to be false, these same images became artefacts of deception, illustrating the fragility of visual authority in the age of machinic vision[40].

Here, the machinic image functions as a disembodied witness, endowed with an aura of objectivity that overrides the testimony of human observers. The human witness, once the guarantor of moral truth in the postwar era, is demoted to a fallible, localised observer, while the satellite, drone, or algorithm assumes the status of an incorruptible viewer. The camera's "view from nowhere"[41]is reimagined as a planetary apparatus of truth, detached from affect, ethics, or accountability.

Artists such as Harun Farocki, Hito Steyerl, and Trevor Paglen have explored the aesthetic and political consequences of this shift. In *Eye/ Machine* (2001–2003), Farocki set footage of automated targeting against industrial inspection systems, revealing how both operate under a logic of *instrumental vision*: images made for machines rather than humans. For Farocki, operational images mark a crisis in representation—an era in which "images are no longer directed at us, but at other images"[42]. Steyerl extends this analysis in *In Free Fall* (2010)[43], showing how digital images circulate as unstable commodities and weapons, entangled in global economies of attention and violence.

The epistemic consequence of this machinic regime is what Weizman calls a *"dark epistemology"*: **a condition in which visibility itself becomes a site of manipulation[44].** The opacity of algorithmic systems, which is classified as software, proprietary datasets, and restricted sensors, produces an environment where the very tools of observation conceal their own conditions of production. Harst identifies this as the *"post-optical" moment of forensics*, where the human no longer stands before the image but must instead interpret *data traces* generated by unseen machines[45].

39 Ibid.,39-40.
40 Eyal Weizman, *Forensic Architecture: Violence at the Threshold of Detectability* (New York: Zone Books, 2019), 72–73.
41 Thomas Nagel, *The View from Nowhere* (New York: Oxford University Press, 1986).
42 Harun Farocki, *Eye/Machine I* (Germany, 2001), video.
43 Hito Steyerl, *In Free Fall* (Germany/Austria, 2010), video, 32 min.
44 Eyal Weizman, *Forensic Architecture: Violence at the Threshold of Detectability* (New York: Zone Books, 2019), 78.
45 Joachim Harst, "Virtual Investigations: Revising the Evidential Paradigm in Law,

This machinic turn introduces a new **crisis of witnessing**, defined by three interrelated paradoxes:

- **Images without viewers:** If images are generated and processed autonomously, can they still "witness" in any meaningful sense?
- **Objectivity without ethics:** If machinic vision claims neutrality while encoding institutional bias, what happens to the moral dimension of testimony?
- **Evidence without presence:** If the production of images is divorced from human experience, can they bear the weight of historical truth?

Farocki's "operational images" thus stand as the aesthetic and ethical counterpart to Weizman's "forensic threshold": both designate the moment when seeing becomes an act of computation. The witness, which was once embodied in the speaking subject, is now dispersed across sensor networks, algorithms, and databases. Yet, as Harst reminds us, even this machinic form of vision depends on interpretation and mediation: the forensic investigator, the programmer, the analyst; all serve as translators between the logics of data and the human demand for meaning[46].

In this context, witnessing is no longer a singular act of presence or speech but a distributed epistemic process. It unfolds across multiple interfaces and actors/human and nonhuman, each occupying a position within what Weizman and Fuller call "an investigative ecology," where truth emerges not from individual perception but from the collective labour of verification[47].

The crisis of human testimony thus culminates in the paradox of machinic vision: an image that sees too much and yet explains nothing. What is gained in resolution is lost in relationality. The challenge for contemporary investigation is therefore not to recover the primacy of human sight, but to render visible the infrastructures where technical, political, and aesthetic are constructed.

VII. Digital Forensics and Counter-Investigation

As Eliot Higgins explains in *Bellingcat: Truth in a Post-Truth World*, the rise of participatory media fundamentally redistributed who can produce and circulate images:

Literature and the Arts," *Wiener Digitale Revue* 5 (2024): 40–41.

46 Joachim Harst, "Virtuelle Investigationen: Transformationen des Indizienparadigmas zwischen Sherlock Holmes und Forensic Architecture," in *Medienkomparatistik* 4 (Bielefeld: Aisthesis, 2023), 42.

47 Matthew Fuller and Eyal Weizman, *Investigative Aesthetics: Conflicts and Commons in the Politics of Truth* (London: Verso, 2021), 14–16.

"..With blogging starting in 2000, and social media starting in 2004 and YouTube, which I think started in 2005, now the tools for media production have been given to the people formerly known as the audience."

(***Bellingcat: Truth in a Post-Truth World***, 00:06:49–00:07:07).

The emergence of digital forensics marks the latest reconfiguration of our evidential paradigm. In this regime, truth is neither discovered nor witnessed but *constructed* through the aggregation, synchronisation, and visualisation of data. Civil society actors like journalists, artists, architects, and online communities can now deploy forensic tools once reserved for states and militaries. These practices, grouped under the umbrella of open-source intelligence (OSINT) or counter-forensics, define a new epistemic field in which verification itself becomes a political act.As Joachim Harst writes, this transformation represents "a virtualisation of the evidential paradigm," where traces no longer exist as stable residues of events but are "*artefacts generated by processes of collection, modelling, and computation*"[48]. The contemporary investigator does not find evidence in the world; they build environments in which evidence can appear. Databases, 3-D models, timelines, and interfaces function as procedural machines of truth, assembling fragments into forms that can be seen, compared, and debated.

This constructivist practice echoes Eyal Weizman's description of Forensic Architecture as the *"production of evidence",* a methodology that unites technical reconstruction with political advocacy[49]. Forensic Architecture's investigations into drone strikes in Pakistan, the 2020 Beirut Port explosion, or cases of police violence are built from thousands of heterogeneous traces: videos, sound recordings, satellite images, architectural plans, forensic reports, experts' reports and witness statements. These elements are synchronised within digital environments where the spatial and temporal relations between fragments can be tested. "To uncover the real," as Weizman and Matthew Fuller put it, "we must *make* the real"[50].

One emblematic case discussed by Harst is Forensic Architecture's investigation of a drone strike in Miranshah, Pakistan. By triangulating photographs, videos, and satellite imagery, the team reconstructed a three-dimensional model of the bombed building. Within this model, investigators identified what Harst calls "*negative traces*"—absences in the spatial field

48 Joachim Harst, "Virtuelle Investigationen: Transformationen des Indizienparadigmas zwischen Sherlock Holmes und Forensic Architecture," in *Medienkomparatistik*, vol. 4 (Bielefeld: Aisthesis, 2023), 36-38

49 (Weizman, *Forensic Architecture*, 2019, p. 64)

50 Matthew Fuller and Eyal Weizman, *Investigative Aesthetics: Conflicts and Commons in the Politics of Truth* (London: Verso, 2021), 12.

that corresponded to bodies vaporised by the blast[51]. These voids, though immaterial, became central pieces of evidence. The model thus operated as both an analytic instrument and a forum of appearance, transforming what Keenan and Weizman earlier described as "the staging of truth in public"[52] into a digital, collective process.

Such practices expose a crucial epistemological shift: the trace has become *relational*. Its meaning arises only through its connections to other data points and the procedures that bind them. Harst emphasises that digital traces are "*event-objects rather than material residues*," existing within computational assemblages that continually update and recalculate[53]. Evidence is no longer a fixed index of the past but a *dynamic configuration* that must be constantly maintained and rendered.

Yet, this proliferation of machinic traces also reproduces the ambiguities of forensic aesthetics. The polished visualisations of Forensic Architecture—rendered through cinematic animations, virtual walkthroughs, and sound analyses—risk projecting an *"aesthetics of objectivity"* that conceals the interpretive labour behind them[54]. Weizman acknowledges this tension: counter-forensic imagery must persuade publics and courts while resisting the very illusion of neutrality that state forensics deploys[55]. The credibility of such evidence, therefore, depends not on erasing mediation but on making mediation visible—turning transparency itself into an investigative principle.

Digital counter-forensics occupies a paradoxical position within the longer genealogy of evidence:

- It retains the detective's logic of inference, yet operates through computational environments rather than direct encounters with material traces.
- It incorporates human testimony, yet redistributes witnessing across platforms, databases, and networked publics.
- It relies on the forensic authority of objects, yet makes those objects legible only through visualisation and model-building.
- It uses machinic images as evidence, yet turns those same operational images into objects of critique and contestation.

51 Joachim Harst, "Virtuelle Investigationen: Transformationen des Indizienparadigmas zwischen Sherlock Holmes und Forensic Architecture," in *Medienkomparatistik* 4 (2023): 39–40.
52 Thomas Keenan and Eyal Weizman, *Mengele's Skull: The Advent of a Forensic Aesthetics* (Berlin: Sternberg Press, 2012), 62.
53 Joachim Harst, "Virtuelle Investigationen: Transformationen des Indizienparadigmas zwischen Sherlock Holmes und Forensic Architecture," in *Medienkomparatistik* 4 (2023): 41.
54 Joachim Harst, "Virtuelle Investigationen: Transformationen des Indizienparadigmas zwischen Sherlock Holmes und Forensic Architecture," in *Medienkomparatistik* 4 (2023): 40–41.
55 Eyal Weizman, *Forensic Architecture: Violence at the Threshold of Detectability* (New York: Zone Books, 2019), 75.

This distributed practice forms what Weizman and Fuller describe as an "**investigative commons**"—a collective infrastructure in which facts are co-produced by human and non-human agents across multiple platforms[56]. In this ecology, witnessing becomes a shared, iterative process rather than a singular act of memory or observation. Traces are continually revised, recorded, and re-visualised as they circulate among activists, artists, lawyers, and algorithms.

At the same time, the counter-forensic field must operate within what Weizman calls a "dark epistemology", a media environment saturated by disinformation, deepfakes, and algorithmic manipulation[57]. Here, truth is not self-evident but *performatively verified*. The challenge **is no longer simply to expose hidden facts but to design protocols of trust that can survive in adversarial conditions**. Transparency, reproducibility become as crucial as accuracy itself.

In this sense, digital forensics completes the long arc that began with Ginzburg's evidential paradigm. The detective's interpretive gaze has expanded into a planetary apparatus of sensors and data networks; the solitary witness has become a distributed collective; the forensic object has been virtualised. What persists across these transformations is the trace—not as a stable imprint of the real but as an epistemic figure continually reinvented by new technologies of seeing and knowing. Each regime, whether detective, witness, forensic, machinic, or digital, redefines what counts as evidence, yet all share the same **ethical and aesthetic task: to make violence visible and to make truth perceptible within the unstable architectures of the image.**

56 Matthew Fuller and Eyal Weizman, *Investigative Aesthetics: Conflicts and Commons in the Politics of Truth* (London: Verso, 2021), 14–16.
57 Eyal Weizman, *Forensic Architecture: Violence at the Threshold of Detectability* (New York: Zone Books, 2019), 78.

2
Methodological framework: from systemic critique to micro-epistemology

An introduction to experimental systems and Rheinberger.

I. Reconfiguring the Evidencial Paradigm

The shift from Ginzburg's evidential paradigm to Forensic Architecture´s virtual investigations can be summarised as a shift from a system that finds traces and masterfully connects them to a system in which traces are translated into a data space and used to create other findings.

In the previous chapter, we have seen how in *Clues: Roots of an Evidential Paradigm* (1979), Ginzburg described a mode of knowledge characterised by the reading of traces, the deciphering of small signs that allow an investigator to infer hidden causes. His key figures, such as the doctor, the connoisseur, and the detective, interpret the world through conjecture and analogy rather than deduction. This "conjectural paradigm" rests on *indexical reasoning*: the belief that every event leaves a material residue which, when properly read, can reveal the invisible[58]. The epistemological model is retrospective: the trace is a fragment of the past, awaiting decipherment by a skilled interpreter.

Yet as Joachim Harst notes, this conception must be revised for the digital era, where the evidential field has become algorithmic, networked, and dynamic[59]. Digital forensics no longer operates within closed, static archives, but within recursive arrangements of tools, data, and hypotheses: these knowledge systems can generate new information through iteration and recombination.

Here, Hans-Jörg Rheinberger's theory of *experimental systems* provides the necessary epistemological bridge. For Rheinberger, scientific experimentation is not the confirmation of a prior hypothesis but rather the *emergence of new objects of knowledge* within evolving constellations of instruments, materials, and conceptual frameworks[60]. Knowledge, in this view, arises from the experimental system itself, through the recursive interplay between what is known and what becomes knowable.

It is not a coincidence that Harst explicitly draws on Rheinberger to describe the transformation of the evidential paradigm in Forensic Architecture's work. He argues that the laboratory and the investigative model now coincide:

58 Carlo Ginzburg, "Clues: Roots of an Evidential Paradigm," in *Clues, Myths, and the Historical Method* (Baltimore: Johns Hopkins University Press, 1989), 101–103.
59 Joachim Harst, "Virtual Investigations: Transformationen des Indizienparadigmas zwischen Sherlock Holmes und Forensic Architecture," in *Medienkomparatistik* 4 (2023): 6–10.
60 Hans-Jörg Rheinberger, *Toward a History of Epistemic Things: Synthesizing Proteins in the Test Tube* (Stanford, CA: Stanford University Press, 1997).

> "Digital investigations function as experimental systems in Rheinberger's sense, configurations that generate new facts through iterative visualisation, testing, and reconfiguration"[61].

Although Harst does not fully develop this epistemological connection further, it is worth pursuing in greater detail.

Again, this epistemic shift is vividly evident in Forensic Architecture's OSINT investigations. These are arrangements of individual pieces of evidence within a digital architectural model, also referred to as an image-data complex. In their reconstruction of the 2020 Beirut Port explosion, for example, hundreds of public videos were synchronised in time and mapped spatially to triangulate the source and sequence of the blasts. Each video, in isolation, was indeterminate, a partial witness. Only when aligned algorithmically within a digital model did these fragments acquire evidential force. What emerges here is not the discovery of a pre-existing fact, but the experimental production of a shared truth-environment, one with a collective forum in which the evidential field itself is continuously tested and refined.

In this sense, how does OSINT embody both Ginzburg´s and Rheinberger's epistemologies? OSINT operates at the intersection of Ginzburg's and Rheinberger's epistemologies: it begins from traces and clues, but it also establishes an experimental setting in which those traces can be made knowable. The trace thus shifts from being a passive sign to being an active medium, something through which the "real" is experimentally constituted.

This convergence explains the distinctive nature of Forensic Architecture's "production of evidence"[62]. Their investigations operate within open, collective experimental systems—epistemic assemblages that combine algorithmic tools, human witnesses, and aesthetic mediations. Each model, map, or animation is simultaneously a technical experiment, a public performance, and an ethical proposition. The production of truth they present is inseparable from its visualisation, and it is quite curious to see how Rheinberger sees in this a universal law of knowledge production, as we can see from the quote of Ernest Cassirer [63] that opens his book Split and Splice[64]:

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61 Joachim Harst, "Virtuelle Investigationen: Transformationen des Indizienparadigmas zwischen Sherlock Holmes und Forensic Architecture," in *Medienkomparatistik* 4 (2023): 41.
62 Eyal Weizman, *Forensic Architecture: Violence at the Threshold of Detectability* (New York: Zone Books, 2019), 64.
63 Ernst Cassirer, Philosophy of Symbolic Forms, vol. 3, Phenomenology of Knowledge, trans. Ralph Manheim (New Haven, CT: Yale University Press, 1953).
64 Hans-Jörg Rheinberger, *Split and Splice: A Phenomenology of Experimentation* (Chicago, IL: University of Chicago Press, 2023).

"No matter whether we are concerned with the ideal or the real, the mathematical or the empirical, with nonsensuous or sensuous objects, the first question is always not what these are in their absolute nature or essence, but by what medium they are conveyed to us; through what instrumentality of knowledge the knowing of them is made possible and achieved."[65]

II. Applying an epistemological perspective: limits of the current Forensic Architecture framework

In Investigative Aesthetics, M. Fuller and E. Weizman present a new philosophy of investigation, in which human and non-human actors participate in a complex network that is both "sensing" (registers information) and "sense-making" (reads and contextualises the information). From this network of actors, which they define as an aesthetic system, new representations of facts emerge. These new investigative systems welcome new entities, communities, frameworks or elements; while also unsettling Western-centred epistemologies and reconfiguring the relationship between power and the construction of truth through narrative. Against the idea of objective, universal representation of facts, a new investigative system should help challenge these overarching mainstream narratives and give space to the marginalised. From a more technical perspective, these systems could be described as situated and perspectival.

An interesting example in this direction is the case of the Death Alley[66]. Death Alley is the ominous name given to the portion of territory on the Mississippi River between Baton Rouge and New Orleans, a heavily industrialised "petrochemical corridor" that is also the settlement of black communities, which suffer greatly from the pollution of the nearby industrial complexes. To help these communities fight back and protect themselves, Forensic Architecture mapped the burial sites of once-enslaved cotton-field workers. These cemeteries are protected by state law and cannot be destroyed or owned by anyone other than the living descendants of these communities.

The case studies combined techniques of regressive cartography, scientific wind simulations and documentary interviews, among others. As a result, they documented the environmental harm caused by the industrial facilities, identified the locations of many burial sites, showed the unlawful expansion of these companies onto the burial sites, and collected direct testimony from the people living there. The communities of the Death Alley

65 Cassirer, Ernst. The Problem of Knowledge: Philosophy, Science, and History since Hegel. Tradotto da William H. Woglom e Charles W. Hendel. New Hain which human and non-human actors participate in a complex network that is both "sensing" ven: Yale University Press, 1950.

66 We are referring to the case study Environmental Racism in the death alley, by Forensic Architecture, 28 June 2021. https://forensic-architecture.org/investigation/environmental-racism-in-death-alley-louisiana

eventually gained media coverage that informed public opinion and provided legal leverage, which they mobilised against the expansion of the petrochemical facilities.

This case offers a good overview of how aesthetic systems work to generate knowledge and narrative; it shows how heterogeneous media and techniques come together to create a new type of system. In Weizman and Fuller's account, an aesthetic system is not subject to a fixed limitation in terms of what it may comprise. Yet this apparent openness is implicitly mediated by the presence of trained experts, whose role helps structure and validate the system.[67].

While this framework has many methodological implications which could potentially enrich the field of visual investigations, it also has some limitations, especially when addressing the technical processes that lead to the generation of knowledge. In other words, the systemic critique implied by Weizman and Fuller justifies the creation of hybrid investigative systems, but does not provide any solution on how to evaluate their internal coherence and the reliability of these systems.

A better understanding of the problem can be achieved by looking at the distinction that Hanna makes between *external* and *internal objectivity*: the first representing "an entity, process, magnitude, construct which "existence and properties are independent of its representation"[68] (the object is totally independent from subjective interferences), the latter, by contrast, is concerned with the processes or methods by means of which science investigates the world; so it might also appropriately be called "methodological objectivity": he internal objectivity lies in the capability of a system to create a consistent and stable outcome.

Looking again at Weizman's and Fuller's theory, we can recognise a critique of external objectivity, as a non-subjective representation of facts and their weaponisation by actors in power, but we cannot find a framework where the methodology of an investigative system can be scrutinised. Without a framework for establishing the internal objectivity of an investigative system, the consistency of its methodology seems to rest solely on ethical grounds.

What happens then when more than one agency, with similar moral and ethical values, investigates the same incident and reaches very different conclusions? And how do we navigate this sea? How do we determine the

67 Taking from Lorraine Daston & Peter Galison, *Objectivity* (New York: Zone Books, 2007).
This type of system operates through what they call "trained judgment" — where expertise and interpretive skill are considered the epistemic virtues of this epistemic regime. In other words, a trained judgment regime guarantees the objectivity of these methods.
68 Joseph F. Hanna, "The Scope and Limits of Scientwofold: on the one hand, it justifies the type grafting at stake in aesthetic systems; on the other, it provides a lens to analyse systems individuallytific Objectivity," Philosophy of Science 71, no. 3 (2004): 340.

standards by which the reliability of one body of work can be judged against another? Can we adapt and use a framing that would enhance our capacity to analyse OSINT case studies? We want to determine whether these case studies can be evaluated epistemically.

The problem of the internal consistency of an investigative system will be framed by evaluating its epistemic reliability: an evaluation that focuses on the relations among technological tools, methods, practices, and the knowledge-generation process at stake within it.

In this text, the theoretical framing we will use to analyse methodologies is borrowed from Hans Jörg Rheinberger's theory of experimental systems in biology.

Experimental systems are discrete, basic units at the core of scientific research that can generate unexpected nuances. What characterises them is their technical and technological nature, in which various stabilised processes/techniques are adopted in pursuit of open-ended outcomes. The goal of such systems is to generate a technical definition (an enumeration of technical processes) of their object of inquiry, which Rheinberger calls the epistemic thing.

For Rheinberger, experimentation in modern science is concerned first with the creation of systems capable of generating novelty; only subsequently do such systems become stabilised as methodologies that can be adopted within new experimental settings. The advantage of this framework is double, on one hand it justifies the type grafting at stake in aesthetic systems, on the other it provides a lens to analyse systems singularly.

III. From a systemic perspective to a situated one: the advantages of micro-epistemology

In the previous chapter, we introduced the need to address Forensic Architecture's practice from a different angle because of the limitations of its current theoretical framework. As we mentioned, Fuller´s and Weizman's framework fails to address how their systems work methodologically, implying that what validates them is the expertise of trained experts and their moral-ethical values.

The theoretical framework presented by Fuller and Weizman is inscribed in Science and Technology Studies (STS), post-Latour and Kuhn. STS is a critical field of study grounded in a sociological matrix; it pursues a critique of science and technologies as systems embedded in society and, therefore, in its economic, racial, and gender inequalities. Traditionally, STS looks at the general system: how scientific fields relate to politics and society, and how inequalities manifest within it. Moreover, it pushes for a less Western-centric discourse in science.

STS has provided conceptual tools and analytical frameworks that Forensic Architecture and OSINT practices more broadly have drawn upon to expose structures of oppression and support marginalised communities. In doing so, it has helped illuminate the relationship between hegemonic representations of fact, the mobilisation of scientific authority, and processes of discrimination. Although STS has been fundamental in framing some of the most relevant OSINT practices -for example, Counter-Forensic - as we already mentioned, it does not offer much help if applied to specific cases in which OSINT operates, and this can become extremely problematic in a field that displays a vast array of heterogeneous investigative practices.

The Rheinbergerian framework proposed here operates on a different conceptual ground: what Gaston Bachelard, within the philosophy of science, called micro-epistemology. Micro-epistemology does not understand the history of science as a linear progression. Instead, it attends to the moments in which particular experimental systems and theories emerge. Its scope of analysis is therefore narrower, concentrating on one system at a time. Although Kuhn's theory of ***paradigm shift*** already broke with the idea of linear scientific progress, in both STS and philosophy of science it still situated its focus at a macroscopic level. With Bachelard, the scale changes completely, causing broader social and systemic dimensions to fall into the background.

By leaving out the macro aspects of science, it is possible to focus on the structure of experimental processes, which are, according to Rheinberger, at the core of scientific advancement since the XX century. For a microepistemologist, aesthetic systems, as experimental ones, could be better understood if analysed in their "uniqueness" and in the unpredictable hybridisations between these systems.

Micro-epistemology means concentrating on a particular scientific momentum and tracing the technological and methodological arrangements that allow scientific novelty to emerge.
For us, looking at aesthetic systems from this perspective entails looking at the development of specific frameworks, the ways in which they become stabilised procedures, and what type of knowledge generation processes they imply.

3
Defining the Framework: Mark Duggan Case Study

The chapter highlights the structural differences between classic forensics and FA's approach using the case study on the killing of Mark Duggan.

I. Methodology

This chapter adopts a comparative methodological approach through a series of case studies to examine how contemporary visual and open-source investigations generate knowledge. Rather than treating each investigation primarily in terms of its conclusion, the chapter analyses the structure of the investigation behind it: the types of evidence collected, the technical procedures used to process that evidence, the role of modelling and expert interpretation, and the epistemic logic through which findings are produced. The central aim is to move from the level of claims to the level of method, asking not only what each investigation concludes, but how it arrives there, with which epistemic regime.

The chapter is organised into two parts. It begins with the case of Mark Duggan, which serves as a methodological entry point because it offers unusually detailed documentation of Forensic Architecture's methodology and enables direct comparison with more conventional forensic reports. This first section identifies the internal structure of an investigative system and clarifies the distinction between established forensic procedures and more open, model-driven forms of inquiry. The second part turns to the Al-Ahli Hospital explosion, a case in which multiple organisations and independent researchers (we will focus on 10 of them) investigated the same event using overlapping but significantly different methods. By comparing these case studies — including media investigations, NGO reports, state claims, and computational analyses — the chapter maps the methodological scaffolding and distinguishes between investigations grounded primarily in conjectural reasoning and those that rely on modelling as a primary means of knowledge generation.

To ensure comparability across these different cases, each investigation is presented through the same analytical structure. Each case study is summarised by its publication context and claim, its central question, the evidence and data inputs it mobilises, its methodological breakdown, and its underlying epistemic paradigm. In the case of Al-Ahli, this qualitative analysis is further supported by a set of comparative tables that synthesise the main variables across investigations, including methodology, actor type, stance distribution, and epistemic paradigm. These tables are meant to provide a systematic overview of the case study and to make cross-case comparison more explicit.

Taken together, these case studies allow the chapter to address a broader methodological question: how should OSINT and visual investigations be evaluated when different actors, working from partially shared evidence, produce conflicting conclusions? The structure of the chapter, therefore, moves from a close analysis of one investigative system to a comparative analysis of several, using the conceptual framework developed in the previous chapter to assess the epistemic paradigms, methodological coherence, and evidentiary logic at stake in each case.

II. Analysis of an investigative system: The case of Mark Duggan

On the 4th August 2011, English police shot Mark Duggan to death. The victim, a black man, was under the surveillance of a controversial police squad that was monitoring gun crimes in black communities in London. Police officers reported that the man had a gun, and the death was the result of a split-second decision one of the officers was forced to take. As an immediate reaction, the death of Duggan gave rise to riots and protests that spread across London and other British cities. As a member of the black community shot by a white policeman, he became the symbol of police racism and, more generally, of institutional racism. To worsen the situation, the legal case was conducted in a manner that notoriously favoured the officers. The London Independent Police Complaints Commission (IPCC) conducted a very murky investigation that did not lead to any meaningful findings on the murder. This partially deliberate lack of evidence led to the officers' acquittal in 2013.

Concerns about the lack of transparency and obfuscation strategies of the IPCC were even noted by the Judge in the final report of the case:

"*As I have indicated, there was intelligence relevant to Mark Duggan's death which the jury could not see. Exceptionally, the Senior Investigating Officer at the IPCC was permitted to see it. However, a senior police officer in an independent police service, from whom the IPCC thought it necessary to get an expert opinion, was not so permitted. That prevented her from forming a fully informed view about the planning of the operation. I would have liked to put her report before the jury and to call her to give evidence, but did not do so because she had not seen the intelligence picture. Furthermore, the IPCC is plainly being hampered in its task by not having the benefit of her expertise. [...]*"[69]
The report ends in a rather worrying note:
"*These limitations not only give rise to understandable suspicions in the minds of those not party to the intelligence but also plainly create a risk that an intelligence-led operation which results in death will not be fully investigated so that lessons may be learned.*"[70]

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69 Inquest Touching the Death of Mark Duggan: Summing Up (London: Her Majesty's Courts and Tribunals Service, 2014),32,https://www.judiciary.uk/wp-content/uploads/2014/06/Duggan-2014-0182.pdf
70 ibid.,32

Given this development, the family of the victim reached out to Forensic Architecture in order to run an investigation on the killing of Mark Duggan, which was based on the evidence presented in the original court case and on later forensic reports, commissioned by Mark Duggan's family and Forensic Architecture.

The new case study was built around the fundamental question **"How did the gun reach the ground?"** The gun in question was found on the grass over a fence at an approximate distance of seven meters from the body of the victim, thus implying that Mark Duggan threw the gun himself before he was shot to death. However, from the officer's testimony emerges that many of them have not seen the victim tossing the gun; the policeman who shot Mark Duggan even said that he saw the gun in his hand and one moment after it disappeared. The Forensic Architecture case - as the court case before - was built around these incongruences, and defining whether Mark Duggan was able to toss that gun or not would have implied that there had been a police cover-up.

The question served as the foundation of the methodological scaffolding, helping to define how the case would be constructed. In this specific case, 3D reconstruction of the environment and of the various human actors and their movements was considered the best choice. As the next step, we see the translation of the data into a new medium — specifically, all the evidence was translated into the 3D scene, allowing for the data to be cross-referenced and compared into the same space, "so that a pattern", or in this case, new findings," could emerge"[71].
Eventually, FA's investigation was presented to reopen the case, but the evidence was deemed insufficient, and the request was denied.

Through this translation of the medium, FA supposedly managed to create new evidence by combining and matching different bits of information within Blender (open-source 3D creation software). These further enabled the team to process the data using available tools, such as animation, body tracking, and VR. In this way, they were able to gather new findings and eventually came to the conclusion that the evidence suggests that the officers could have moved the gun after Mark Duggan was already dead.

71 Rheinberger, Split and Plice, 14

Case Study analysis:

FA offers extensive documentation on the case, covering the original legal case, what they were trying to contest from that verdict, and the methodology they developed to create evidence against IPCC investigations. Our focus here is not to examine every single piece of evidence, but rather to understand the methodology used.

Evidences:

For this investigation, FA used the open-source material that is still fully available on the British National Archive website. Along with the police testimony, there were video and photographic material made by civilians, collected evidence and the following scientific and forensic reports:

Evidence

- Footage (BBC, civilian witnesses)
- Police Testimony (9 officers were present at the crime scene)
- Civilians Testimony
- Scientific Reports
- Expert Reports
- Evidence from the crime scene (the gun, Mark Duggan´s Jacket)

Forensic reports

- **Post-Mortem Examination** by Forensic pathologist Professor Derrick Pounder on 19 August 2011, at the request of Mr Duggan's family. Professor Pounder subsequently provided a detailed statement to the IPCC and to other experts commissioned by the IPCC, including Professor Clasper.
- **Biomechanical report** on Mark Duggan's Movements when the shot happened, by biomechanical engineering expert and consultant orthopaedic surgeon Professor Colonel Jonathan Clasper, commissioned by Forensic Architecture. Dr. Clasper was asked to prepare his report based on the assumption that Mr. Duggan was in possession of a gun at the time that he was shot.
- **Biomechanical report** by Dr. Jeremy Bauer (Bauer Forensics), commissioned directly by the MD family in the context of the civil claim (2019).
- **Analysis of witness B footage**[72], by Imagery Analyst Mr. David Charles Thorne (former police officer)
- **Analysis of available video evidence,** by Mr. Clive Richard Burchett (LGC Forensics), commissioned

72 Witness B refers to a civilian that took a long video with their phone of the crime scene. It was one of the main footage evidences of the case, together with the BBC footage.

by Judy Kemish, solicitor to the inquest.

The main footage evidence came from witness B, who filmed from the building close to the crime scene, with the police cars on one side and the fences and trees on the other, which severely constrained the view.
Police testimony was one of the main pieces of evidence; the depositions of the various officers were contradictory on how facts happened and whether the victim had a gun in his hand, whether he threw it or not.

Usually, the development of a judicial case goes as follows: first all the evidence is collected being footage or testimony or elements from the crime scene, then all the available testimony is reorganized (depositions are saved and compared, footage is chronologically matched by experts), next steps are the scientific/forensic reports which already used the available information to reconstruct important elements of the crime scene and events.

To better understand the differences between standard forensic reports and Forensic Architecture's reports, we will take a closer look at one of the reports, namely the one from Jeremy Bauer (Biomechanical expert). The documents state the qualification of Dr. Bauer, then proceed to enumerate the given material. The reports aim to understand the approximate body position of Mark Duggan while he was being shot, and whether he was able to throw the gun beyond the fence, as some police testimony reported.
At first, Dr. Bauer reports all the available information on the autopsy of the body made by Mr. Simon M. Poole on 5th August 2011; he also references a 2D reconstruction of the scene (Fig.2) based on the footage and police testimony.

He then proceeds to describe the methods used to determine under what conditions Mark Duggan could have thrown the gun over the fence.
As the first thing, Dr. Bauer defines the physical framework that will be used, framing the problem as follows: "How much energy was needed to throw the gun that far?"

> "The fundamental physics of projectile motion were used to determine the initial velocity and trajectory required to toss the gun over the wall, fence, and/or the surrounding vegetation from the area where he was shot, such that the gun landed and came to rest in the grass on the south side of the wall and fence."[73]

73 Bauer, Jeremy J. Report on the Death of Mark Duggan. Claim No. HQ12X02226, Pamela Duggan & Others v The Commissioner of Police for the Metropolis. Seattle: Bauer Forensics, LLC, June 21, 2019conducted, namely by using 3D point clouds of the crime scene and 3ds Max (a well-known.

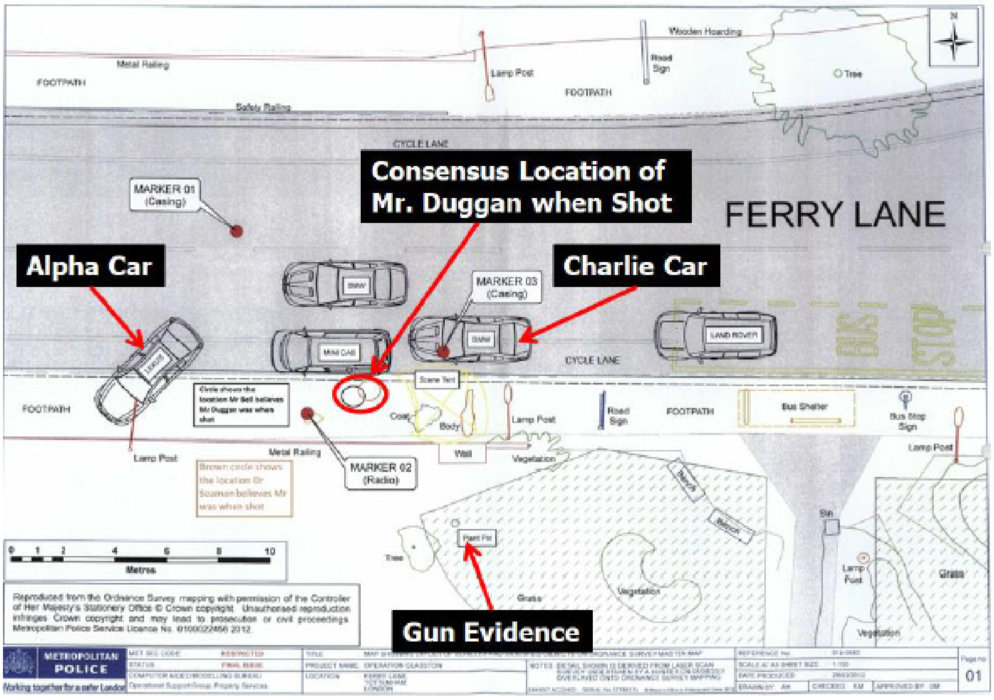


Fig.5: 2D reconstruction of the scene

Then defines how the simulation part of the experiment will be enacted, namely using 3D point clouds of the crime scene and 3D Studio Max (a famous software for 3D CGI) to animate and simulate. (Fig.5)

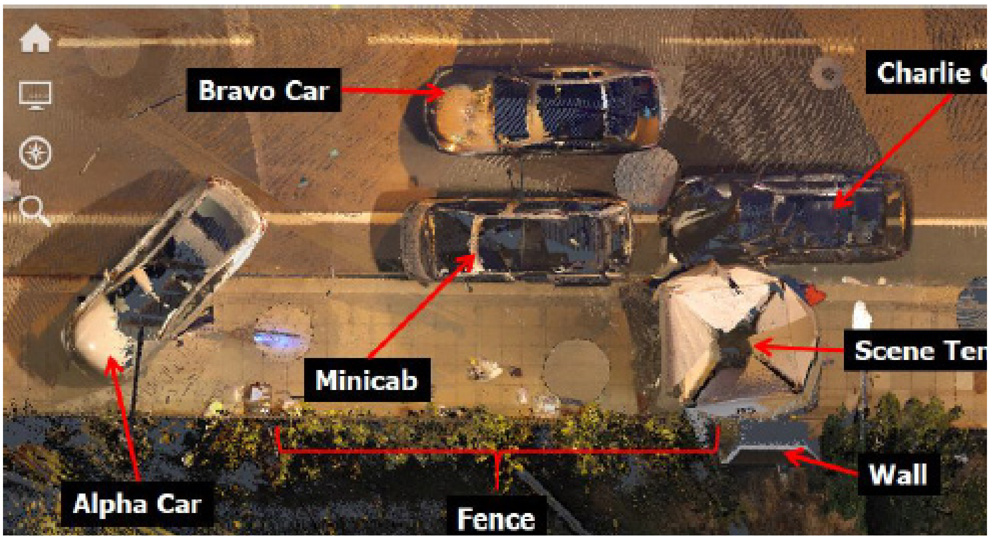


Fig.6: Scene capture from laser scan processing software (Autodesk Recap v. 6.0) showing a top-down view.

A surrogate of the body of the victim is recreated in 3D using Adobe Fuse CC (v2017.1.0), so that studies on projectile paths and the posture during the shots could be done in order to define later whether Mark Duggan was capable of throwing the gun that far. (Fig.6)

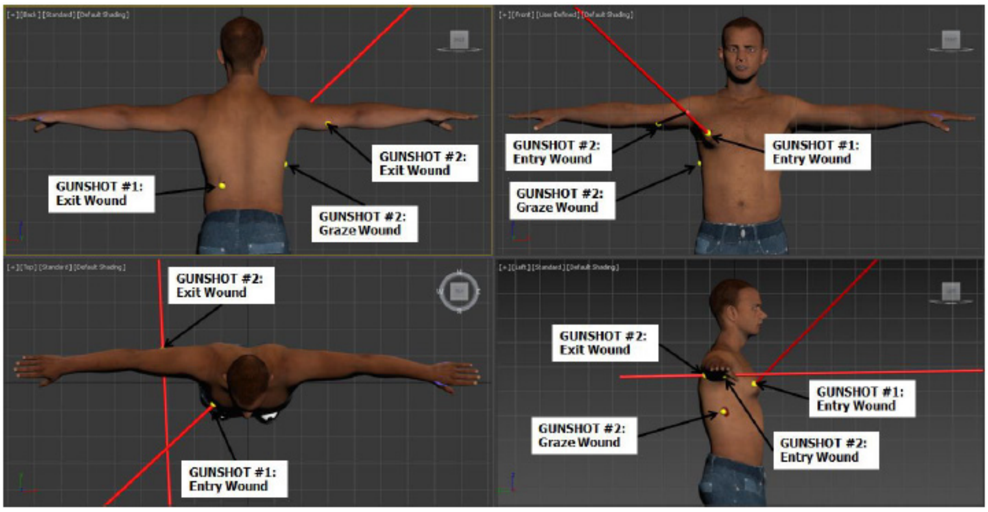


Fig.7: Snapshot of the surrogate 3D body of Mark Duggan. Body made in Adobe Fuse CC.

quoting directly from the report:

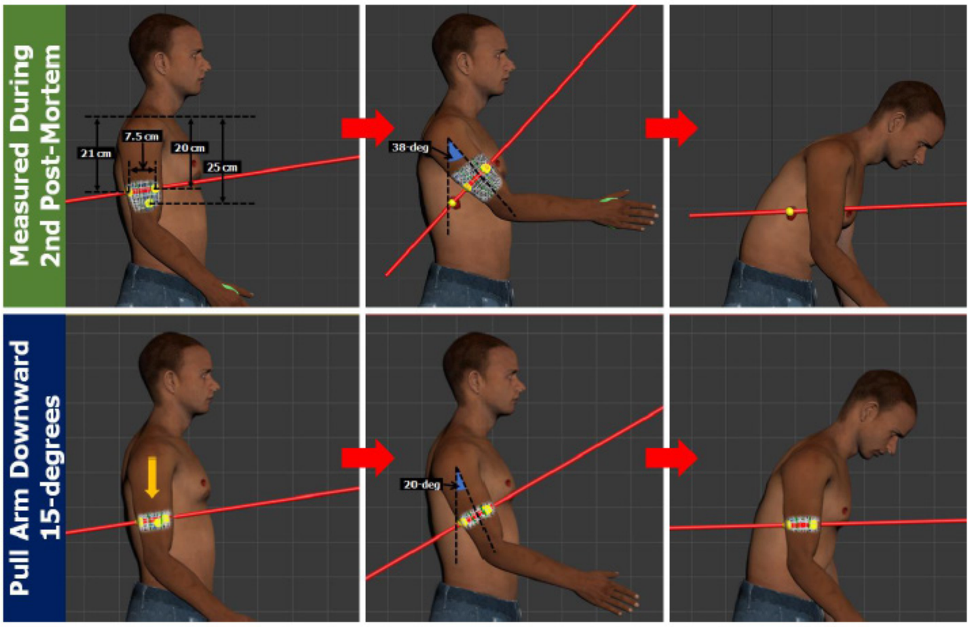
"Thus, to determine Mr. Duggan's likely posture after having been shot, for the purposes of determining his likely orientation upon throwing a gun to the area where the gun was documented, the paths of the two bullets that struck Mr. Duggan were marked on the 3D surrogate (Figure 5). Five spherical markers were placed on the surrogate at the specific anatomic locations associated with bullet wounds identified in Mr. Pounder's second post-mortem report. Two cylinders were then created to represent the two bullet paths. The cylinders were connected between the respective entry and exit wounds. The torso and right arm were then positioned to angle the bullet paths such that they were consistent with having originated from V53's position on the pavement, east of Mr. Duggan. Once positioned in the scene, the right arm of the surrogate was oriented such that it was fully extended toward the top of the fence. This position of the hand was considered the point where the gun may have been released had Mr. Duggan tossed a gun after he was shot by V53."[74] *(Fig.7)*

Once the 3D surrogate is imported into 3D Studio Max, Dr. Bauer can map all the information he gathered: The autopsy, informing on the entry and exit point of the bullets, the deposition of the police officer V53, and the location of the gun. (Fig.7)

In subsequent steps, the body's position during the 2 shots was reconstructed, indicating that the torso was flexed forward. (Fig.7) For the second shot, two different reconstructions were done, the first following the second post mortem and the latter with a rotation of 15° of the arm. (Fig.7)

74 Bauer, Jeremy J. Report on the Death of Mark Duggan. Claim No. HQ12X02226, Pamela Duggan & Others v The Commissioner of Police for the Metropolis. Seattle: Bauer Forensics, LLC, June 21, 2019.

Fig.8: Reconstruction of body position



Eventually, the surrogate was placed into the point cloud of the crime scene (Fig.9), now with one rather good approximation of Mark Duggan´s body position if he actually managed to throw the gun.

Fig.9: Reconstruction of body position while throwing the gun in 3D Studio Max

"Given the direction of the two bullet paths, had Mr. Duggan tossed a gun over the fence, he would have had to toss the gun directly to his right side (Fig.9), with a large sweeping motion of his arm, or with a motion consistent with throwing a flying disc, or "Frisbee", directly to his right side, and likely from a posture where he was bent forward. Mr. Duggan was standing approximately 7.1 m from where the gun evidence was documented when he was shot. The horizontal distance from the gun to the middle of Mr. Duggan's hand upon raising his arm from his side was approximately 6.2 m. The height difference between the area where the gun evidence was documented and the height where Mr. Duggan would have released the gun was approximately

2.0 m. Given the height of the fence and Mr. Duggan's likely position on the pavement with respect to the fence when he was shot, he would have had to toss the gun with his right hand at a minimum speed of 6.7 m/s (22.0 ft/s) at an angle between 31 - 40."[75]

In the last part of the report, considerations are made on the muscles' conditions and the overall body reactivity after the 2 shots. With a rather detailed analysis, Dr. Bauer comes to the conclusion that Mark Duggan was not in the physical condition of throwing the gun at that distance.[76]

III. The construction of the case study and the method of Forensic architecture

As we can see, the initial material used by Forensic Architecture is rather heterogeneous, ranging from a very basic indexical type of evidence (the jacket) to layered scientific reports.
Moreover, most of the reports (like the Bauer one) are not considered as valuable or objective as first-hand evidence, for the jury can reach such a conclusion only after a careful comparison of all proofs together; only then, when a coherent and exhaustive view of the event emerges (or not), can a report be considered reliable. As we said previously, in this court case, the available evidence was deemed insufficient; the police officer was found innocent, and the case closed.

Forensic Architecture's role in this setting was then to bring evidence adequate for a juridical environment, while also reconfiguring the available data so that new evidence could emerge. Moreover, Forensic Architecture plays an important role in denouncing human rights violations by state actors; thus, certain types of linguistic and visual rhetoric are employed when presenting the case, and, independently of the development of a court case, the public can be informed. We have to keep these considerations in mind to understand how the investigation employs methodologies and tools very similar to those we've seen used in Dr Bauer's report, while also adopting new methodologies and, at times, a different rhetoric. We can assess that the praxis of Forensic Architecture is centred on the tension among these 3 elements: established and renowned methods, methodological and technological innovation, and a human rights agenda (thus a more comprehensible rhetoric to reach a wider audience).

75 Bauer, Jeremy J. Report on the Death of Mark Duregarding the muscles' condition and the overall body'sggan. Claim No. HQ12X02226, Pamela Duggan & Others v The Commissioner of Police for the Metropolis. Seattle: Bauer Forensics, LLC, June 21, 2019.

76 I conclude, to a reasonable degree of scientific and biomechanical certainty, that: 1) Mr. Duggan was bent forward, toward V53, when he received both gunshot wounds; 2) Mr. Duggan likely could not have tossed, or even held, the gun after being shot by V53; and 3) Had Mr. Duggan been able to toss the gun after being shot, numerous officers would have witnessed the swinging arm motion required to toss the gun to the grass from the area where Mr. Duggan was shot.
Bauer, Jeremy J. Report on the Death of Mark Duggan. pp.19.

As written above, Forensic Architecture´s investigation was meant to reopen the case, and it was developed around the question ”How did the gun reach the grass?” This was considered a central element of incoherence between the scientific report we have analysed and the police deposition.

Similar to what we have already seen, the Forensic Architecture team proceeded to 3D-scan the incident scene; they used Reality Capture to generate the point cloud and the 3D mesh. The Photogrammetry served as a base to reconstruct a highly precise 3D scene in Blender.[77] Subsequently, important assets such as the vehicles were placed, and for the sake of fidelity, the scene was populated with the same foliage present in the real place, following overall reconstruction procedures and tools proper to architectural workflows.[78]
The 3D surrogate of Mark Duggan was made with MakeHuman Software, subsequently rigged[79] in Cinema 4D and transferred to Blender, where the overall scene was constructed, following the crime scene 2D reconstruction present in the evidence (Fig.5). Using the available post-mortem reports they marked on the surrogate the entry and exit points of the bullets with a technique known as "vertex painting" and finally tried to reconstruct the position of Mark Duggan according to the shotlines. (Fig.10)

So far, the methodologies adopted by Forensic Architecture are the same as those used in the Bauer report; the main differences are the software used (i.e., Blender instead of 3D Studio Max) and the "look" of the scene (more polished and anonymous in FA´s case). Until this point, the methodologies used represent the state of the art in modern forensics; moreover, the reconstruction by FA is explicitly a copy of the Bauer report. And not only in Bauer´s report, but we also find the same methodologies used in other reports reapplied here. This statement holds for the method used to estimate the location of the officers W42 and v53 and the synchronisation of all available footage (witness B and BBC).

77 from the Forensic Architecture report:"Our digital model of the real-world site of the incident (Ferry Lane, Tottenham) is constructed from a photogrammetric survey of the area conducted by members of FA on 8 March 2019. We refer to this layer of the model, which includes accurately-dimensioned reconstructions of urban infrastructure, street furniture, and vegetation, as the 'site'."
Forensic Architecture. "The Killing of Mark Duggan: Report and Methodology." Forensic Architecture, Jan. 2020, pp 45, content.forensic-architecture.org/wp-content/uploads/2020/06/2020.06-Report-The-Killing-of-Mark-Duggan.pdf.
78 The site layer is subsequently populated by models of the objects (such as vehicles) that were present at the time of the event. These digital objects are constructed using architectural modelling techniques, and by close reference to the Witness B footage, aerial plans prepared by the MPS97 and photographs taken at the time.98 By assembling and cross-referencing these models in 3D space we can precisely corroborate the dimensions of the relevant objects, resulting in a highly accurate model.
ibid.,45
79 In the 3D sphere, to rig something means to create a skeleton that makes your 3D model move according to your needs, for instance making your 3D human model move like a real human.

Fig.10: Blender scene with reconstruction of the bullet trajectories with the vertex painting technique.

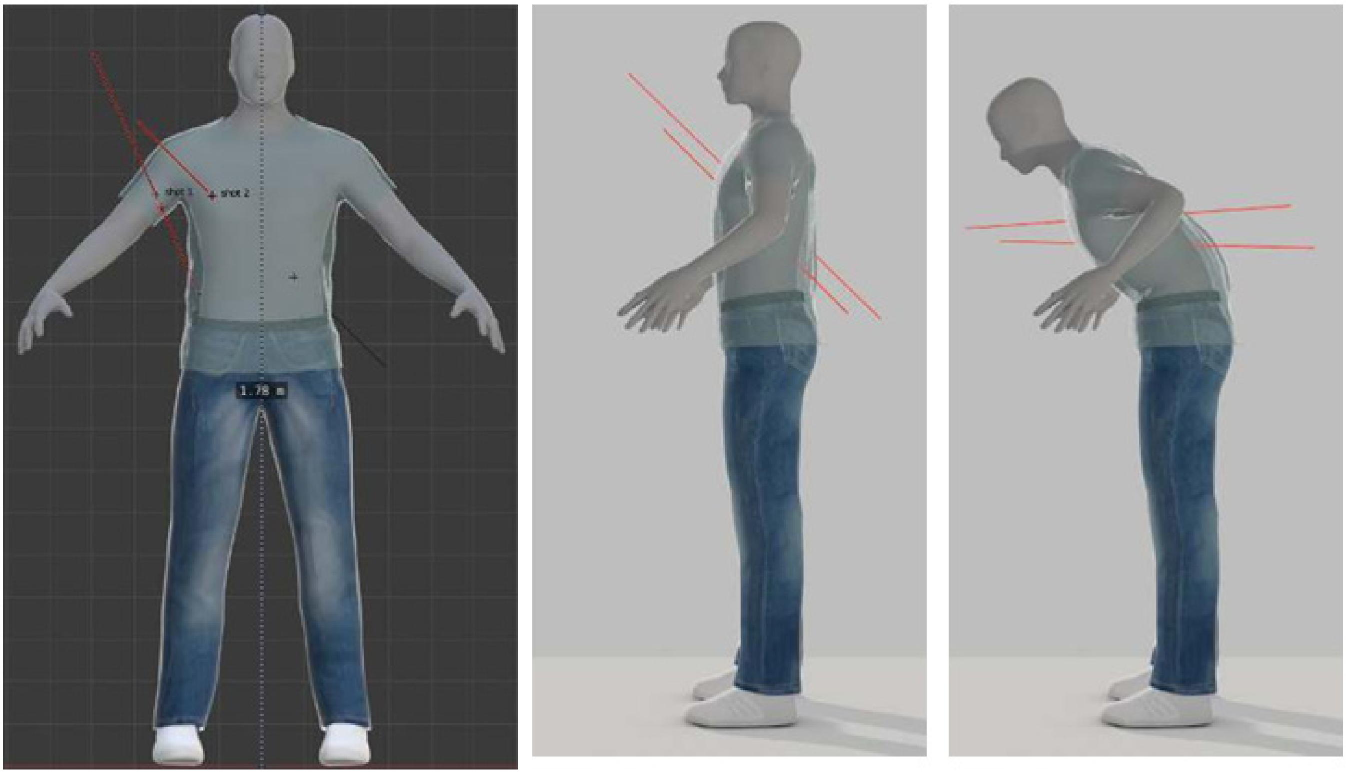
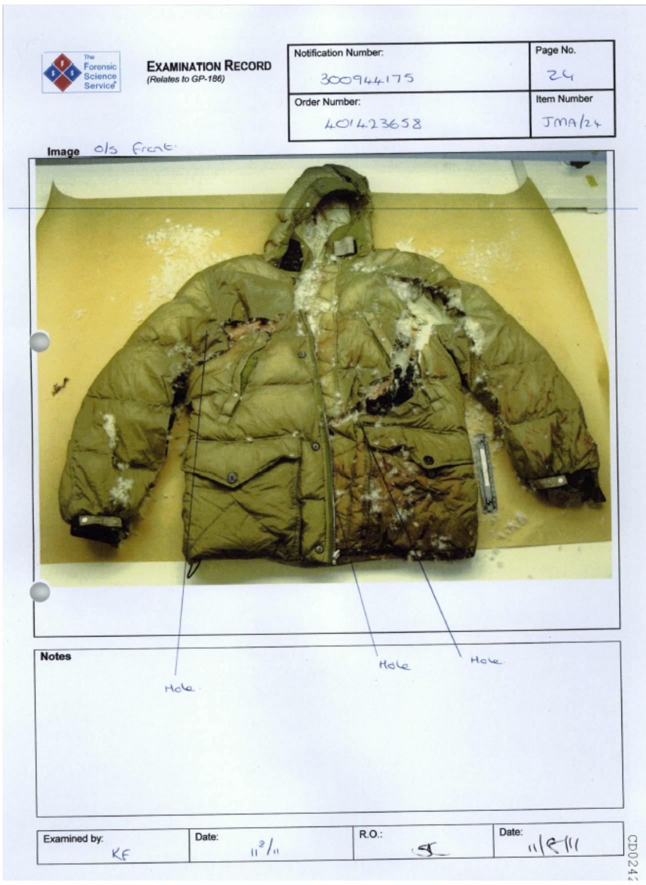


Fig.11: Jacket of Mark Duggan


The Forensic Science Service

EXAMINATION RECORD
(Relates to GP-186)

Notification Number: 300944175	Page No. 24
Order Number: 401423658	Item Number JMA/24

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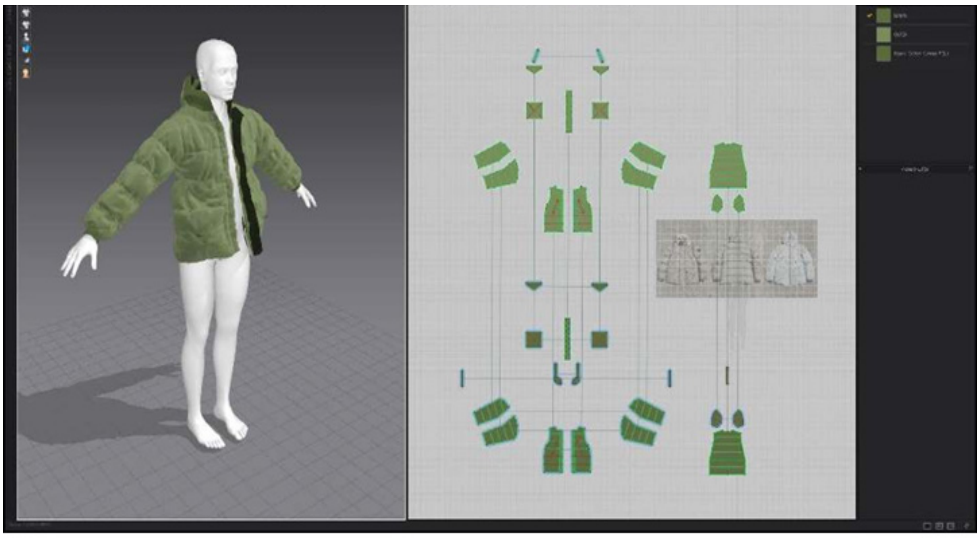
Fig.12: Reconstruction of Jacket on 3D software Marvellous Designer



The first methodological difference was the addition of the victim's jacket (Fig.11), which also presented bullet holes. Reconstructed in 3D with Marvellous Designer (Fig. 12), it was then placed on the animated surrogate body, allowing them to scrutinise whether the holes in the jacket were consistent with the body movement reported in the officer's testimony. More precisely:

"This was necessary because of the damage to the lower left corner pocket of Mr Duggan's jacket, and what this implied about the position of Mr Duggan's left arm at the time of V53's second shot" [80]

A method called "projection mapping" (Fig. 13) was used to project witness B's footage onto the crime scene directly. This way, the 3D of the crime scene was matched to the footage and "the observed movements of the officers were visually related to other features of the site, such as vehicles and street furniture. This made the process more accurate and mitigated the challenges posed by the poor quality of the footage, particularly its low pixel resolution, the effects of image compression, and the lack of 'anti-aliasing'.

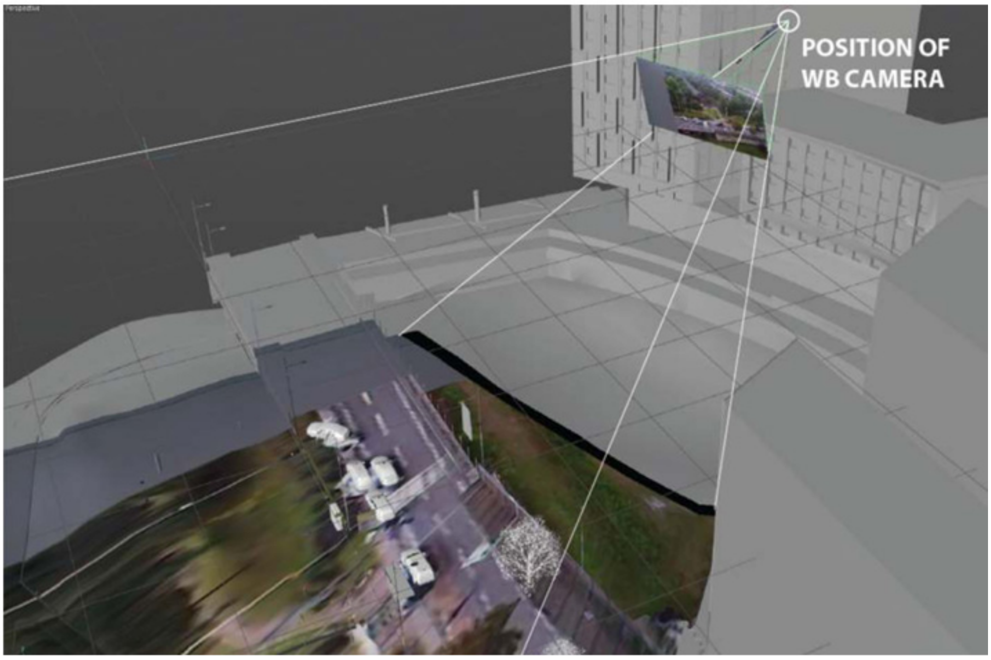


Fig.13: Projection Mapping of witness B footage done in Blender.

As a results to this addition it was possible to track and visualize the blind spots of the scene from witness B's perspective (Fig.14), which further helped defining the movements of the officers and the time they spent inside of blind spots, thus making possible the argument that some of the officers could have been able to move the gun into the grass.[81]

80 ibid.,50.

81 Important to note that photomatching is a quite established technique in the movie and entertainment industry, which allows to place 3D elements in real footage.

Fig.14: Tracking of the blind spots in Blender thanks to projection mapping.

The testimonies of the officers were organised in a spreadsheet "in which every row constituted a separate account (e.g. W70's witness statement of 7 August 2011), and every column constituted a 'moment' (e.g. Mr Duggan moves towards V53 and W70)."[82] This rearrangement was then used to reenact all the testimony in the 3D scene, making discrepancies in the testimony emerge more clearly.

Adding a more approximate methodology, Forensic Architecture speculated on the manoeuvre (Hard-Stop) the police used to stop the Cab where Mark Duggan was travelling. In order to do so, they used a police training video (CO19) that illustrates how to perform a "Hard-stop" maneuver, an aggressive, high-risk vehicle containment procedure where officers use multiple patrol cars to surround a suspect vehicle, forcing it to stop immediately. Adjustments to the timings, locations, and pacing of the vehicles and officers in their site model were made, "*by taking the training video as a baseline, and taking account of elements of those officers' accounts of the 'hard stop'* "[83]

Lastly, we have the adoption of a VR headset to view through each officer's eyes. This last methodological (and technological) addition played an important role in the Forensic Architecture report because it allowed us to challenge the testimony of some officers (e.g. some said they did not see Mark Duggan throwing the gun, while the VR reconstruction showed it was impossible not to). (Fig.16)

82 ibid.,46.
83 ibid.,55.

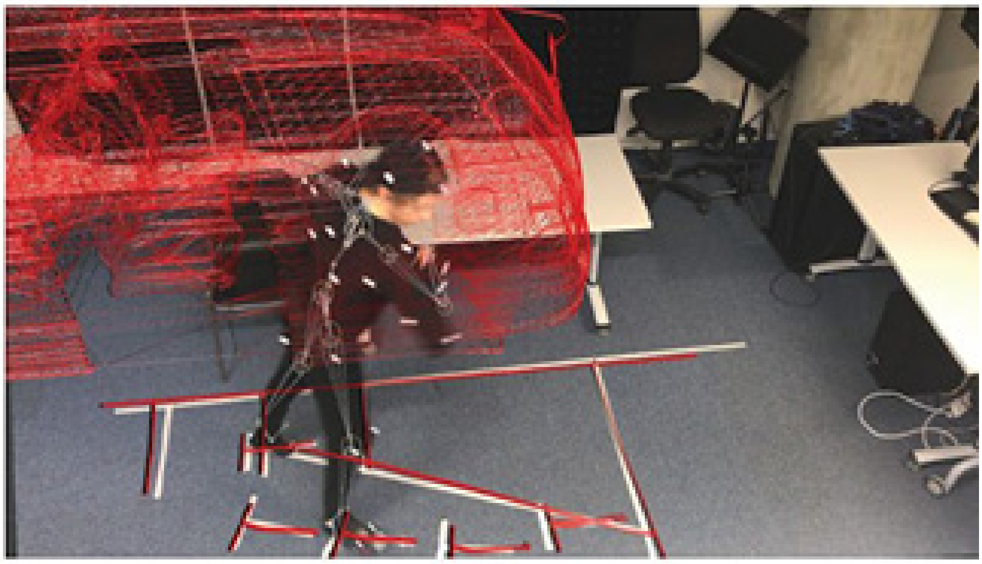


Fig.15: Forensic Architecture researcher acting as Mark Duggan with Motion Track suit.

Fig. 26. A still from our reconstruction of V53's approximate perspective during the period of the shots, if Mr Duggan had thrown the gun after V53's second shot (Scenario S1c).

Fig.16: viewpoint of one of the officers, VR reconstruction.

This report represents firstly a summary of the main evidences and methodologies used during the original court case (by quoting the previous findings as well as reproposing the same methods), secondly a more flexible scaffolding in which all the available evidence could be referenced and remapped into one another (projection mapping of witness B footage and emergence of the blind spots in the 3D reconstruction) and thirdly the adoption of experimental methodologies and technologies which aim to make new evidence emerge (the adoption of a VR headset).

In opposition to what we saw in the Bauer Report, where the aim of the report is narrow (defining if was mechanically possible for Mark Duggan to throw the gun under the given conditions), the finding holds thanks to multiple scientific and experimental angles (anatomy, biomechanics engineering, 3D reconstruction using testimony and post-mortem reports) and academic quotations strengthen the argumentation, Forensic Architecture's report has the ambition to represent an overall picture of the events. Even if the question asked is similar to the one from Bauer, "How did the gun reach the grass", the findings presented by FA do not contest only one claim (like the impossibility of Mark Duggan to throw the gun at that distance in Bauer's case) but a rather big number of previous findings[84]. At the beginning of the report these findings are presented: 1) the victim was unable to toss the gun 2) evidence shows that police officers could have tossed the gun 3) the sequence of the gunshots testified by officer V53 was incorrect 4) IPCC had no right to conclude that officers could not move the gun based on witness B footage 5)Imagery analyst commissioned by IPCC failed to synchronize the footage properly.

This approach can hardly be assimilated into a forensic report; if that were the case, the aim of the investigation would have been narrowed down, and more time should have been spent in justifying a claim from a multiplicity of aspects[85]. Also, from a rhetorical perspective, the language used in Dr Bauer's report was more technical and aimed to present scientific work, while Forensic Architecture tries to be clearer and more discursive.

However, the most relevant difference for this thesis lies elsewhere. In the Bauer reports, we see how narrow experiments are conducted to generate reliable findings, which then must be compared and related to other findings so that a thesis can be developed and a verdict reached. Here, the final epistemic goal is reached through conjectural reasoning.

For Forensic Architecture, a different paradigm is at stake: one based on the idea of emergent knowledge.

The translation of evidence into data and its mapping into different data spaces (from testimony to 3D animation, from 2D footage to 3D projection, etc.) enables processes in which new knowledge emerges directly from the models, sometimes even unexpectedly. [86]
Even if FA used tools and methods similar to those used by Dr Bauer, their focus lies in the possibilities that the techno-methodological scaffolding they developed can generate nuances beyond the limitations of a more institutional but narrower framework. In contrast, Dr. Bauer uses the very same tools as, implying that they are largely used in this field and their use guarantees stability and consistency in the process.

84 ibid.,9.
85 see Bauer report in the previous chapter.
86 In their report Forensic Architecture explicitly says that through their investigation 3 additional findings were made, thus meaning that they were not expecting to find them. ibid.,9.

If we frame Forensic Architecture's approach as model-led, we can better understand why and how a more stable, institutionally accepted framework is partially sacrificed in favour of unconventional tools (such as VR) and methods.

In this light, forensic sciences offer a solid base into which additional frameworks and tools can be grafted, and the knowledge generation process can shift from an evidential, conjectural one to an emergent one.

Considering all of this, Forensic Architecture's work cannot be properly evaluated within the boundaries of modern forensics and instead requires a different, more appropriate lens.

IV. Kindred spirits - FA & Rheinberger

While Forensic Architecture's methodology partially coincides with that of modern forensics, the major difference lies in the principle that guides their knowledge generation. If we were to frame this in Rheinberger's terms, we would say the main difference lies in the gesture. Constructing a system based on emergent knowledge is, to a certain extent, more similar to what modern experimental sciences do. That is to try out different frameworks and established and new procedures until a new type of finding emerges, or, to summarise with Rheinberger, modern science's "*gesture*" is to *"make the invisible visible"*[87].

"*The whole effort is designed to reveal structures and processes to our eyes or, more generally, to make them accessible to our senses, to make manifest those things that are not disclosed by unmediated observation and thus are not immediately evident.*"[88]

Revealing microscopical structures, manifesting DNA or amino acid sequences, are ways of unveiling in life sciences. In the context of Forensic Architecture, the gesture of unveiling concerns understanding events that are spatially and temporally remote, which manifest only through small bits of evidence.

In the Mark Duggan case study, we acknowledge how the available evidence is reused and recontextualised, for example, by adding witness B's footage directly into the 3D scene. We also see the adoption of a new technology, the VR headset, to show the officer's point of view.
Other glimpses of a possible comparison between Forensic Architecture's investigative approach and Rheinberger's experimental systems can be found in Harst. He points out that in open-source investigations, we have a similar"epistemological revolution" to the one described by Rheinberger in the natural sciences, where the investigative framework moves from theory-led to data-led research.

Other similarities can be found by looking at both Rheinberger´s and Weizmann-Fuller's theories, such as the relationship that scientists and FA researchers have with the technological environment in which they are immersed. For both (scientists and OSINT professionals), it is valid Weizman's and Fuller's statement that new technologies are tools that amplify our sensing faculties while also opening new possibilities of sense-making[89]. And for both, Rheinberger's definition of the relation between research and technology holds, when he writes that for modern sciences, the very possibility of knowledge production "*depends on the*

87 Rheinberger, Split and Splice,11
88 ibid.,11.
89 Matthew Fuller & Eyal Weizman Investigative Aesthetics:Conflicts and Commons in the Politics of Truth,(London, Verso, 2021)

media-technological landscape that subsumes the very procedures that are at stake"[90]. In other words, the production of knowledge lies in the procedures and technical means which technology enables, again "*the sciences as we know them today only exist in and through this media-technological landscape.*"[91]

To make a parallel with what we saw in the previous chapter, it is impossible to imagine Forensic Architecture's findings as valid, if not sustained by the technological apparatus they used (complex 3D environments in which a multiplicity of different data could be translated through photomatching, 3D scans, animation techniques and so on). In this landscape, the constant advancement of technological means drives a tendency to continually test and refine procedures and tools, building upon established frameworks, expanding them, or even replacing them, much like in scientific experiments. As we have seen above, established forensic frameworks are grafted onto one another, even with experimental tools such as VR glasses.

Another aspect to consider and to compare regards the processes at stake during the experimentation. Rheinberger highlights two macro processes, namely magnification/reduction and acceleration/deceleration. He explains that to expose and register new findings, a scientific experimental system needs to adopt methods which can make the phenomenon graspable. For Rheinberger, these procedures can be reduced to time dilation and scaling procedures. As for FA, different media fragments can be mapped onto a 3D model or other models. Here, time can be reversed, replayed, slowed down, or sped up, and both wide and narrow perspectives can help analyse an incident.These procedures naturally emerge when data is mapped into a new data space, especially when the data space takes the form of a 3D model.

Forensic Architecture and natural science procedures cannot be reduced to their technology tools, regardless of how the technological landscape affects them. These disciplines rely on the researcher/investigator's capacity to adapt tools and methods to address new inquiries. Also, Rheinberger does not believe that the realm of experimental systems falls into the definition of techno-sciences, as "*instruments by themselves are not the moving forces of the experimental 'goings-on." It is their embedment in experimental systems that counts.*"[92]. Although today's science is understandable only through the technological landscape it is developed in, we must differentiate between two things. A "*technical product*"[93] (namely a machine that fulfills its purpose), and the epistemic thing which is an open ended object, that " *is not technical in itself, although it grants the "goings-on of technics."* "[94]

90 Rheinberger, Split and Splice,12
91 Ibid.,12
92 Rheinberger, Towards a History of Epistemic Things, 36
93 Ibid.,32
94 Ibid.,32

Fig. 17 by author Bricoleur

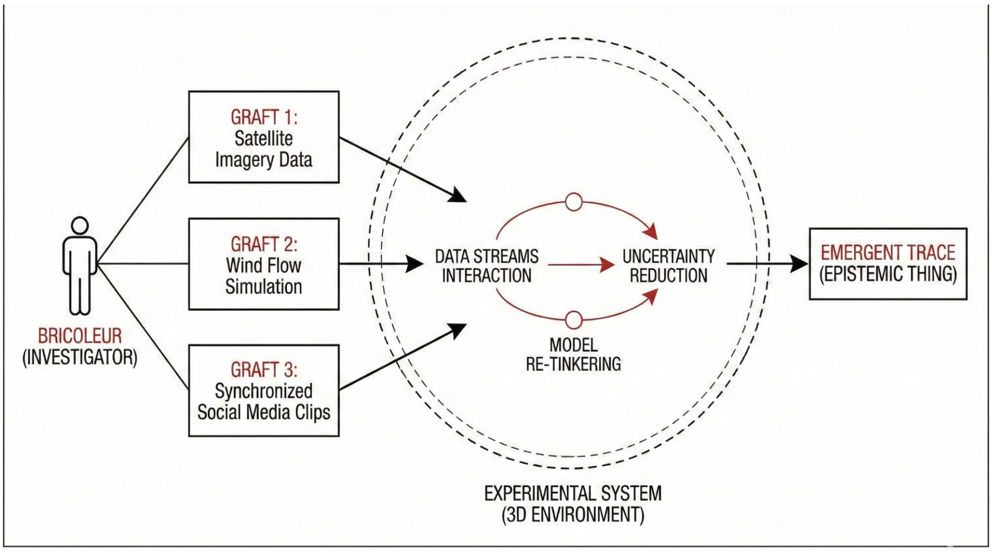


Figure 4. The grafting of multiple investigative methods within an experimental system.

Scientists, then, "*are, first and foremost, bricoleurs (tinkerers), not engineers",* as the nature of the experimental ensemble is non-technical and its emergent properties are not reducible to the sum of its parts. [95] The tinkerer aspect matches with the hybrid and heterogeneous nature of experimental systems; in the same way, open-source intelligence researchers are never operating from a solely technical perspective; instead, they take from many fields and construct hybrid systems. In practice, the OSINT field in which Forensic Architecture operates is a combination of different fields and frameworks. Drawing our conclusions, we can infer that the core gestures of modern experimental sciences also fit the Forensic Architecture practice.

Namely:

- **The OSINT environment depends on the media-technological landscape that subsumes the procedures at stake**. Tools and methodologies are constantly renewed and put under scrutiny.
- **Knowledge generation in OSI involves the use of technological tools to address space-time inaccessibility in the case studies, such as remote sensing, data cross-referencing, data visualisation,** and 3D reconstructions.
- Open-source investigations rely on technological tools, but at their core are hybrid systems that seek to solve complex, open-ended questions rather than fulfil a purpose. Therefore, **scientists/investigators are considerable tinkerers.**

From these definitions, we can add other characteristics that, according to Rheinberger, belong to experimental systems, and we argue, also belong to the open source investigative frameworks:

95 Ibid.,32

- **OSINT and experimental systems should be flexible enough to add other frameworks and technologies.**
- **The system accuracy should not rely on a single valid method of inquiry**; instead, it should rely on a net of elements that together are tackling the object of inquiry from different perspectives, thus, valid knowledge might emerge.

V. Core Terminology: Epistemic Thing and Technical object

As we approach the realm of experimental systems and examine how Forensic Architecture's practice intersects with it, we must consider the more technical elements that constitute the core of Rheinberger´s take on modern experimental science. At the core of these practices are "***epistemic things"*** and "***technical objects"***. The first indicates the thing to be discovered, which can be defined only through the material processes that allow scientists to register it; the latter is the discovered thing, which has become a technical element in further inquiries due to its technical and processual nature. In other words, an ***epistemic thing*** can be known only through the technical apparatus that registers it. Once the epistemic thing has been discovered, the experimental system that made this possible can be considered stable enough to serve as a technical process for other experiments. For instance, In life sciences Epistemic things are "*are material entities or processes—physical structures, chemical reactions, biological functions— that constitute the objects of inquiry*"[96], more generally *"epistemic things embody what one does not yet know"*[97], it is not an abstract concept it has instead a material element and it is knowledgeable only through the processes that makes it emerge, in fact, Rheinenberger says about it" *you cannot do better than explain what the new object is by repeating the list of its constitutive actions. [...]The proof is that if you add an item to the list, you redefine the object, that is, you give it a new shape*."[98]

As an example, Rheinberger brings the enzymatic sequencing of DNA, a process that defines the exact order of nucleotides in a DNA molecule. This process creates a series of DNA fragments of varying lengths with the same terminating molecule, which are then separated by size using gel electrophoresis, allowing the sequence to be read. As enzymatic sequencing was itself an epistemic thing that became known, it shifted in use by becoming a technical object, in other words, the same process that synthesised DNA strands, once the DNA sequences were defined, became a tool adopted in many labs as a part for further research with new objects of inquiry (epistemic things).

96 Rheinberger, Towards a History of Epistemic Things, 28
97 Ibid.,28
98 Ibid.,29

In the case of Mark Duggan, the epistemic thing is the sequence of actions that brought the gun into its final location. It is an epistemic thing in the sense that it is knowable only through the material elements we have to make experience of it (reports, first-hand evidence, testimony) and through the procedures and tools utilised to make further knowledge emerge. The definitive structure of the system depends heavily on the initial question, ***"How did the gun reach the grass?"*** Technical objects could be considered the established techniques used (projection mapping, 3D scanning, etc.). Of course, technical objects in this field are not limited to forensic procedures; they could be any established process.

To conclude, in both experimental systems and FA's investigative system, epistemic things and technical objects are major elements. Firstly, in both systems, objects of inquiry are knowable only through the procedures and technical means that make them graspable[99]. Secondly, in both systems, we see how stabilised knowledge-generating procedures can be used in different scenarios and be grafted into different systems.

VI. A comparison of the processes of knowledge production: Traces, data, and models

At the outset of both the evidential paradigm (Ginzburg) and the experimental system (Rheinberger), we find the trace. In both cases, the trace serves as an index pointing to something not directly visible. Also, both share a momentum in which the trace initially manifests itself as an *"asemic kernel"*[100] that needs to be stabilised and stored in some way: thus the emergence of data (in Rheinberger) and of evidence (in Guinzburg).

Major distinctions in the nature of a trace between forensic settings and experimental systems lie in the technical conditions that generate it and in the role it plays in the production of knowledge. While in Rheinberger the trace is the result of a techno-epistemic event, where the very nature of the trace depends on the material conditions that *renders it visible,* in Ginzburg the trace is something discovered by the acumen of the investigator and it is not necessarily bound to the adoption of technological systems to be grasped; in juridical course what eventually counts are the conjectural reasoning.
Traces in experimental systems are actively fabricated, whereas in classic forensic contexts they are originally discovered; they most likely need to be found and recognised by the investigator and cannot be created in a complex experimental setting. Even though experimental scientific methods are used in modern forensics, they are never used at the first stage. Traces in forensics cannot be crafted.

99 I.e. The reconstruction of the body position of Mathey do in modern sciences; thus, all the nuances we have from Forensic Architecture come from the rthey do in modern sciences; thus, all the nuances we have from Forensic Architecture come from the reassortment of pre-existing data across alsoto eassortment of pre-existing data across,to „intometadata-richrk Duggan is only possible through 3D software that allows to map the police testimony and other types of evidence altogether.
100 Rheinberger, Split and Splice, 13

Moreover, where Rheinberger's system generates data internally, Ginzburg's investigator constructs coherence across a pre-given array of evidence. OSINT and Forensic Architecture, as well, directly inherit this dependency from the pre-existing indexical evidence, since the epistemic virtues that define mechanical objectivity are at the very core of these practices. In other words, the preexistence of an indexical trace subsumes all investigatory fields with no exception; even when these traces are contested or considered partial and misleading, any investigation must at its core hold on to them.

Again, these investigative systems do not deal with the generation of indexical traces as it happens in modern sciences; thus, all the nuances we have from Forensic Architecture come from the reassortment of pre-existing data into different media/data spaces. Moreover, many digital traces are not "asemic kernels" in Rheinberger's sense; OSINT and FA as well, rarely find themselves working with this type of raw traces, they most likely work with footage material, satellite imagery, which is already a rather elaborated and metadata-rich type of data, which can be easily translated in different data spaces. To stress this one more time, Rheinberger's initial "raw" indexical trace is never created in modern forensic and OSINT, and it is hardly present (as a raw trace) in their investigations. Even though traces have very different meanings, more similarities emerge when we look at how these two systems (Forensic architecture investigative systems and experimental systems) deal with data and models.With data we mean those traces that have been contextualised and made readable and organizable, from raw signs to readable images. By models, we mean the result of mapping certain data into a data space of any kind, so that patterns or new data could emerge.

For Rheinberger, once a trace is stabilised, it becomes data. The data can be translated across media, reconfigured, and assembled into what we usually call a **model**, which could potentially bear epistemic novelty. It may generate new questions, expose patterns, or even become the starting point for another experimental trajectory. In the evidential paradigm, the process is different. The trace may be stabilised and become data, and the data could even be mapped into a data space and used for scientific/forensic procedures, but is eventually presented as evidence among others, which are then connected through conjectural reasoning by an investigator able to make sense of the elements.

As we remarked previously, the Forensic Architecture report presented substantial findings from a complex reconstruction of the incident that mapped all available data into a single medium. Their findings were not presented as the result of the investigators' conjectures but rather as knowledge that emerged from their model-making practice. Again, we believe that, from this angle, Forensic Architecture represents a substantial shift from modern forensics and a significant part of OSINT practices, and that this shift lies at the core of the main technical similarity between FA

investigative systems and Rheinberger´s experimental systems.
If Rheinberger´s theoretical framework of scientific modelling gives us a better understanding of how FA´s investigative systems work, then his analysis of the type of knowledge creation in these models should also hold.

As mentioned above, scientific Models assume that data (traces stabilised and stored), can be translated into different media; thus, we get a model that results in a data aggregation. This process also works with different types of data that can be mapped in the same data space. Moreover, models could also become the raw material of other models.
The result is a bottom-up perspective, in which we get the illusion of seeing the whole phenomenon, which is never the case, as models are the process of negotiation *"that is phenomenon-oriented on the one hand, and historically determined on the other."*[101] If the biggest weakness is *" to forget the illusory moment inherent in them",* on the other hand*, their strength is that they form a flexible scaffold that can react sensibly as a whole if its constituent data are altered, exchanged, or reconfigured. In other words, they form a system in data space."*[102]

Since they are naturally reductive representations, Rheinberger stresses that models are better when compared with other models; if models relate consistently to each other, for instance by showing similar behaviours, then the results are considerably more reliable. This aspect plays a major role in our thesis: while in traditional forensics and OSINT the validity of the conclusion ultimately depends on the strength of the evidence presented, in scientific models the epistemic value comes from the possibility of comparing several models with one another. FA's paradigm shift could lead to major changes in how OSINT could eventually work.

The latest historical events highlighted a major problem in today's investigative landscape: the difficulty of determining which case study has the most reliable thesis when multiple actors are working on the same investigation and with the same evidence but reach different conclusions (The Al-Ahli case). In such a situation, relying solely on the authority of one investigative agency among others is not enough.
Having an analytical lens that helps us distinguish which paradigm is at stake would already help us place these two types of investigative methods: one based on conjectural reasoning and one based on the emergent properties of models.

As we near the end of the chapter, we need to make further considerations to better position our claim and proactively address some problems. Firstly, when talking of models, there is the risk of also including computer simulations. In Rheinberger's theory, these models hold a peculiar place: because they could be totally independent from indexical traces, for

101 Split and Splice, 26
102 Ibid.,26

everything could be directly created inside of the simulation. These practices, which emerged from the need to surpass the limits of lab experimentation,[103] also represent a category of scientific experimental system. However, we should not mistake any computational model for a computer simulation. There are cases in which computational power is needed to speed up certain processes; thus, computation does not add any epistemic value *per se. There are* also cases in which knowledge generation has no connection to external factors and depends entirely on computer simulation.

As we wrote, OSINT practices are closely tied to the evidential paradigm, as they depend on pre-existing traces, and so does Forensic Architecture. Given that it is hard to imagine computer models being built entirely autonomously from external inputs.
Another way to frame it is to consider Forensic Architecture models as the grafting of different methods onto the architectural modelling practice from which their modelling practice originates. Second: for Rheinberger, models are a data space where the registered information can be associated and eventually make emerge some pattern or tendency. There is no premeditation in the sense that the researcher is trying to prove something, even when a model represents cellular behaviour at a molecular level, where a certain level of abstraction and cognitive intervention are needed. The goal is to map the data into the model faithfully. Rheinberger assumes that every data cluster used to build a model is trying to represent "facts" as accurately as possible. This implies that models might differ in quality, but any scientific model's ultimate goal entails an ethical/moral posture that governs transparency and researchers' commitment not to lie.

The same cannot be assumed in forensic investigations (and specifically when dealing with human rights violations), where data could be manipulated in favour of one of the parties, possibly the one with greater power. Hence, some actors could deliberately provide false evidence, undermining a case study's validity at its core, but generally, OSINT is structurally equipped to handle such situations. Additionally, comparing models could potentially make fake information less effective.

To conclude, Models in Rheinberger's conception seem to be the most fitting type of system when discussing Forensic Architecture's works specifically. When we are saying so, we are keeping in mind that models are considered in the most canonical interpretation given by Rheinberger; we are not trying to compare FA models to computer simulations.

103 Here referencing the paper from Peter Galison "Computer Simulations and the Trading Zone," in From Science to Computational Sciences: Studies in the History of Computing and its Influence on Today's Sciences, cured from Gabriele Gramelsberger (Zürich: diaphanes, 2011), 97–130.

4
Case studies: Applying the Framework

This chapter applies the framework defined in the thesis. By using a comparative case study approach, we analyse how contemporary visual and OSINT investigations produce knowledge, shifting the focus from final claims to the methods and epistemic structures behind them.

Fig 18: Photograph taken in 1916 of al-Ahli Hospital, described here as 'the English Hospital in Gaza'. Image source: Library of Congress, www.loc.gov/item/2007675298 (Accessed on 8 October 2024). Not for reproduction.

I. Framing the Problem

"Palestinian officials say an Israeli strike caused the blast, while Israeli military say it was caused by a failed rocket launch by militant group Islamic Jihad." - Bellingcat[104]

On October 17, 2023, an explosion at Al-Ahli Al-Ma'amadani Hospital in Gaza killed hundreds of people. Within hours, competing attributions emerged: Israeli airstrike, misfired Palestinian rocket, interceptor malfunction; each advanced by actors with different institutional positions, among which we find: IDF, Hamas, Palestinian authority, NGOs like Human Rights Watch, investigative journalists, Bellingcat, Forensic Architecture and independent researchers.

104 https://www.bellingcat.com/news/2023/10/18/identifying-possible-crater-from-gaza-hospital-blast/

While Hamas and the IDF accused each other, the other agencies investigating had a very hard time trying to define whose missile caused the massacre. Not only was the area of the incident not directly accessible, but most of the footage was not of much help (visibility was reduced because the incident happened at night), and, even worse, there were no traces of any missile remains. Because of the lack of proof, OSINT agencies were not able to bring any relevant findings; however, two main interpretations emerged:

- The incident was likely caused by a misfired Hamas or Islamic Jihad missile, the main evidence being the low scale of building structural damage, the absence of missile remains and the rather small missile crater.
- The incident was caused by an Israeli missile, with the main evidence being a report from Forensic Architecture based on a model they developed for the case study.

At the time, most OSINT investigators considered the first interpretation the more plausible. Overall, the biggest problem the case highlighted was the impossibility of actually defining who, among the investigative agencies, was to be trusted, and how even to evaluate that. The public discussion degenerated into accusations of partiality, favouring one side over the other, thus implying a lack of objectivity in the investigations.

The Al-Ahli case study is particularly relevant to us because it showcases the two epistemic paradigms we described in the previous chapter. On one side, we have an epistemic paradigm based on evidence, trained experts and conjectural reasoning, on the other one based on modelling and emergent knowledge. In the next pages, our goals will be the following:

- Show how these investigations operate in different epistemic paradigms (they have different knowledge generation processes)
- Understand the limits, advantages and disadvantages of the model-led investigations and conjecture-led ones.

Summary of Investigations:

Between October 17th 2023 and the 19th September 2025, the different Actors have been conducting investigations into this case using different methodologies and approaches to how they investigate and reach their conclusions. A comprehensive comparative overview of all ten investigations, including their data inputs, methodological approaches, epistemic paradigms, and coclusions, is presented below.

There is a substantial length difference in the analysis between model-led and conjecture-led approaches. This difference depends on three main factors:

- Conjecture-led approaches present similar, if not the same, footage and techniques. Therefore we tried to avoid repetition.
- Methodologies applied in model-led approaches are codependent, one step is not understandable without explaining the one prior.
- Most of the conjecture-led investigations do not explain in detail why and how experts reach a certain conclusion, or how their applied techniques function.

Summary

“The IDF has concluded an After Action Review and can confirm that the Islamic Jihad was responsible for the Gaza hospital blast that took place at the Al-Ahli Hospital in Gaza on October 17th, 2023.”

The IDF published its account on October 18, 2023[105], right after the blast, making it the earliest among all investigations. Their claim was predictable: a misfired rocket launched by Palestinian Islamic Jihad struck the hospital car park. The evidence cited was radar data and an intercepted telephone conversation between two militants. In the upcoming weeks, the Earshot team proved that the conversation had two different audio channels (one for each militant) merged via editing software such as Audacity, and that some audio noise was added to blend the channels. The Earshot report, presented by Channel 4, concluded that this intercepted call could not be used as evidence.

The Central Question

Asking what question the IDF’s investigation holds is pointless. The IDF acts as if it is not presenting an investigation but rather facts. This attitude is supported by their political power, alliances, extremely advanced surveillance technology, and world-famous intelligence, among other factors.

The IDF’s spokesperson claimed access to classified sensor data that, if genuine, would in principle resolve the question independently of any visual or acoustic analysis.

Fig 19: IDF Aerial Footage: Before and After

Investigation

IDF presented the following evidence:

- **Military sensor data**: radar tracking of projectile

105 Israel Defense Forces, "Al-Ahli Al-Ma'amadani Hospital: Initial IDF Aftermath Report," October 18, 2023, https://www.idf.il/en/mini-sites/israel-at-war/all-articles/al-ahli-al-maamadani-hospital-initial-idf-aftermath-report-october-18-2023/.

trajectories in the relevant time window

- **Alleged audio intercept**: a recorded conversation claimed to be between Palestinian militants discussing the misfired rocket
- **Radar and launch detection:** IDF presented direct instrumental evidence of a rocket launch from within Gaza and its subsequent trajectory toward the hospital area.
- **Intercepted communications analysis:** offered a second independent line of evidence, in the form of apparent admissions by the parties responsible.

- **Classified intelligence evidence:** A bunch of printed images presented by the spokesperson, satellite images with red circles and lines drawn on them, presenting the evidence, without any context.

Nothing among their evidence was justified, contextualised, or explained. Their claim relies on the assumption that they possess an objective overview of the facts.

Epistemic paradigm

IIDF´s praxis establishes institutional authority over the facts rather than inviting scrutiny of the reasoning behind them. For instance, the radar data and audio intercept were not released in a form that permitted independent verification. For instance, the radar data and audio intercept were not released in a form that permitted independent verification. IDF's account demands acceptance or rejection based solely on institutional trust.
Overall, we cannot be sure whether any investigative methodology was deployed at all.

This is not, in itself, disqualifying: intelligence agencies routinely cannot disclose the full basis for their assessments. According to the criteria of our framework, the IDF does not fall under our two epistemic paradigms. Their evidence can not be considered reliable from an external-objectivity standpoint, as it was later demonstrated that they presented falsified evidence (e.g.: intercepted call). It was also impossible to assess through which methodologies and tools they sustained their evidence and claims. Utilizing our framework, we can conclude that IDF's assertions lack the necessary components to be regarded as trustworthy from both epistemic approaches (conjecture-led and model-led)

Case Study 2: Forensic Architecture

Investigation 1:
https://forensic-architecture.org/investigation/israeli-disinformation-al-ahli-hospital
Investigation 2:
https://forensic-architecture.org/investigation/when-it-stopped-being-a-war
Video Reference:
https://i.vimeocdn.com/video/1836943588-51ef76a0b4c6d-d528d8710ac3eb23060f2583029538dcb3c3c273a7b-8be51e1a-d_1920x1080?&r=pad&region=us

This Video was used in multiple investigations, further mentioned as the "Balcony Video".

Summary

Forensic Architecture published their initial findings on February 26, 2024, more than four months after the blast, and long after the initial media cycle had closed. The investigation was subsequently updated on September 19, 2025, incorporating the situated testimony of Dr Ghassan Abu-Sittah, a British-Palestinian surgeon who was operating inside Al-Ahli Hospital at the moment of the explosion.

FA has additionally documented a broader pattern of Israeli attacks on Gaza's healthcare infrastructure, identifying a four-phase process by which hospitals have been systematically forced out of service since October 2023.[106] This might explain the lack of a proper report on this case and the scattered documentation across their website and X account. Overall, documenting the systematic destruction of Gaza's infrastructures by the IDF holds a significant priority under many aspects.

In the very first weeks after the blast, FA and Earshot entered into a joint collaboration presented on social media. This was quickly deleted, due to a rushed conclusion of the investigation. What remains of it is Earshot´s Doppler curve analysis.

The Central Question

The original case study focused on finding out whether an intercepted Hamas missile caused the explosion. The IDF spokesperson, Daniel Hagari, suggested that the explosion and fire could have been caused by propellant residue inside the missile.
To prove such claims, the investigation was built around the question:

106 Forensic Architecture, *A Cartography of a Genocide: Israel´s conduct in Gaza since 2023*, 2024.
https://forensic-architecture.org/investigation/a-cartography-of-genocide

“Could the Hamas/Islamic Jihad intercepted missile shown in the AL Jazeera footage, be the cause of the incident?”[107]

The update on the incident, which goes by the name: “***W***hen it stopped being a war: the situated testimony of Dr Ghassan Abu-Sittah”[108], does not present a central question. In this case, the situated testimony can be more reasonably framed as a documentary technique. However, the testimony of Dr Abu-Sittah carries evidentiary weight and presents the opinion of a trained expert.

The Doctor observes that the type of wounds caused by the explosion is consistent with a fragmentation bomb, a type of weapon not possessed by Hamas or Islamic Jihad. This testimony strengthens FA’s thesis that the incident was not caused by a misfired Hamas missile. Overall, the case aims to sensitise public opinion on the brutality of Israel’s criminal behaviour. The testimony ends with Dr Abu-Sittah saying, “Here is where it stopped being a war and became a genocide”.

Investigation

Israeli Disinformation: Al-Ahli Hospital, 2024:

The main technique deployed by Forensic Architecture was 3D trajectory analysis.

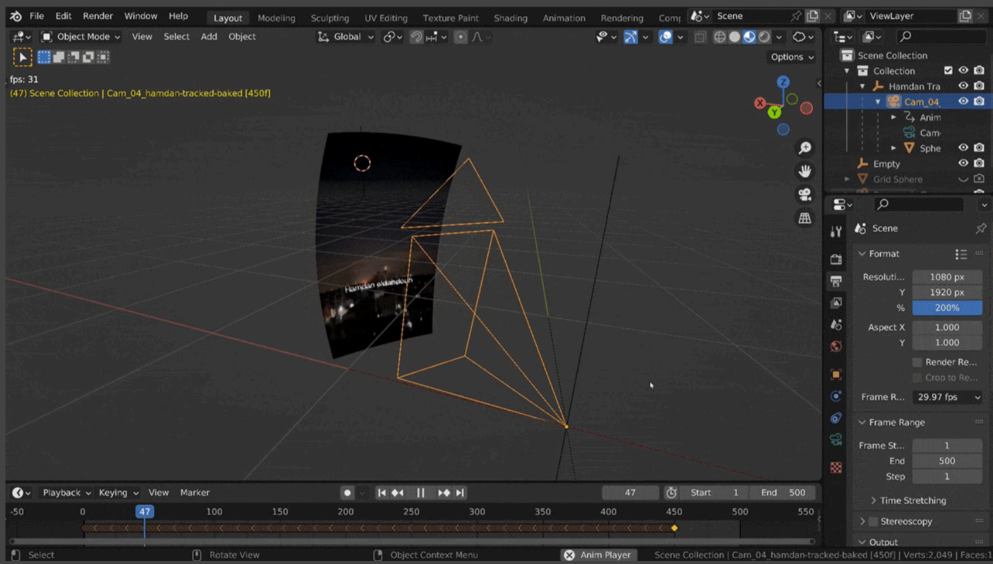
FA uses 2 videos capturing the explosion of an intercepted missile in mid-air: one shot by an Al Jazeera journalist in Gaza (Known as the balcony video), the other from a stationary camera in Tel Aviv.

Fig.20: Camera movement tracking using static object in the footage, made in Blender

107 Forensic Architecture, *Israeli Disinformation: Al-Ahli Hospital*, 2024. https://forensic-architecture.org/investigation/israeli-disinformation-al-ahli-hospital

108 Forensic Architecture, *When it stopped being a war: the situated testimony of Dr Ghassan Abu-Sittah*, 2024. https://forensic-architecture.org/investigation/when-it-stopped-being-a-war

Fig 21: 3D Trajectory analysis- matching of camera movement according to the footage movements.



First, FA tracked the motion of the Al Jazeera camera (Fig. 21) by identifying anchor points (Static lights in Gaza's buildings) that could be tracked individually. Using these anchor points, the camera movement was reconstructed in 3D (Fig. 22).
Subsequently, a line was drawn between the camera view and the missile visible in the balcony video (Fig.22). The same technique was applied to the stationary video (Fig.23).

Fig 22: Missle trajectory

The two cameras and their trajectories were geolocated, and the lines from the two videos intersected, indicating the missile's 3D location. In this way, FA was able to estimate that the Hamas missile was intercepted at an altitude of 5/5.7 km, thus meaning "that any fragments from this explosion would have reached the ground 31s later in freefall".[109] The hospital blast occurred only 8 seconds later, meaning

109 Forensic Architecture, Israel disinformation: Al-Ahli Hospital, 2024 https://forensic-architecture.org/investigation/israeli-disinformation-al-ahli-hospital

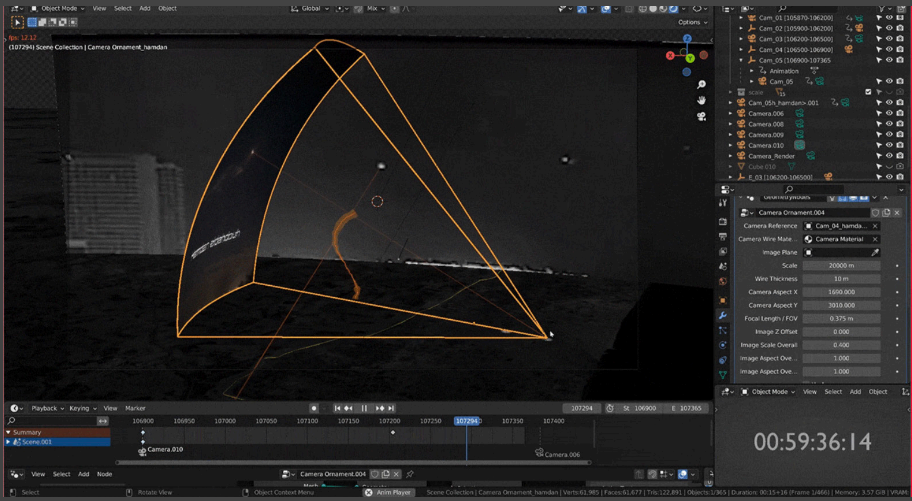


Fig.23: Matching of missile trajectories

that this missile could not have been responsible for the incident.

The same technique was then used to track the trajectory of all the 17 rockets shot by Hamas, which were visible in another video (Fig.20). The aim was to estimate the location and speed at which each missile finished the propellant; the results showed that missiles were at "1km altitude about 4km west of al-Ahli, and travelled a comparable distance at a comparable speed: between 1.8-2km and 500-630m/s"[110], thus suggesting that all rockets burned a similar amount of propellant. Further consultation with an anonymous expert confirmed that the propellant was already over at that distance.

To summarise, thanks to 3D triangulation techniques, Forensic Architecture concluded that none of the Hamas missiles seen in the video could have hit the hospital; moreover, the possibility of the damage being caused by missile propellant was discarded. FA noted that the first finding (Hamas missile distance) matches the New York Times and Washington Post findings, whereas the second finding (on the rocket propellant) contrasts with the conclusions of Human Rights Watch, the Washington Post, the BBC, and the Associated Press.

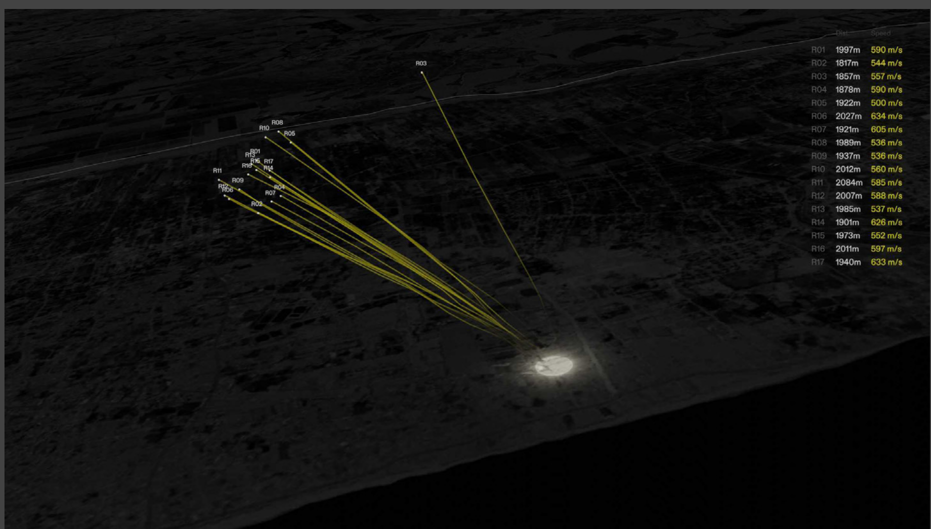


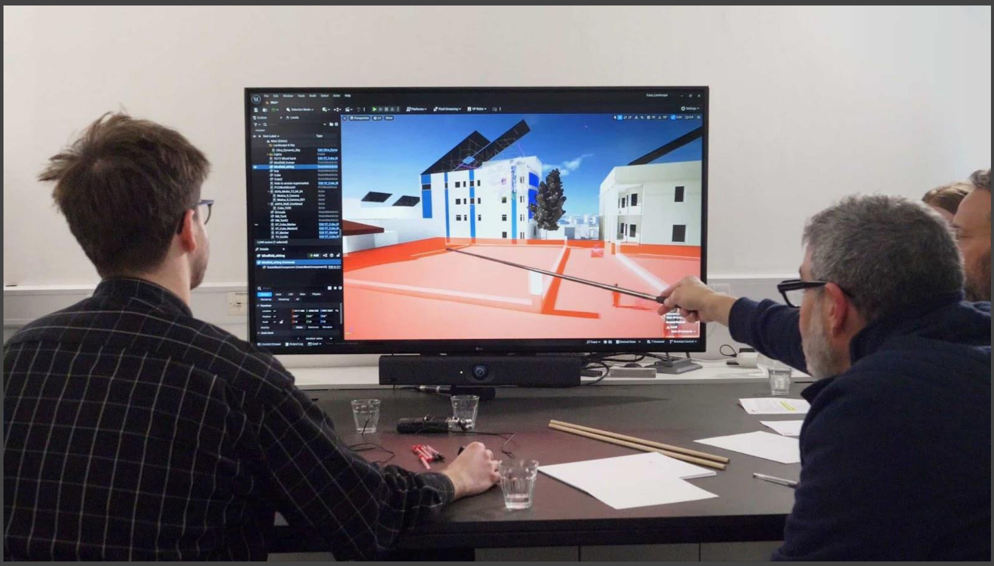
Fig.24: 3D Reconstruction of Hamas missiles' trajectory

110 ibid.

When it stopped being a war: the situated testimony of Dr Ghassan Abu-Sittah, 2024

Fig.25: A still image from a video of the aftermath of the blast recorded by Dr Ghassan Abu-Sittah, placed within FA's 3D reconstruction of al-Ahli Hospital. (Forensic Architecture, 2024)

Fig.26: Dr Abu-Sittah recounts his experience of the night of the blast while a researcher navigates the 3D reconstruction of the al-Ahli Hospital compound. (Forensic Architecture, 2024)



For this case study, 3D tools were used in support of the situated testimony of Dr. Abu-Sittah.

Summary

Dr Abu-Sittah is describing the architecture of the site's buildings and his movements on the night of the incident (Fig. 25). Meanwhile, an FA researcher, following his testimony, is placing walls and objects in the 3D scene (recreated in Unreal Engine[111]). The camera inside Unreal Engine follows the physician's movements. We note, from the smooth interpolations of the camera movements, that part of these videos has been rendered later. The testimony is enriched with the photo-matching of footage into the 3D model, a technique often used in open-source investigations.

Eventually, the doctor will give his professional opinion on the nature of the wounds, assessing that they were consistent with a fragmentation missile: the cuts were sharp, clean and without dirt. Nothing in the Hamas arsenal could have caused these kinds of wounds.

Epistemic paradigm

In the first investigation, we again see how the evidence is mapped into Blender so that something new could emerge. These new findings concern the distance of the Hamas missile when intercepted (1km high and 5km away from the hospital), and that all Hamas missiles had burned all of the propellant.

The processes at stake in this case study are compatible with a model-led investigation: there is the translation of data into the same medium (Blender) and the emergence of new findings. We should also note that the last finding on the missile propellant is reinforced by the opinion of a trained expert.

For what concerns the second case study is concerned, it cannot be considered a model in our framing. Here, FA´s methodologies are not used to create models in Rheinberger's sense; instead, the focus is on Dr Abu-Sittah's testimony and its emotional value. However, a new piece of evidence emerges from Dr Abu-Sittah´s expert medical opinion. This new evidence, coming from a figure who is both a witness and a trained expert, is simply presented as an additional piece of the puzzle; therefore, it is consistent with an epistemic paradigm based on conjectural reasoning.

111 Famous Game Engine

Case study 3: Earshot

Summary

As previously stated, Earshot and Forensic Architecture initially collaborated on a case study. Nowadays, we can only find their investigation through indirect sources (Channel 4) or from earshot's social media pages.
From their X (former Twitter), Earshot presented an investigation on the Al-Ahli Hospital; their attempt was to disprove IDF´s thesis that the missile responsible for the incident came from the southwest, fired by the Islamic Jihad. Their proof consisted of an analysis of the missile's Doppler curve from 2 different video recordings: one taken 150m south-east of the hospital, the other 1500m north-west.

The Central Question

The investigation was built around the question: From which direction was the missile shot?

Investigation

The Doppler effect curve is considered a reliable indicator of an emitter's trajectory relative to the receiver.

Depending on the position of the receiver, the curve would look differently: a rise on the curve indicates that the emitter is accelerating towards the receiver, whereas a fall indicates that the emitter is moving away from the receiver.
The Doppler effect in the balcony video (150m south-east) shows that the missile first moved closer to the receiver and then moved away from it. (Fig.27) A less pronounced Doppler curve in the other video indicates that the missile was accelerating toward the receiver until impact. (Fig.28)
According to the IDF's press release, the missile was fired from the southwest, implying that the Doppler curve in the balcony video should steadily increase, unlike what the spectrogram shows. The possible remaining trajectories, according to the Doppler signature, are north-east, east or south-east. (Fig.29) The results from the second video further confirmed Earshot's finding.

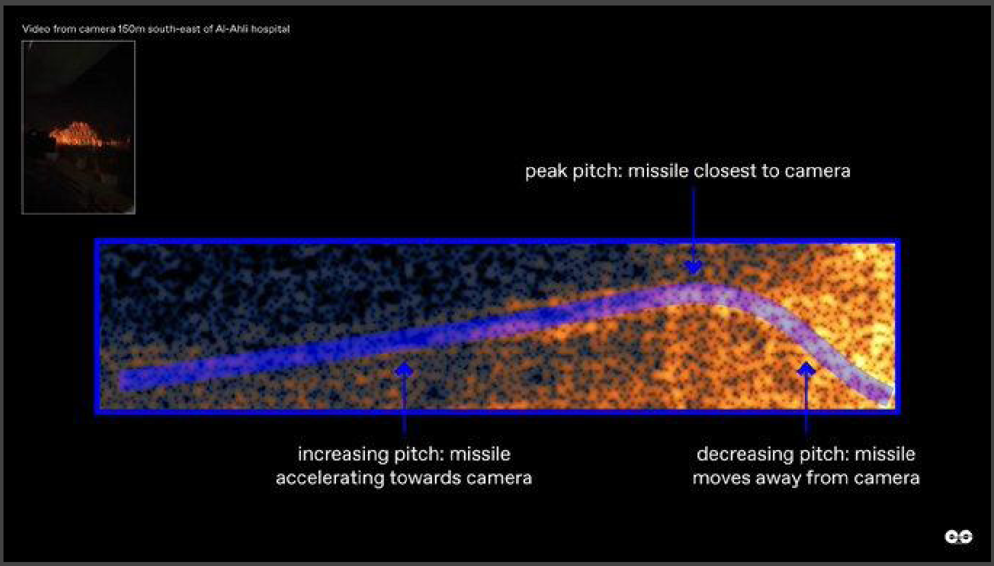


Fig.27: Doppler curve balcony video shot 150m south-east of the hospital

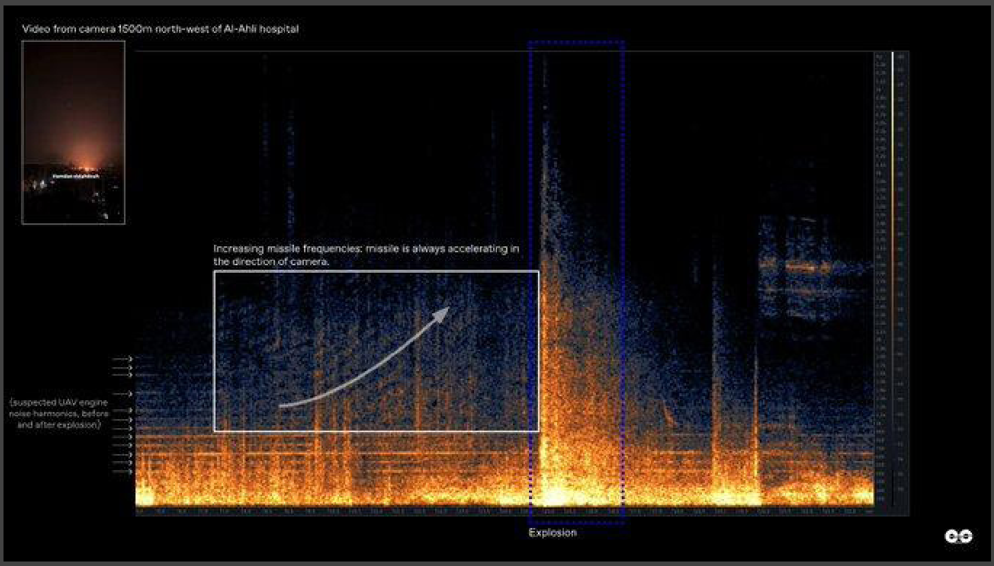


Fig.28: Doppler Curve of video shot 1500m north-west of the hospital

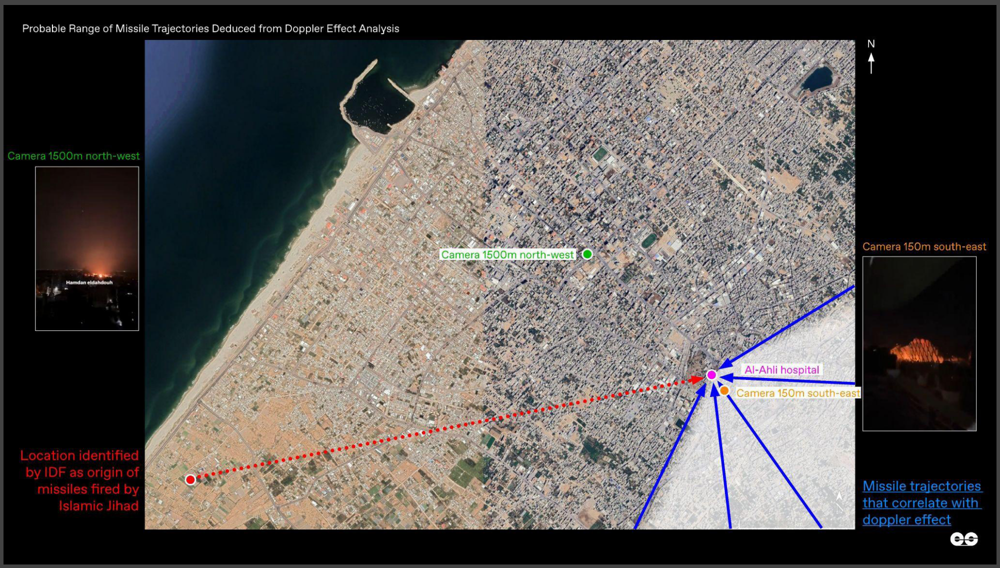


Fig.29: Doppler Curve of video shot 1500m north-west of the hospital

Epistemic paradigm

Earshot´s system is compatible with an epistemic paradigm based on modelling and emergent knowledge. The data (in this case, the audio track of the video footage) is translated into a different data space (Cartesian plane measuring time in seconds and frequency in Hz). From this translation, a Doppler curve emerges, and its behaviour can be compared with the missile's supposed trajectory or with other simulated Doppler signatures.

The methodology deployed by Earshot is well-established and shows no specific adjustments for this case study; in this sense, they applied a very stable technique (technical object).

While data is emergent, we do not observe changes in the techniques deployed, in contrast to what usually occurs in experimental systems.

Additional notes

Earshot findings were criticised by HRW, who addressed the model as too reductive.

Criticism of their method also came from Michael Kobs, an independent investigator. Kobs noted that their model could only work if the emitter travelled on the same plane as the receiver (i.e. an ambulance car travelling on a flat road will be in the same plane as the receiver)

Case Study 4: Human Rights Watch

Summary

HRW published their findings on November 26, 2023[112], positioning themselves as a corrective to both the IDF's confident attribution and the initial chaos of breaking news coverage. They argued that the evidence eventually pointed towards a misfired Palestinian rocket.

The Central Question

What does the physical damage at the site tell us about the munition that caused it?

Investigation

- **Ballistic expert opinion**
- **Video geolocation** was used primarily to establish the timing of the explosion relative to other recorded events and to orient the visual evidence spatially within the site.
- **Satellite imagery analysis** enabled HRW to establish the crater's physical parameters with reasonable precision by comparing them against known impact signatures for different munition types. According to their ballistic experts, none of Israel's missiles was compatible with the crater of the explosion, the absence of any missile remains suggested that Hamas might have tried to hide them.

Epistemic paradigm

HRW operates within a well-defined institutional framework. Its reports are addressed simultaneously to legal audiences, governments, and the general public. This entails that HRW should rely on facts; they cannot speculate nor rush to conclusions if evidence is not enough to formulate a proper hypothesis.
However, for Al-Ahli, they took a stance on the IDF's side.
This case study did not present differences in the methodology compared to their other investigations. Overall, their investigative practice, which heavily relies on evidence, verification, fact-checking and experts' opinion, is assimilable to a knowledge paradigm based on conjectural reasoning.

112 *Human Rights Watch*. 2024. "Gaza: Findings on October 17 al-Ahli Hospital Explosion," May 6, 2024. https://www.hrw.org/news/2023/11/26/gaza-findings-october-17-al-ahli-hospital-explosion.

Case Study 5: The New York Times

Summary

The NYT Visual Investigations team published on October 25, 2023[113], making them one of the earliest organisations to run an open source investigation. They concluded that the evidence was more consistent with a misfired rocket from Gaza than with an Israeli airstrike.

The Central Question

Does the visual and physical evidence better match the profile of an Israeli airstrike or a misfired rocket?

Investigation

- **Satellite imagery analysis** was used to establish the state of the hospital grounds before and after the blast, providing a baseline against which claims about the scale and nature of the damage could be evaluated.
- **Video geolocation** enabled the team to situate the launch footage spatially relative to the hospital and assess whether the trajectory was geometrically plausible.
- **Blast and damage comparison**

Epistemic paradigm

The NYT's conclusions carry enormous public weight by virtue of the institution's credibility, yet the investigation was rather fast, perhaps too fast to be accountable, being released only a few days after the blast.

Overall, the way they conducted the investigation is not different from what HRW did. We have verification protocols, analysis of the crater with the help of ballistic experts, footage analysts and geolocation techniques. Again, the knowledge paradigm is based on conjectural reasoning: Evidence is presented, methods are used, and eventually everything is organised into a single chain of thought.

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113 "Gaza-hospital-israel-hamas-video." 2023. New York Times. October 25, 2023. Accessed October 25, 2023. https://www.nytimes.com/2023/10/24/world/middleeast/gaza-hospital-israel-hamas-video.html

Case Study 6: BBC Verify

Summary

BBC Verify published on October 26, 2023[114]. This was two days after the blast, with a subsequent update on October 26 incorporating further analysis. The investigation was produced by a team that included on-the-ground reporting by BBC journalist Rushdi Abualouf and contributions from the BBC's visual journalism unit. BBC Verify has found insufficient evidence to reach a conclusion.

The Central Question

Can the available audio-visual evidence establish when and from where the projectile was launched, with sufficient precision to support an attribution?

Investigation

- Video geolocation of footage.
- Sound and timing comparison of footage
- Ballistic expert opinion on the crater

Epistemic paradigm

This case study also relies on methodologies, trained experts and evidence, connected through conjectural reasoning.

114 Palumbo, Paul Brown Joshua Cheetham, Sean Seddon and Daniele. 2023. "Gaza Hospital: What Video, Pictures and Other Evidence Tell Us About Al-Ahli Hospital Blast." October 19, 2023. https://www.bbc.com/news/world-middle-east-67144061

Case Study 7: Bellingcat

Summary

Bellingcat published on October 18, 2023[115]. Among all publishers, Bellingcat is the only group that did not support any of the hypotheses. They firmly maintained that there was insufficient evidence to reach a conclusion.

The Central Question

What does the crater tell us about what caused it, and what range of munitions is consistent with its physical characteristics?

Investigation

- Satellite imagery analysis
- Video geolocation was used to confirm the crater's location and relate it to the broader site geography.
- footage analysis
- 3D scan of the crater made from footage

Epistemic paradigm

Overall, the Bellingcat case study also falls into the category of a conjecture-led investigation.

115 Team, Bellingcat Investigation. 2026. "Identifying Possible Crater From Gaza Hospital Blast - Bellingcat." Bellingcat. February 11, 2026. https://www.bellingcat.com/news/2023/10/18/identifying-possible-crater-from-gaza-hospital-blast/.

Case Study 8: The Washington Post Visual Forensics

Summary

The Washington Post Visual Forensics team published their investigation on October 26, 2023, nine days after the blast. Conducted by specialists (Evan Hill, Meg Kelly, and Imogen Piper), the investigation drew on geolocation analysis, audio forensics, and damage assessment. WaPo reached the same conclusion as Forensic Architecture regarding the Al Jazeera balcony video, identifying the projectile as an Iron Dome interceptor missile launched from inside Israel rather than a misfired Hamas rocket. This diverged on the overall attribution, concluding that the hospital explosion was more consistent with a misfired Palestinian rocket than an Israeli airstrike. It is important to mention that the expert analysis wasn't documented.

The Central Question

Does the visual and physical evidence better match the profile of a misfired Palestinian rocket or an Israeli airstrike, and what does the Al Jazeera balcony video actually show?

Investigation

- **Triangulation**
- A video from Netivot and Bat Yam, two videos in Israel, approximately forty miles apart to triangulate the launch origin to a point nearly three miles southwest of al-Ahli Hospital. Rocket travel time estimates from two independent experts placed the hospital arrival between 26 and 45 seconds after launch, consistent with the 44-second window between the barrage and the explosion.
- For the Al Jazeera video, WaPo used the same footage alongside a Gaza City rooftop video to triangulate the projectile's origin to a point approximately 2.5 miles inside Israel, near a site whose satellite imagery was consistent with an Iron Dome battery. Five experts independently identified the projectile as a ***Tamir interceptor missile*** based on its non-ballistic, snaking trajectory. The seven-second interval between the midair explosion and the hospital explosion was sufficient to rule out any physical connection between the two events, as debris would have needed to travel at supersonic speeds exceeding 500 meters per second to reach the hospital in time.
- **For the blast site:**

- **Crater Analysis:** More than six experts independently reviewed imagery of the scene. The crater measured nearly three feet wide, with a spray-like fragmentation pattern extending in one direction. Buildings surrounding the courtyard sustained only light damage, while more than a dozen vehicles in the parking lot were incinerated, one flipped onto its roof. No munition remnants were visible in any imagery. Expert consensus held that the damage was inconsistent with an Israeli air-strike, which would have caused significantly greater structural destruction. Marc Garlasco, a former Defence Department battle damage assessment analyst, noted the large amount of localised thermal damage as more consistent with a munition carrying substantial propellant falling prematurely and igniting than with a directed strike.
- Audio analysis of the balcony video, conducted by Rob Maher of Montana State University, identified an increasing frequency in the incoming projectile's acoustic signature, indicating acceleration consistent with vertical descent under gravity, more consistent with a malfunctioning rocket than a horizontally moving projectile. This was further used in other investigations.

Epistemic paradigm

WaPo's investigation operates entirely within a conjecture-led epistemic paradigm. Evidence is collected, geolocated, and submitted to expert opinion, then connected through a chain of abductive reasoning toward a probable conclusion.

Case Study 9: Michael Kobs

Summary

Michael Kobs, an independent researcher, published a technical report attempting to determine the missile´s trajectory through computational analysis of the Doppler curve recorded in the “balcony video”[116]. He concluded that the ammunition came from the direction of an Israeli fighter jet; he achieved this conclusion by ruling out

The Central Question

Given that the Doppler curve of the balcony video is independent of the frequency of the emitter, the question asked is: **from which angle was the missile shot?**

Investigation

Knowing that the Doppler curve shape is independent from the emitter frequency allows Kobs to simplify the amount of variables at stake in his computational model.

As he writes:**”**the shape of the curve is independent of the emitter frequency. This means that if the Doppler effect rises to 2kHz at an emitter frequency of 1kHz and then falls to 500Hz, the same relationships apply to all frequencies. A 2kHz emitter frequency would therefore oscillate up to 4kHz and then drop to 1kHz in exactly the same time under the same conditions with regard to the trajectory. On a logarithmic scale, the curve therefore always looks the same.[117]”(Fig.30)

Departing from this assumption, it is possible to select one point of the curve (i.e. the peak) and adjust the frequency accordingly, in order to have the trajectory coincide perfectly.[118]

117 Michael Kobs, "Reverse Engineering the Projectile Trajectory from the Doppler Effect: Al Ahli Baptist Hospital Attack," Scribd, 2023, p.5, https://www.scribd.com/document/721217432/Reverse-Engineering-Al-Ahli-Hospital-Attack-En.

118 ibid, 5.

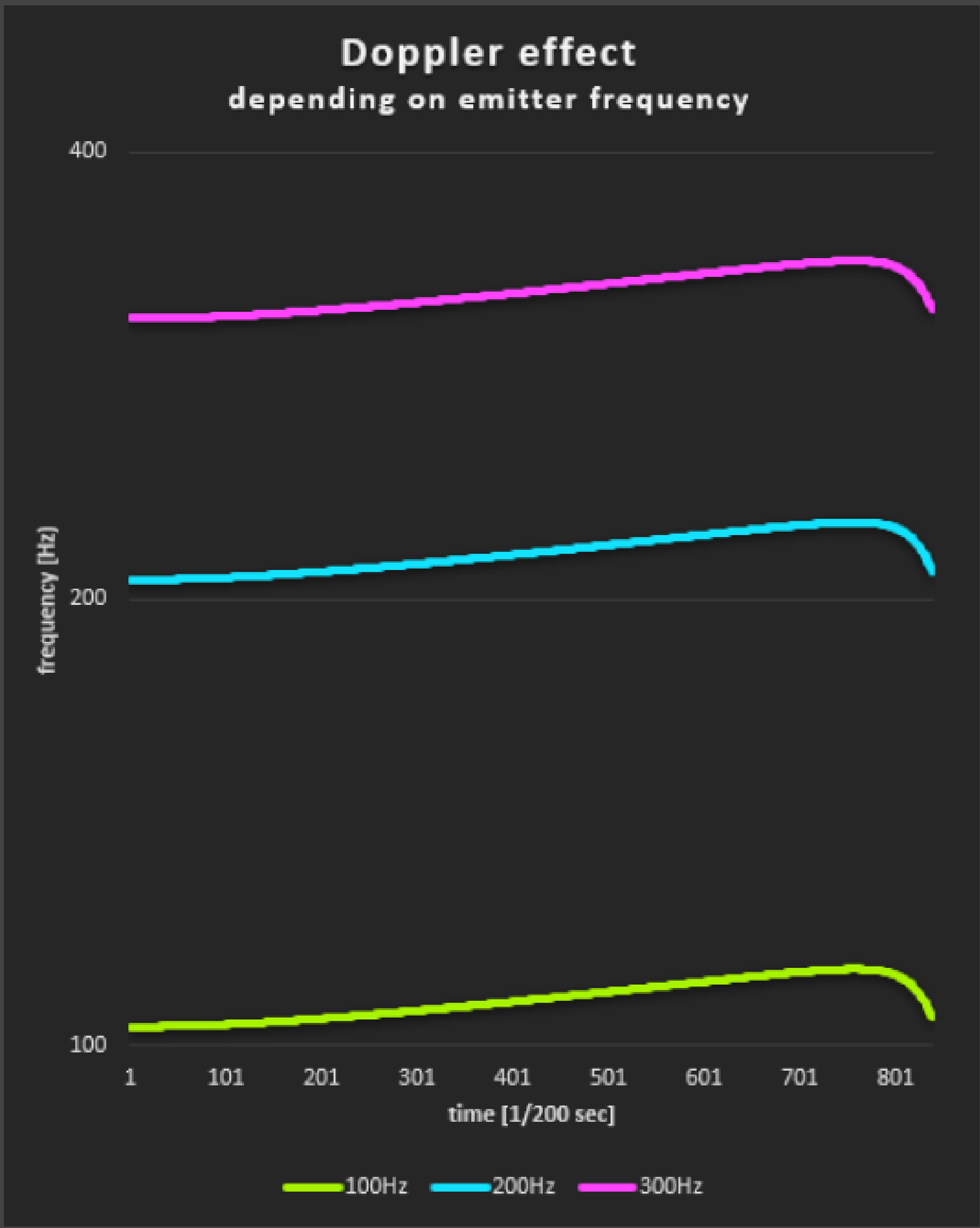
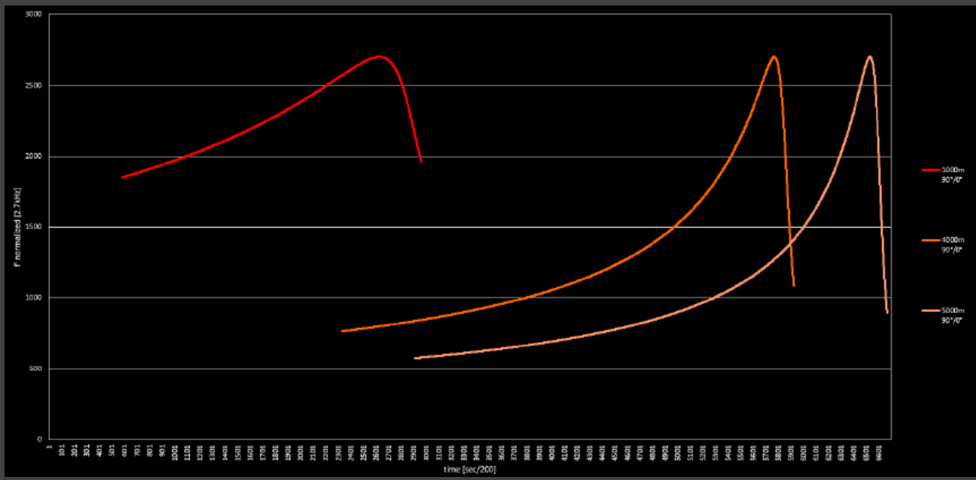


Fig.30: Image showing how the Doppler curve looks the same regardless of the emitter frequency.

Given that the peak of the curve is at 2600 Hz, Kobs notes the Doppler signature would change depending on impact angle, azimuth (the angle of rotation of the impact angle around the impact point, lying on the axis impact point-observer), initial altitude of the missile and its speed.
The picture below shows how, by assuming a perfectly vertical drop (perpendicular to the ground), the altitude influences the Doppler curves' scaling[119]. (Fig.30)

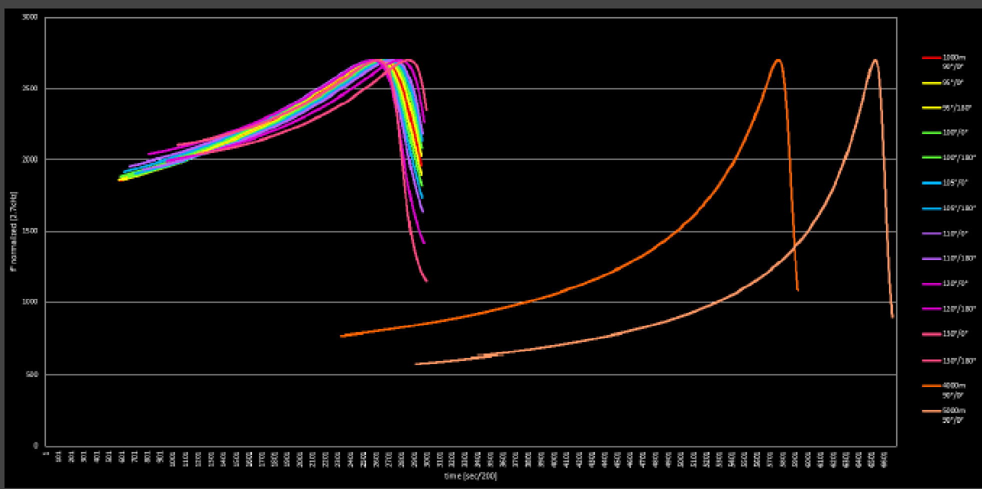
119 ibid.,6

Fig.31 behaviour of the curve, given the same frequency peak, a perfect perpendicular angle from the ground and different initial altitudes



Kobs also delineates the model´s limits: if the emitter is flying perpendicularly to the receiver (which means perfectly on the right or on the left of the line between the observer and the crater), then it becomes extremely difficult to define the impact angle, because the Doppler curves would have extremely similar behaviours (Fig.31).

Fig.32 Comparison of Doppler curves of missiles with different impact angles but flying perpendicularly to the observer



Another limitation of the model is how similar the Doppler curves behave between the peak and the lowest point. If, for instance, the impact angle is at 90 degrees, the information given by the curve between peak and low would be insufficient. In this scenario, the only way to define the azimuth is by adding the rise of the curve. More generally, the rise behaviour can indicate how long the missile has been in flight as well as where it was flying relative to the observer.

In summary, Kobs model cannot define the impact angle for missiles travelling perpendicular to the observer (azimuth 90°); the addition of the rise of the curve is fundamental to defining the azimuth when the missile's impact angle is close to being perpendicular to the ground.

According to these considerations, Kobs delineates the 3 decisive points of the Doppler curve for the model (Fig.33):

- Time and frequency of the impact
- Time and frequency of the maximum
- Time and frequency of an early point before the maximum

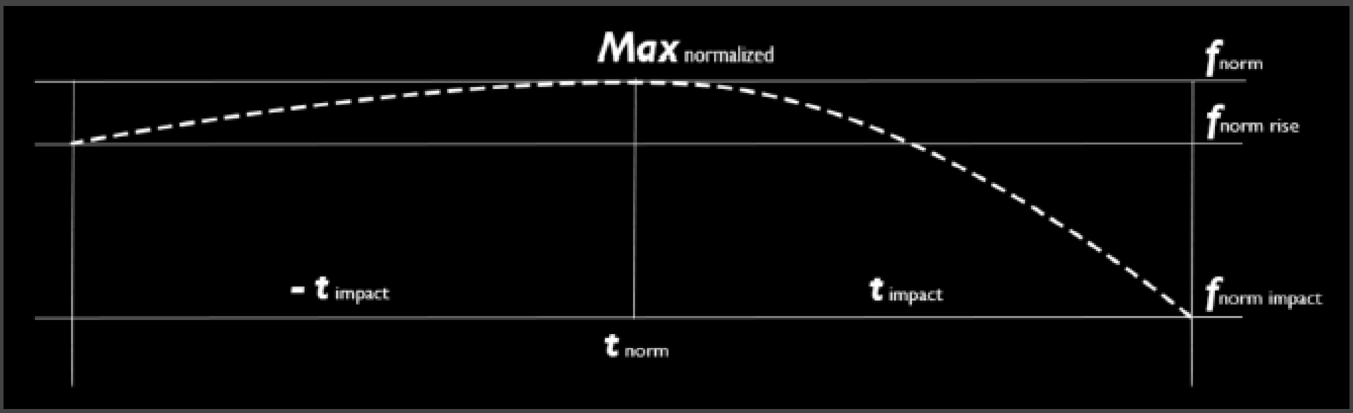


Fig.33: Subdivision of the curve and definition of the important variables to consider for the investigation

The following steps have Kobs implementing data derived from the available evidence on the case:

- The distance of the observer from the Hospital (+-160 m)
- Forensic Architecture and Earshots impact angles (90°,103°green)(Fig.32)
- IDF and HRW impact angle (85°, blue)(Fig.32)
- Angle derived from fighter jet position at the time (34° pink)[120] (Fig.33)

120 The angles are calculated relatively to the camera-crater axis, where 0° is the the angle between camera and impact point to

Fig. 34: trajectories azimuth according to the main investigations (FA,Earshot,HRW,IDF,) plus the fighter jet

Additional data, such as the altitudes of the Israeli plane and the interceptor, are determined using 3D triangulation techniques. The model shows that the previously estimated altitude of 6000m-7000m was wrong; therefore, the height of the objects is adjusted at roughly 4000m above the ground. Kobs does not provide an extensive explanation of how he applied the triangulation technique, nor does he specify which source he is disproving. It seems that Kobs assumes the triangulation technique as being a reliable process, something akin to a technical object.

Once the simulations are run, the computer model discards all hypotheses except the fighter-jet hypothesis, whose Doppler curve perfectly matches that of the balcony video. Quite interesting for us, is the surprise that derives from the finding, which indicates that the result emerged from the simulation, rather than being deduced by Kobs himself:

" This result unexpectedly confirms exactly the flight direction of 35° relative to the line of sight as confirmed by the fragmentation traces around the crater as well as the impact angle, which I had given as about 110°[121]"

Lastly, one more model is presented: a 3D model of the Al-Ahli hospital (reconstructed in Blender) using the photo-matching technique. The trajectory of the missile (obtained from the computer simulation) is drawn with a purple

121 ibid.36.

spline, and the likely shrapnel scattering is derived from the angle of impact. The model's shrapnel scattering is then confirmed by the photo-matching technique. (Fig. 34)

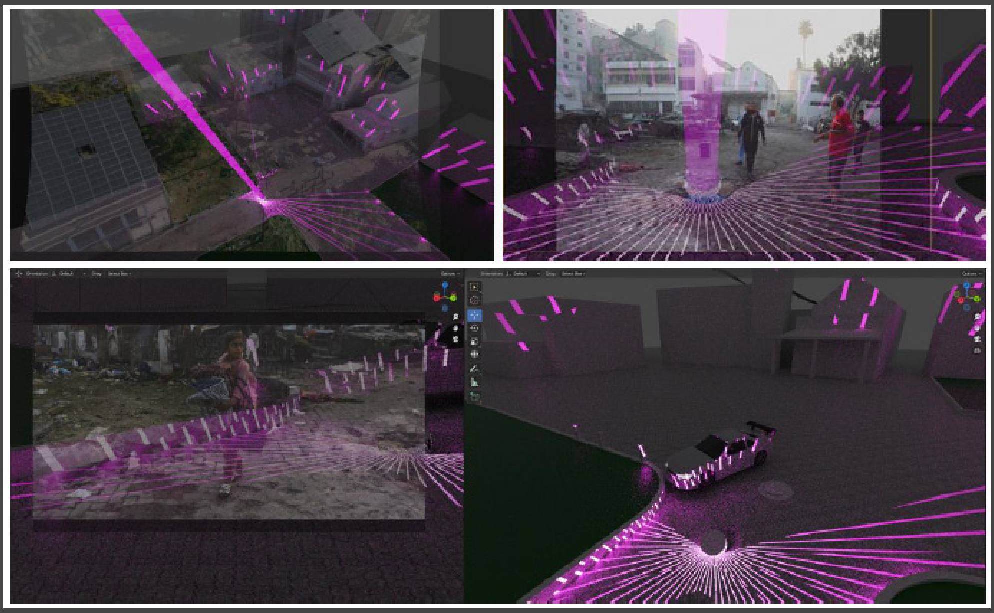


Fig.35: 3D reconstruction of Al-Ahli hospital, photomaching techniques, mapping of the missile trajectory and projection of the shrapnel scattering.

Epistemic paradigm

Kobs knowledge generation process is extremely compatible with Rheinberger´s models. We see the adoption of 3 different models: Doppler curve simulation model, 3D triangulation from footage, and 3D reconstruction of the missile trajectory + photo-matching. The result of each model is then mapped to the next, and overall, the surprise of Kobs highlights how the finding emerged rather than being derived from conjecture.

Kobs investigative system is compatible with the notion of the epistemic thing, which, in this case study, is represented by the missile's trajectory, azimuth, speed, and impact angle. These unknown objects manifest only through the computational techniques Kobs adopts. Triangulation techniques and photo matching are used as stabilised techniques/processes, without being overly explained, suggesting that they could be considered as technical objects.

To conclude, the limits and risks of this system are very clear and follow Rheinberger´s definition: models are limited and should not be mistaken for a global representation of a phenomenon. In this specific case, the spectrogram is acknowledged to be subject to noise, the air resistance model is explicitly simplified, the terminal velocity of the specific munition is unknown (but not really important for the evaluation), and the model does not work for specific limit cases (i.e. the Missile traveling perpendicular to the observer).

Case Study 10: Maher Arar Blog

Summary

Maher Arar Blog is an independent group of open-source investigators. We cannot assure whether they are a reliable source. The number of team members is unclear, and we cannot verify their qualifications. We stumbled across their article through Bellingcat´s Discord channel on Gaza; someone in the group suggested it as the most complete among the articles written about the Al-Ahli incident.

Maher Arar Blog has published two articles on the topic. For our scope, we will focus on the second, but we will briefly report on the content of both.

The first one critiques the low investigative effort of AP, NYT, Washington Post, Wall Street Journal, and Le Monde, which unanimously sided with the IDF claim. This thesis is sustained by the introduction of new elements to the investigation:

- 1) The inconsistency between Al-Ahli damage and the destructive power of Hamas missiles
- 2) Another type of missile, the GBU-39, present in the IDF's arsenal, could have caused the rather small crater of the explosion. These missiles have been sold to Israel by the US since 2021; there are videos showing the IDF using them, and their destructive power is consistent with the Al-Ahli damage.
- 3) GBU-39 missiles are known for their carbon-made shell, which totally disintegrates on impact. This could account for no remnants of the bomb being found on-site.
- 4) Hamas missiles' sound signature differs from the one heard in the balcony video.
- 5)The video footage shows two aircraft, which could have fired the missile; this investigative direction was not considered in the other investigations.

Through calculations, they also reach the following conclusions:

Hamas missiles could never have hit the hospital in the right time frame. Through the footage analysis, they conclude that the planes were travelling at 175 m/s and were flying at an altitude of roughly 4500m. From that altitude, a missile fired from one of the aircraft would have reached the hospital within a compatible time frame.

The Second investigation follows and expands on Michael Kobs one. The computational model developed by Kobs is expanded by adding new dimensions: the drag coefficient (Air resistance of the missile), the missile's mass, and its radius. The system is built on the assumption that the missile was fired from an aeroplane, as suggested in their previous article (as well as in Kobs´s one). The model results show that the Doppler curve was compatible with that of a GBU-39 missile.

The Central Question

The question around which the second investigation revolves is heavily dependent on Kobs model: **by using computer simulations, is it possible to reproduce the Doppler shift curve shown in the spectrogram to within a certain acceptable accuracy?** Specifically, the system is designed to determine whether the Doppler curve is compatible with 2 missile types: GBU-39 (and its GBU-39A/B variants) and GBU-53 (aka StormBreaker), including its GBU-53B variant.

Investigation

The case study is part of an article that presents a mix of collected evidence, conjectures, and a model-led investigation. While we can compare Kobs´s paper to a scientific paper, where the scope is narrow, the methodology extensively explained, and the model viable for inspection; the Maher Arar blog article, only presents the results of its system, moreover these findings are mixed with arbitrary assumptions that don't add anything to the investigation, such as the claim that they believe the missiles were remote piloted.

They claim to be very competent in their investigation; however, compared to others, their article is rather messy and mixes claims, critiques and investigations without a hierarchy. On the other hand, they were also the only among all the investigative agencies to point out that no ballistic expert talked about GBU-39 missiles, owned by Israel.

Starting from Kobs´s model, they try to add other dimensions to it, namely:

- Drag coefficient
- Missile´s mass
- Missile´s Radius

Some variables from their previous investigation (height at the initial time, angle of the approaching aircraft) are here considered constants.

What they need to find is not the possible trajectory anymore, but if the missile type is compatible with the Doppler signature of the balcony video.

Kobs treated the emitter frequency as nearly constant, while the trajectory was the variable to be defined. This investigation inverts the parts: given the trajectory as constant, the variable is the missile frequency, which must match the Doppler signature.

Following Kobs, the model's limits are defined: the type of problem they want to solve is called a "multi variable minimization problem. There is no analytical solution and as such these types of problems are typically solved by numerical methods in which only one variable is allowed to change at a time."[122]

Error margins are defined; if the interval for the fall/rise time of the Doppler curve differs by 10ms from that of the balcony video, the results are discarded.

As reported above, two missiles are selected for the simulation: GBU-39 and GBU-53. The first is notoriously in IDF possession, while the latter is not known for the latter.

The results showed that the GBU-39 missile is compatible with the Doppler curve of the balcony video. Further simulations are done to try out whether a Hamas missile is compatible with their parameters.

Four scenarios are simulated:

- The Hamas missile was hit, but did not change trajectory
- The missile was hit and changed its trajectory
- The missile misfired and suddenly changed trajectory
- The missile, after a certain time, starts falling vertically onto the hospital

The aim of the simulation was to define what drag coefficient and speed (among other variables) was needed for the missiles to reach the hospital at the given time, creating that specific Doppler signature. The results were incompatible with the actual drag coefficient of Hamas's missiles (which is pretty high) and with the missile's speed.

The article also shows video footage from CNN, apparently showing the wings of the GBU-39 missile.(Fig.36, Fig.37)

Epistemic paradigm

This case study combines conjecture-led and model-led investigation. While the use of both paradigms is not necessarily incompatible, some of the claims may be unreliable, given that Maher Arar's blog is independent, and we cannot guarantee the quality of their work.

Nonetheless, it is curious how their approach towards Kobs´s model confirms the epistemic paradigm based on models and emergent knowledge. Kob's system is used as a technical object; the object of inquiry changes, so does the overall system. The computer simulation is adapted to define what type of missile hit the hospital.

Maher Arar blog brings evidence that was not considered by other agencies, thus it comes to different conclusions in what seems to be a conjecture-led process; subsequently, these conjectures are reinforced by setting up an investigative system where data emerges from knowledge.

Overall, some findings and evidence are strong; however, these strong parts of the investigation are sometimes overshadowed by some

122 Maher Arar, "Doppler Shift Analysis Leaves No Doubt An IDF Aircraft Is Behind The Al-Ahli Hospital Massacre- Part-II," Maher Arar Blog, April 23, 2024, https://ararmaher.wordpress.com/2024/04/23/Doppler-shift-analysis-leaves-no-doubt-an-idf-aircraft-is-behind-the-al-ahli-hospital-massacre/

arbitrary claims, like that the IDF probably used guided bombs, and by the tone which suggests that they do not need to follow the rules that agencies must follow to maintain credibility.

Fig.36: CNN footage of the crater, marks of the wing made by Maher Arar blog

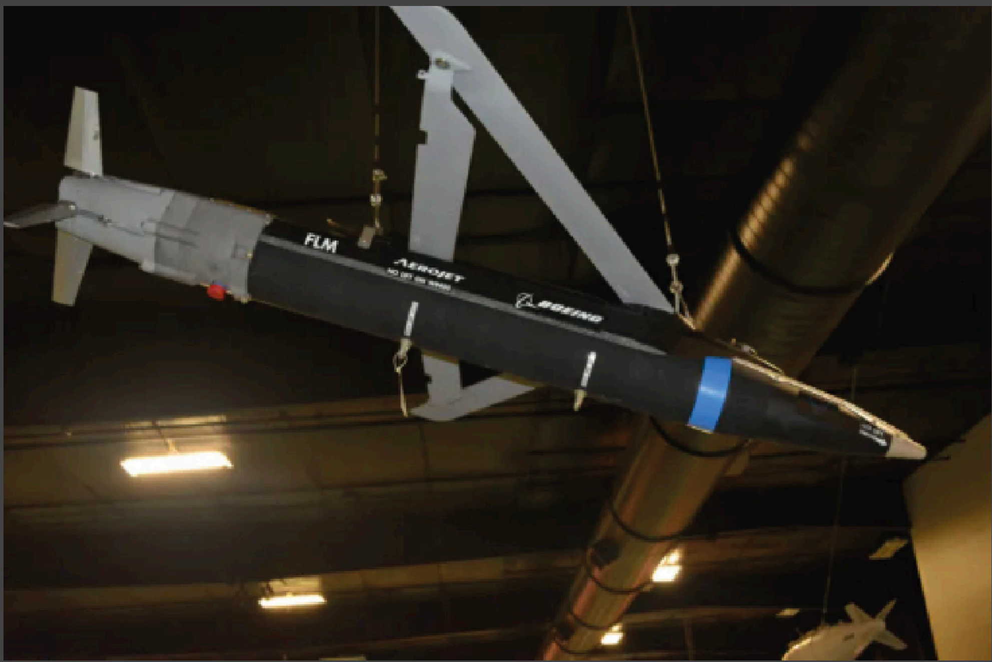


Fig.37: photo of a GBU-39 missile

Combined Web-based Platform for all 10 Cases

Informed by our research, we created a Web-based platform, generalising our method into practice. The web-based platform and repository were implemented as a lightweight, static web-based system developed iteratively alongside the research process. Our aim was not to treat it as a finished product, but rather as a space where we can organise our data, materials, and test visualisations, and draw conclusions from comparisons and connections. It is important to mention that the Web-based platform is a prototype that we have iterated on many times. It is important to emphasise that the platform remains a prototype. It was not developed as a finalised archival or monitoring system, but as an exploratory and iterative research artefact. Its form and functionality in parallel with the investigation, and many of the design decisions emerged directly from ongoing engagement with documents, time-lines, and analytical categories rather than from predefined technical requirements.

Working on the platform during our research enabled us to reflect on information about the Al-Ahly case that we could communicate, structure, and categorise transparently. Transforming findings into a visual interface required moments of deciding which actors and findings belong together or relate to one another, and which materials to foreground or leave peripheral. While the investigation relied primarily on close reading, document analysis, and theoretical interpretation, the platform provides a way to spatialize and visualise the outcomes of our data collection. In this sense, this visual counterpart to our heavily text-based investigation complements the research by offering an alternative mode of access and orientation, particularly regarding the density of information. In developing the interface, inspiration for both structural and visual design choices was drawn from existing platforms that seek to investigate and communicate human rights violations through spatial, temporal, and evidentiary interfaces such as Forensic Architecture.[123]

These examples informed not only questions of visual language but, more importantly, approaches to organising, compressing, and conveying complex bodies of information in a coherent and legible manner. In particular, they demonstrate how heterogeneous materials, such as documents, timelines, numbers, and statistics, can be assembled into interfaces that support investigation rather than mere presentation. While the scale and scope of the present platform differ significantly, this reference provides important orientation.

123 https://forensic-architecture.org/investigation/a-cartography-of-genocide

Overall Architecture

Comparative Analysis: Summary of data

Methodology Comparison Matrix

Organization	Video	Satellite Imagery	Crater/ Damage Analysis	Audio Analysis	3D Modeling	Trajectory Analysis	Triangulation	Computational Modeling
Forensic Architecture	x	x	x	x	x	x	x	x
Human Rights Watch	x	x	x	–	–	–	–	–
New York Times	x	x	x	–	–	–	–	–
Washington Post	x	–	x	–	–	x	x	–
BBC Verify	x	–	x	x	–	–	–	–
IDF	–	–	–	–	–	–	–	–
Bellingcat	x	x	x	–	–	–	–	–
Michael Kobs	x	–	–	x	x	x	x	x
Earshot	x	–	–	x	–	x	–	x
Maher Arar	x	–	–	x	–	x	–	x

Stance Distribution + Epistemic Paradigm

Conclusion	Organizations	Epistemic Paradigm
Palestinian Rocket	HRW, NYT, IDF, WaPo	Conjectural Reasoning
Israeli Munition (Fighter Aircraft)	Forensic Architecture, Earshot, Michael Kobs, Maher Arar	Modeling
Inconclusive / Uncertain	Bellingcat, BBC	Conjectural Reasoning

Publication Timeline

Date	Organization	Actor Type Distribution	Question
Oct 18, 2023	IDF	State Actor	Can military intelligence data establish the origin point and identity of the projectile?
Oct 18, 2023	Bellingcat	OSINT / Research Agency	What does the crater tell us about what caused it, and what range of munitions is consistent with its physical characteristics?
Oct 25, 2023	NYT	Newsroom / Media	Does the visual and physical evidence better match the profile of an Israeli airstrike or a misfired rocket?
Oct 26, 2023	WaPo	Newsroom / Media	Is the IDF's key piece of video evidence what they claim it is?
Oct 26, 2023	BBC	Newsroom / Media	Can the available audio-visual evidence establish when and from where the projectile was launched, with sufficient precision to support an attribution?
Nov 26, 2023	HRW	NGO / Human Rights	What does the physical damage at the site tell us about the munition that caused it?
Feb 26, 2024	Forensic Architecture	OSINT / Research Agency/ Counter Investigation	Could the Hamas/ Islamic Jihad inter-cepted missile shown in the AL Jazeera footage be the cause of the incident?
Oct 20, 2023	Earshot	OSINT / Research Agency	From what direction did the missile arrive?
April 08, 2024	Michael Kobs	Independent Researcher	From what angle was the missile shot?
April 23, 2024	Maher Arar	Independent Researcher	Using computer simula-tions, is it possible to reproduce the Doppler shift curve shown in the spectrogram to within a certain acceptable accuracy?

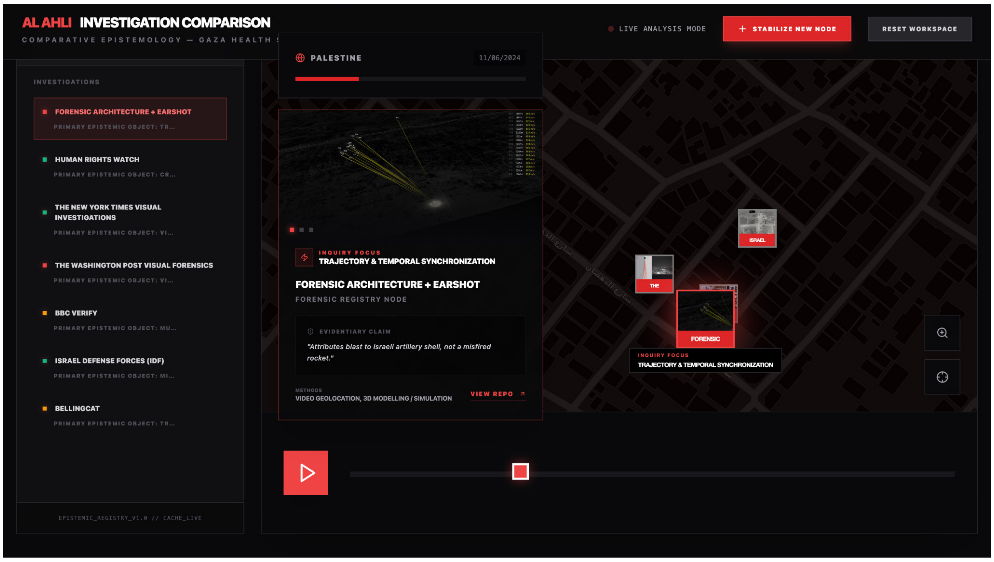
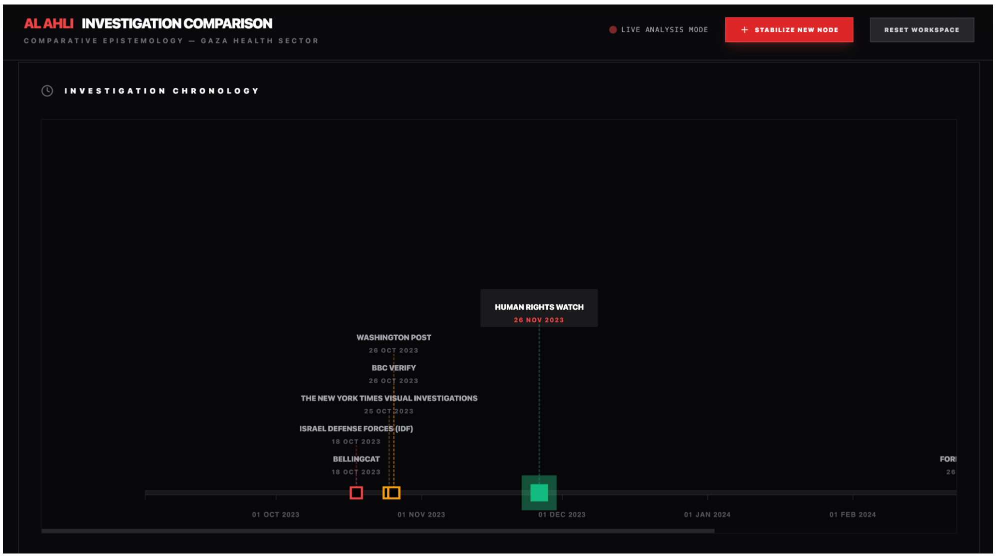


Fig.38 is an interactive web dashboard that dissects how different organisations investigated the Al-Ahli Hospital blast in Gaza (October 17, 2023) and how they reached different conclusions.



Tech Stack

React 19 + TypeScript · Vite · D3.js · Leaflet · Tailwind CSS · Node20

Brief description

The Webpage was built using React 19 with TypeScript, D3.js for network visualisation, Leaflet for spatial mapping, and Tailwind CSS. The source code and full dataset, including the FA-derived map layers and instructions for contributing additional data, are available at Repo. The dashboard is accessible at [https://github.com/Engy-ai/epistemic-play--2-].

It is structured around ten interconnected features, each addressing a different dimension of the comparative analysis. Its underlying spatial data draws partly on open-source material published by Forensic Architecture in their public GitHub repository[124], includes facility status information and map overlays from their continuing record of assaults on Gaza's medical infrastructure. The tool-kit is based on an extensible foundation rather than a fixed dataset, as the data is publicly licensed and intended to be expanded—the repository welcomes new contributions. The pertinent feature descriptions below indicate where FA's released data was used.

The Spatial Registry uses FA's published map layers for hospital locations and facility status throughout Gaza to map the physical locations pertinent to each investigation against public satellite data. Using this spatial data as the basis for the comparative interface is a methodological decision in and of itself; it grounds the analysis in the event's physical geometry and highlights the limitations imposed by occluded angles, camera placements, and accessible sightlines that influenced what each investigator could determine. Spatial access is a structural determinant of the type of knowledge that an investigation can yield, not a neutral circumstance. Furthermore, we wanted to highlight FA's last

The Timeline arranges publishing dates in relation to the event's chronology. In ways that are rarely stated explicitly, the temporal aspect of research is epistemically significant: Forensic Architecture was released 132 days after the IDF, whereas the IDF was published within twenty-four hours. These are not only variations in pace; they also represent various institutional demands on the knowledge-production process, varying tolerances for uncertainty, and varying linkages between study and conclusion. The Trace Evidence Pipeline shows which pieces of evidence were used in what capacity and by which investigations. The paradigm framework takes this divergence as conceptually essential. It highlights instances in which different organisations worked from shared material and reached different conclusions, as well as instances in which variations in conclusions can be traced to differences in the evidence base.

Furthermore, the Investigation Cards provide structured per-investigation documentation: the central question asked, the data inputs used, the methodological approach taken, the conclusion reached, and a paradigm classification corresponding to the analytical categories developed in Chapter 4. They are designed for comparative reference rather than extended argument.

124 Forensic Architecture. "Forensic Architecture." GitHub. Accessed April 13, 2026. https://github.com/forensic-architecture.

The Methodological Network is a force-directed graph in which nodes represent individual methods and edges represent shared use across investigations. It maps the degree to which different investigations share methodological infrastructure despite reaching different conclusions, and makes visible the system-level differences that the paradigm classification captures at a higher level of abstraction.

At last we created several diagrams, as a next step that we embedded in the website as well. These draw connections to the several investigations made.

5
Epistemological Divergence in Open-Source Investigations

Conclusion: This chapter argues that OSINT investigations are not methodologically uniform but split into two fundamentally different epistemological paradigms: conjecture-led and model-led. These paradigms produce knowledge in distinct ways and, strikingly, tend to lead to opposing conclusions about the same event.

I. Discussion

The analysis confirms that conjecture-led and model-led investigations are not simply methodological variants but epistemologically distinct systems. An exception was made for the Maher Arar Blog, which presents both approaches simultaneously.

The most striking finding to emerge is the near-perfect alignment between epistemic paradigm and final attribution. All conjecture-led investigations, except for Bellingcat´s and BBC Verify, concluded that Hamas was responsible, while model-led investigations concluded that IDF was responsible. These different claims could be addressed in many ways, there are good reasons to believe that this clean division depends on the epistemic paradigms at stake, but let us explain why.

For Ginzburg, knowledge emerges from marginal and seemingly negligible details that act as clues to an otherwise inaccessible reality. When the investigations are mainly conjecture-led, evidence, techniques/procedures, and experts' opinions are treated as parallel elements, meaning they are not necessarily intersecting or used together. It is through conjectures that the elements are connected, and eventually a hypothesis is formalised. The hypothesis arises from a chain of reasoning in which all the available elements are connected through seemingly logical inferences: if A, then probably B; if C, then probably D; etc.

In this epistemic paradigm, evidence plays the most important role; ultimately, the chain of relations that conjectural reasoning weaves must revolve around the main piece of evidence: in the case of Al-Ahli, the size of the crater was the primary evidence that led the hypothesis to point to Hamas. Here we see the weight of the evidential paradigm in action. Eventually, conjectural reasoning must make sense of the investigative traces and cannot formulate a picture of the event with too much freedom; it must stick to what the primary evidence suggests. In the same way, in the original Mark Duggan case, the presence of the gun in the grass was a major piece of evidence in favour of police innocence, and any accusation of cover-up would not directly justify the gun's location if not supported by stronger, preferably indicial, evidence.

This positioning has epistemic consequences; if we look again at Hanna's definitions of external and internal objectivity, we clearly see that this epistemic paradigm totally depends on the existence of external objects independent from any human interference.

To summarise, conjecture-led paradigms are characterised by the following attributes:

- Primate of the evidence over the other processes at stake
- Elements (evidence, expert opinions, processes, etc.) used in the investigation are treated independently of each other. Thus, there is nothing similar to Rheinberger´s systems
- The result of the investigation is a claim based on abductive reasoning, essentially a chain of inferences
- This epistemic paradigm relies on external objectivity

On the model-led approach side, we have the following findings. Model-based investigations show that evidence and procedures play precise roles in the investigative process. The sequencing of the processes deployed is fundamental for the development of the investigation, for these elements are interdependent on each other. This interdependency and internal organisation substantially create a system, different from the conjectural paradigm. We can call these systems *investigative systems* or *hybrid systems,* as they represent a conjunction between Rheinberger's experimental systems and Guinzburg's evidential paradigm.

As we have already shown, investigative systems treat evidence as data, which is first mapped into a data space (3D environments, Cartesian planes, etc.) and then processed through various techniques so that new data can emerge. The translation of different types of data into the same medium is what we call a model in Rheinberger's sense; to stress this once more, these models are not only visualisation tools but also systems that make new knowledge emerge.

These iterative processes at stake in model-led investigations can lead to claims that are significantly less dependent on indicial evidence than their counterparts. Instead, the validity of their claims ultimately depends on the limits of the model itself (As we noted with the Doppler models of Earshot and Kobs): what it can sense, to what degree of precision and under what conditions it fails.
This epistemic paradigm relies on the coherence and stability of the methodological framework; in other words, it falls under the regime of internal objectivity.

The definition of internal objectivity has important implications, as it does not entail that an investigative system, if it's consistent, is necessarily right. Instead, it entails that the system can deliver stable outcomes. Therefore, if a hybrid system leads to an incorrect conclusion, its components and processes can be reused and upgraded in other systems, as shown by the evolution of the Doppler curve model from Earshot to Kobs.

As Rheinberger also confirms, Models are upgradable, adaptable and graft-able to other models.

In summary, model-led investigation entails:

- primate of the methodology over the evidence
- evidence, techniques, protocols, ecc, are all woven and sequenced following a precise order, thus they work in a systemic way
- The result of the investigation is a claim based on the results of the model/models
- This epistemic paradigm relies on internal objectivity

Finally, from our reflection emerges that with the introduction of a model, its limits should be carefully explained. It is curious to note that Kobs and Maher Arar blog, both independent researchers, do that, while reknown organizations like Forensic Architecture and Earshot don't. Even though Forensic Architecture always presents their methodologies in detail, they never point out what are the limits of their investigative systems.

II. Conclusion

This thesis departs from a diagnosis that Joachim Harst makes explicit in *Virtuelle Investigation*: the evidential paradigm needs to be fundamentally reconfigured.

"While such investigative practices are often associated with positivist claims to knowledge in popular accounts, it is also necessary to underline their constructivist dimension from a scientific point of view: Reconstructions, visualisations, and simulations produce evidence and, in some cases, produce the traces to be evaluated in the first place (as Rheinberger demonstrated for the experimental system in 2007). Visualisations often enough lead to meaningful evidence that goes beyond their data basis. Thus, today's investigative practices both bear witness to and engage in a progressive virtualisation of investigative knowledge, which has effects on the evidential paradigm and its basic concepts (evidence, trace, reference, cognition) – and thus also on works of art that are connected to it."[125]

Where Ginzburg's investigator reads material traces through conjectural reasoning, the contemporary digital investigator constructs environments in which evidence can emerge, translating heterogeneous data into shared representational spaces from which new knowledge emerges. As Harst argues, in the digital paradigm, the trace no longer refers to physicality but to relational patterns within systems of data and computation; it becomes a procedural construct rather than a discovered remnant. This shift is not merely technological, but epistemological. The investigation of the Al-Ahli hospital explosion exemplifies this transformation: rather than

125 Joachim Harst and Nursan Celik, *"Virtual Investigations: New Perspectives on the Evidential Paradigm in Law, Literature, and the Arts,"* call for papers, University of Münster, 2022, access PDF.

relying on isolated pieces of footage or testimony, groups such as Forensic Architecture reconstructed the event through the synchronisation of multiple videos, 3D trajectory modelling, and spatial analysis, ultimately challenging official narratives by demonstrating inconsistencies in the claimed origin of the blast. Here, evidence does not pre-exist the investigation as a stable object but is produced through the alignment of media, the calibration of viewpoints, and the construction of a model capable of generating verifiable relations among fragments. And yet, as Harst demonstrates through his reading of Forensic Architecture, the OSINT field that has grown up within this reconfigured paradigm has not yet developed the theoretical vocabulary to account for how it differs from its predecessors. This thesis proposes a vocabulary.

This thesis introduces an epistemological framework for OSINT investigations and deliberately focuses solely on this approach: **examining how investigations are conducted, what methods are used, and through which paradigm knowledge is produced.**

What emerged is that two very different paradigms are at stake in OSINT practices, and one of them lacked a proper epistemological framing prior to this work. Conjecture-led investigations: dominant in the field and carried out by its most institutionally established actors. They had never been examined as a knowledge-generating system with defined structural limits and vulnerabilities. Model-led investigations, by contrast, find a natural correspondence with Rheinberger's experimental systems: they translate data into a new representational space, allow findings to emerge rather than be deduced, and carry within themselves a mechanism for self-correction through comparison. The distinction between the two paradigms maps precisely onto Hanna's definitions of external and internal objectivity — conjecture-led investigations ground their reliability in the quality and independence of the evidence; model-led investigations ground theirs in the coherence and stability of the system itself.

Once the operational differences between the approaches were clarified, we concluded that model-led approaches can guarantee internal objectivity, whereas conjecture-led approaches cannot. Hybrid systems have the advantage of being reusable, adaptable, and upgradable. In synthesis, they can better deal with human biases and errors. The same cannot be said for conjecture-led investigations, where the addition of new elements or the acknowledgement of biases does not refine the existing chain but tends to discard it entirely, requiring fresh abductive reasoning from the beginning. This is not a marginal methodological difference. It defines the conditions under which an investigation can be revised, contested, and trusted.

Because of this, conjecture-led investigations can be more vulnerable to fake evidence, whereas model-led ones could potentially expose it. The structural reason for this lies in Rheinberger's account of how experimental systems generate knowledge: because findings emerge from the behaviour of the system as a whole rather than from individual pieces of evidence, a fabricated input must be consistent with the full technical output of the model to go undetected and a considerably higher bar than conjecture-led methods impose. The Mark Duggan case is an example of this: the investigative system eventually showed that police testimony was inconsistent with the physical evidence, a finding that no amount of institutional authority could override once the model had been run. In the Al-Ahli case, Earshot's detection of the edited IDF audio intercept demonstrates the same principle from the other direction: it was not the content of the recording that exposed the manipulation but its acoustic structure, visible only once the audio was translated into spectrogram space, precisely the kind of translation Harst identifies as constitutive of the new evidential paradigm. This resistance to fabrication is particularly important, given that actors like the IDF can counterfeit evidence. It is even more important now than before because of the widespread use of AI to fake footage. We are moving quickly into a space where OSINT is increasingly needed and indicial evidence is less and less reliable.

Apart from their positive aspects, investigative systems also present issues that must be named with equal clarity. As Rheinberger insists, models are not global representations of a phenomenon but partial, bounded epistemic instruments with defined limits and constitutive blind spots. This he calls the "illusory moment" inherent in them, the tendency to mistake the model for the totality of what it represents. Models tend to be persuasive precisely because they seem to represent a phenomenon globally, whereas they have severe limits. Their accountability grows when multiple models can be compared with each other, so that congruences and differences might emerge, and models can be reworked to be more precise. This is not an optional addition to model-led practice; it is, in Rheinberger's framework, the condition of its epistemic reliability.

To properly take advantage of a model-led investigation, we need to be able to compare it with other model-led investigations. Kobs's critique of Earshot's model and its subsequent revised formulation clearly show that without comparison, a model's limits can remain unseen, and that with comparison, the system becomes more precise. This aspect brings up a major problem in today's OSINT landscape. Only a very few renowned actors use model-led approaches, which places them in a position of supremacy compared to other agencies—the still a large technological, technical, and epistemic gap between Forensic Architecture and very few others. The rest of the field is so significant that it makes it very difficult not to be wholly persuaded by their animated videos and exhibitions. This is exactly the danger Harst identifies in the aestheticisation of the evidential paradigm: the rendered image produces conviction through its aesthetic authority, independently of the scrutiny its underlying system deserves. While we believe that their epistemic innovation is highly relevant, we also think that, to work, model-led investigation must be widely adopted. Otherwise, the risk is that being more persuasive than correct will inherit the rhetorical power of the new evidential paradigm without the comparative infrastructure that would make that power accountable.

III. Answering the research questions

R1 asked what the epistemic foundations of open-source investigation are. The answer, developed through Ginzburg, Harst, and Rheinberger, is that there are two sets of foundations, not one, and the field has largely operated without recognising the distinction. Conjecture-led investigations rest on the classical evidential paradigm: abductive reasoning from discrete traces toward the best available explanation, with reliability grounded in the quality and independence of the evidence. Model-led investigations rest on something structurally different, what Harst identifies as the virtualisation of the evidential paradigm, and what Rheinberger's framework allows us to understand as the construction of experimental systems in which knowledge is not deduced but produced through iterative data translation. These are not variations of the same epistemology; they generate knowledge through different processes, carry different capacities for self-correction, and have different vulnerabilities to manipulation.

R2 asked what differentiates the newest approaches, exemplified by Forensic Architecture, from the rest of the field. The answer lies precisely in the relationship between data and finding that Harst describes and Rheinberger theorises. In conjecture-led investigations, data is gathered and then interpreted; the investigator reasons toward a conclusion. In model-led investigations, data is translated into a new representational space — a three-dimensional reconstruction, a spectrogram, a computational simulation — and the conclusion emerges from the behaviour of that space. The investigator's role shifts from interpreter to system-builder. This is not a methodological preference but an epistemological difference, one that carries distinct implications for objectivity, revisability, resistance to fabricated evidence, and the conditions under which findings can be compared and trusted.

R3 asked whether an epistemically oriented approach can yield substantive conclusions about this field. This thesis argues that it can, and that without such an approach, the most important structural differences between investigations remain invisible. Without an epistemological framework, the debate between investigations collapses into accusations of partiality — each side impugning the motives of the other because there is no shared language for evaluating the quality of their methods. An epistemic framework does not resolve the factual question of what happened at Al-Ahli. But it allows us to say something more precise: that the model-led and conjecture-led investigations are not simply disagreeing about facts, they are producing different kinds of knowledge claims through different processes, and those processes have different structural relationships to bias, fabrication, and revision. This conclusion matters beyond the Al-Ahli case and OSINT, as a contribution to the broader epistemology of digital evidence in an age of increasingly contested visibility.

Future implications

In his talk, *Ungrounding: The Architecture of Genocide*, delivered at Tütün Deposu, Istanbul, on 12 October 2025, Eyal Weizman proposed that the era of open-source investigation may be approaching its limits. Institutions, organisations, and individuals are placing less and less significant data online; the open surface from which OSINT drew its material is contracting. Weizman argues for a return to fieldwork, to the gathering of information that only physical presence and human testimony can provide. He and Matthew Fuller frame this through the concept of the "production of evidence," in which producing and knowing are no longer separable acts. The investigator does not find evidence that pre-exists them; they participate in its materialisation. This formulation, which opened the theoretical framework of this thesis, returns here not as a starting point but as a horizon: what began as a description of how Forensic Architecture works becomes, in Weizman's later account, a prescription for how the field must evolve.

This contraction of the open-source surface does not necessarily signal the end of investigations. This may instead signal a necessary reorientation of its scale. Forensic Architecture's own institutional trajectory points in this direction: the organisation has been expanding into new geographic contexts, establishing collaborations and presences in regions closer to the violations they document. An example of that would be Forensics in Germany and Index in France. This gradual decentralisation suggests a shift in the unit of investigation, from the global to the local, from a single organisation attempting to reconstruct events remotely from anywhere in the world, to a network of situated investigators capable of combining physical proximity with model-led methods. A more distributed field, anchored in local knowledge and local presence, would also begin to address one of the central structural problems identified throughout.

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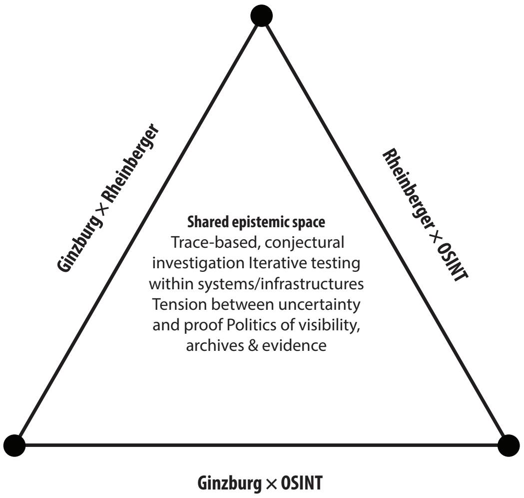
Appendix

A1. First distinctions of the similarities between OSINT - Ginzburg and Rheinberger's Theory

RHEINBERGER
Experimental systems • Epistemic things / technical objects • Lab set-ups as generators of the unknown • Iteration, path-dependence, productive error



GINZBURG
Evidential / indicial paradigm • Clues, traces, marginal details • Conjectural, case-based reasoning • Reconstruction of singular past events • History, psychoanalysis, detective work

OSINT
Open-source investigations • Digital traces: satellite, video, social media • Platform-governed, adversarial data fields • Mixed forensic, journalistic, legal publics • Collaborative, networked work

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Ginzburg × Rheinberger
How clue-reading (Ginzburg) and experimental systems (Rheinberger) overlap.

Both reject purely abstract theory: knowledge comes from handling material traces in concrete settings (archives, labs, case files, images).
In both, seemingly minor details can re-route an entire investigation or research trajectory.
The "case" (historical dossier / lab setup) becomes a machine for generating questions, not just answers.
Time structure tension: Ginzburg looks back to a singular past event, while Rheinberger designs systems oriented toward an open, experimental future.

Rheinberger × OSINT
Thinking of OSINT infrastructures as experimental systems.

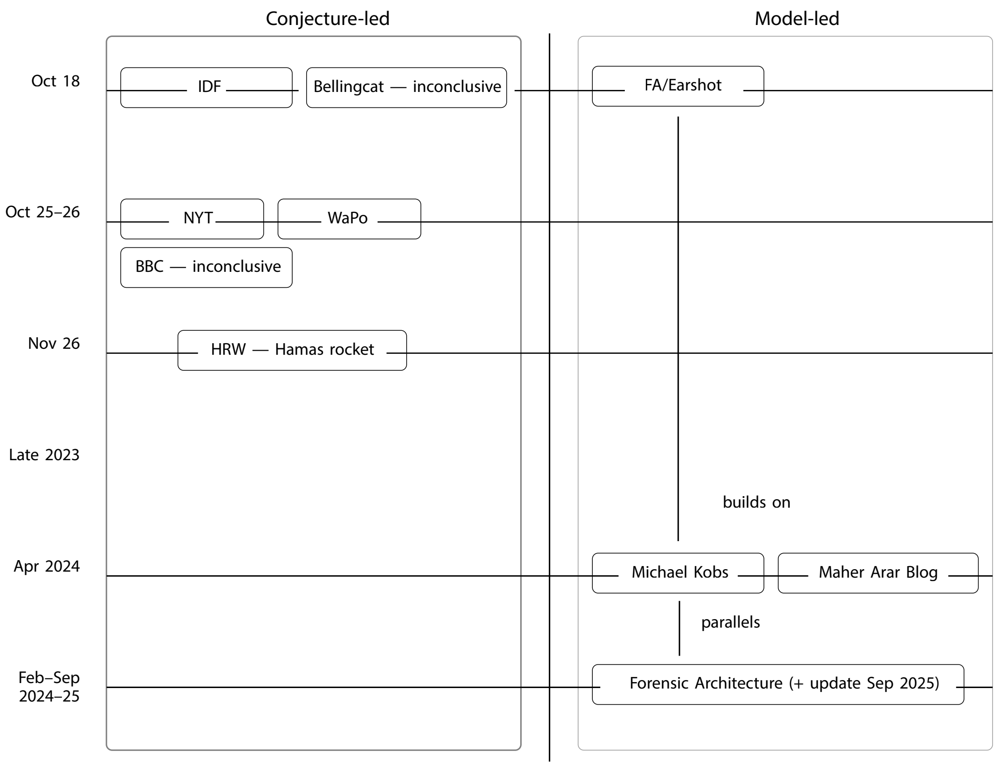
OSINT stacks (APIs, scraping tools, mapping platforms, ML models, collaboration protocols) function like laboratory ensembles that shape what can be seen.
Investigations are iterative and path-dependent: early choices of tools, coordinates, and datasets constrain future questions and findings.
Errors and dead ends (mis-geotags, false positives, model failures) are productive: they reveal the limits of the system and lead to new methods or data sources.
Unlike a closed lab, OSINT systems operate in hostile, platform-controlled environments where data can disappear, be manipulated, or be weaponised.

Ginzburg × OSINT
OSINT as a contemporary version of the evidential / indicial paradigm.

OSINT analysts read tiny digital traces (shadows, reflections, metadata, compression artefacts, interface elements) like Ginzburg's art historians read ears and fingernails.
Both work with incomplete, noisy data and rely on conjectural reasoning, cross-checking clues across multiple sources.
Triangulation (many weak clues one strong claim) mirrors Ginzburg's idea of building a case from scattered indices.
The difference: Ginzburg's paradigm is mostly historiographic, while OSINT is embedded in live conflicts, legal claims, and media battles.

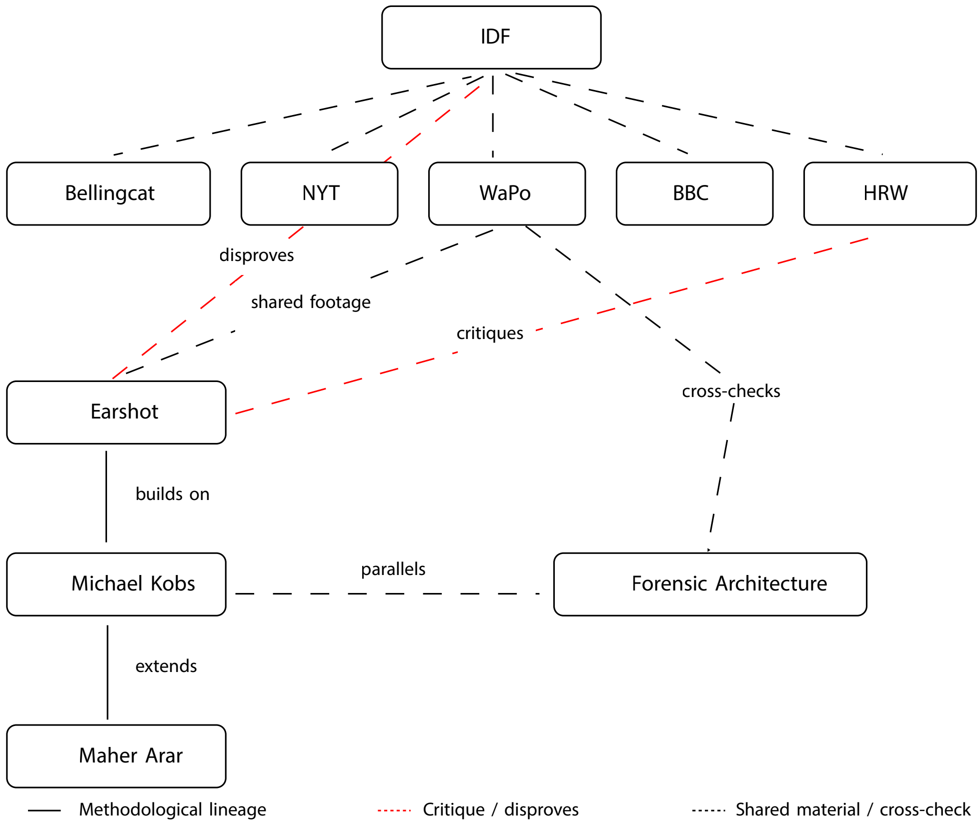
A2. Investigation Comparison

Actor	Published	Paradigm	Key method	Conclusion
IDF Israel Defence Forces	Oct 18, 2023	Institutional	Radar + audio intercept	Hamas rocket
Bellingcat OSINT collective	Oct 18, 2023	Conjecture	Satellite imagery 3D crater scan	Inconclusive
New York Times Visual Investigations	Oct 25, 2023	Conjecture	Satellite, geolocation blast analysis	Hamas rocket
Washington Post Visual Forensics	Oct 26, 2023	Conjecture	Triangulation, audio blast analysis	Hamas rocket
BBC Verify Public broadcaster	Oct 26, 2023	Conjecture	Geolocation, sound timing, ballistics	Inconclusive
Human Rights Watch International NGO	Nov 26, 2023	Conjecture	Satellite imagery ballistic experts	Hamas rocket
Earshot Audio forensics org	Late 2023	Model-led	Doppler curve spectrogram	Israeli origin
Michael Kobs Independent researcher	Apr 8, 2024	Model-led	Doppler simulation 3D photomatch	Israeli jet
Maher Arar Blog Independent blog	Apr 23, 2024	Mixed	Extended Kobs model drag/mass simulation	GBU-39
Forensic Architecture Research agency	Feb 26, 2024	Model-led	3D trajectory Blender model	Israeli strike

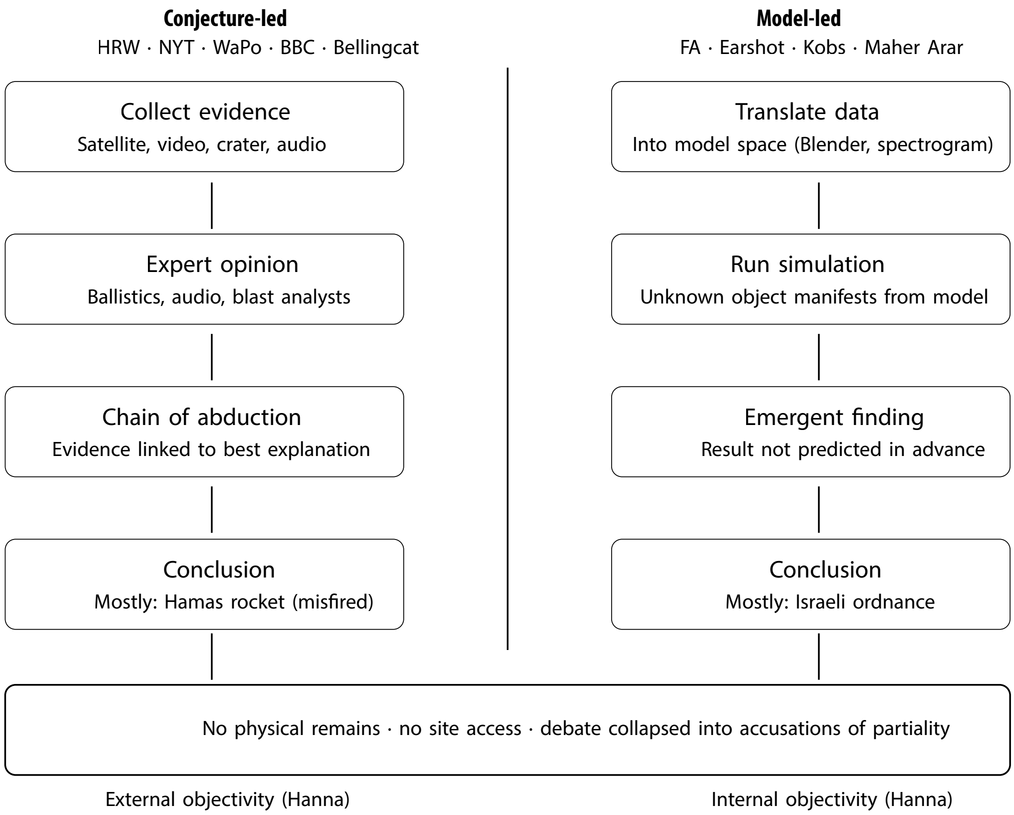


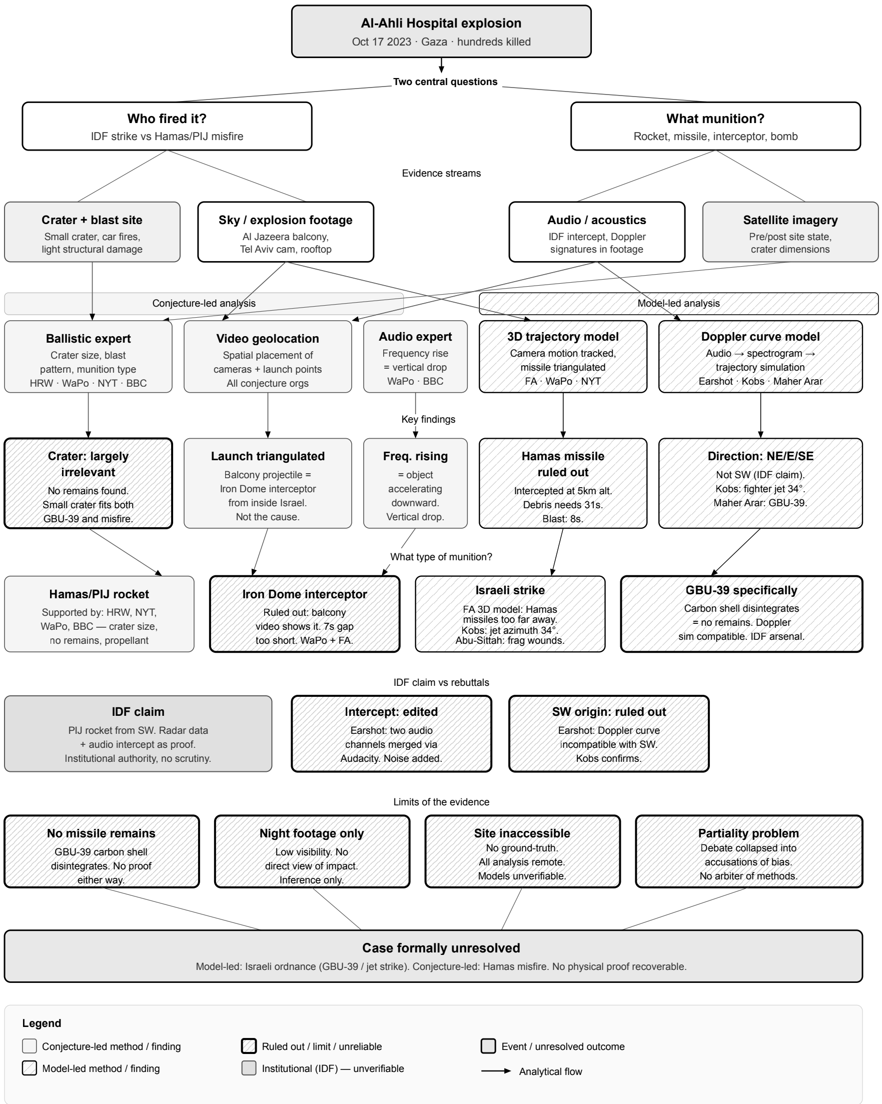
A3. Investigation Timeline

A3. Influence Network



A4. Flowchart




Al-Ahli Hospital explosion
Oct 17 2023 · Gaza · hundreds killed
Two central questions
Who fired it?
IDF strike vs Hamas/PIJ misfire
What munition?
Rocket, missile, interceptor, bomb
Evidence streams
Crater + blast site
Small crater, car fires, light structural damage
Sky / explosion footage
Al Jazeera balcony, Tel Aviv cam, rooftop
Audio / acoustics
IDF intercept, Doppler signatures in footage
Satellite imagery
Pre/post site state, crater dimensions
Conjecture-led analysis
Model-led analysis
Ballistic expert
Crater size, blast pattern, munition type
HRW · WaPo · NYT · BBC
Video geolocation
Spatial placement of cameras + launch points
All conjecture orgs
Audio expert
Frequency rise = vertical drop
WaPo · BBC
3D trajectory model
Camera motion tracked, missile triangulated
FA · WaPo · NYT
Doppler curve model
Audio → spectrogram → trajectory simulation
Earshot · Kobs · Maher Arar
Key findings
Crater: largely irrelevant
No remains found. Small crater fits both GBU-39 and misfire.
Launch triangulated
Balcony projectile = Iron Dome interceptor from inside Israel. Not the cause.
Freq. rising
= object accelerating downward. Vertical drop.
Hamas missile ruled out
Intercepted at 5km alt. Debris needs 31s. Blast: 8s.
Direction: NE/E/SE
Not SW (IDF claim). Kobs: fighter jet 34°. Maher Arar: GBU-39.
What type of munition?
Hamas/PIJ rocket
Supported by: HRW, NYT, WaPo, BBC — crater size, no remains, propellant
Iron Dome interceptor
Ruled out: balcony video shows it. 7s gap too short. WaPo + FA.
Israeli strike
FA 3D model: Hamas missiles too far away. Kobs: jet azimuth 34°. Abu-Sittah: frag wounds.
GBU-39 specifically
Carbon shell disintegrates = no remains. Doppler sim compatible. IDF arsenal.
IDF claim vs rebuttals
IDF claim
PIJ rocket from SW. Radar data + audio intercept as proof. Institutional authority, no scrutiny.
Intercept: edited
Earshot: two audio channels merged via Audacity. Noise added.
SW origin: ruled out
Earshot: Doppler curve incompatible with SW. Kobs confirms.
Limits of the evidence
No missile remains
GBU-39 carbon shell disintegrates. No proof either way.
Night footage only
Low visibility. No direct view of impact. Inference only.
Site inaccessible
No ground-truth. All analysis remote. Models unverifiable.
Partiality problem
Debate collapsed into accusations of bias. No arbiter of methods.
Case formally unresolved
Model-led: Israeli ordnance (GBU-39 / jet strike). Conjecture-led: Hamas misfire. No physical proof recoverable.
Legend
Conjecture-led method / finding
Model-led method / finding
Ruled out / limit / unreliable
Institutional (IDF) — unverifiable
Event / unresolved outcome
Analytical flow

**Now from me to you and you to me:
It has not been easy in the world in the last years, but we managed.**

Fritz:

Now, to the person I wrote this thesis with. There is so much to say, you are really talented, smart and sweet. Every now and then I am astonished by how many things you can do and how well you do them. In addition to this you have always been my best friend in Berlin. Thank you for being Engy.

Engy:

I feel incredibly greatful to have shared this journey with Fritz over the past three years. From working side by side on countless projects throughout our Master's, to traveling together to Winterthur and Linz, presenting our work, and ultimately arriving at this moment. This experience has been deeply shaped by our collaboration. This thesis reflects not only the work itself, but also the process we built together along the way. Thank you for being my best friend, and for making both Berlin and this Master's feel like home.